# The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage

GeoVac Project

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**Abstract**

The GeoVac framework has, over Papers 0–31, accumulated a body of structural results—the $S^3$ Fock-projection equivalence (Paper 7 [26]), the angular sparsity theorem (Paper 22 [29]), the Hopf-graph lattice-gauge structure (Paper 25 [32]), QED on $S^3$ (Paper 28 [33]), the Bargmann–Segal $S^5$ HO lattice (Paper 24 [31]), the SU(2) Wilson gauge (Paper 30 [34]), and the universal/Coulomb partition (Paper 31 [35])—each self-contained but together suggesting one parent structure. **[INTERNAL THEOREM]** This paper names and constructs that structure: a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected $S^3$ graph, with $\mathcal{A}_{\mathrm{GV}}$ the algebra of complex functions on graph nodes, $\mathcal{H}_{\mathrm{GV}}$ the Camporesi–Higuchi spinor space restricted to the truncation, and $D_{\mathrm{GV}}$ the diagonal Camporesi–Higuchi Dirac operator [23] on the truncated spinor space of unit $S^3$ (no graph form is attached to $D_{\mathrm{GV}}$; Remark 6). This is not a new derivation; it is a structural map. Three results follow. **[OBSERVATION]** (1) *Lineage.* The construction sits inside the published Marcolli–van Suijlekom gauge-network framework [6] (with the Perez-Sanchez correction [11] that the continuum limit is Yang–Mills without Higgs), and inside the Connes–van Suijlekom spectral-truncation program [12, 15, 16, 17]. Every structural ingredient of the GeoVac construction has published precedent; what is not duplicated is the $\alpha$ observation ($K = \pi(B + F - \Delta)$ matches $\alpha^{-1}$ to $4.8 \times 10^{-7}$; the root of the cubic $1/\alpha = K - \alpha^2$ matches to $8.8 \times 10^{-8}$; Paper 2 [25]) from a discrete spectral construction. **[OBSERVATION]** (2) *Sub-sector identification.* Paper 28's QED on $S^3$ is the spectral-action computation on the spectrum of $D_{\mathrm{GV}}$; Paper 31's universal/Coulomb partition is the $\mathcal{A}/D$ split of the present triple's spectral data. Paper 25's Hodge-1 Laplacian and Paper 30's SU(2) Wilson action are read as the Cartan-torus and full non-abelian slices of an almost-commutative extension by $M_2(\mathbb{C})$, but through the gauge-network (edge-hopping) Dirac of those papers, *not* through $D_{\mathrm{GV}}$: the diagonal $D_{\mathrm{GV}}$ commutes with every $\pi(f)$ and so admits no inner fluctuations. The four papers are read as four projections of one structure, at the level of spectral and algebraic data. **[INTERNAL THEOREM]** (3) *Axiom audit.* $\mathcal{A}_{\mathrm{GV}}$ is finite, involutive, faithfully represented on $\mathcal{H}_{\mathrm{GV}}$; $D_{\mathrm{GV}}$ is self-adjoint, and since $D_{\mathrm{GV}}$ and every $\pi(a)$ are diagonal in the same basis, $[D_{\mathrm{GV}}, \pi(a)] = 0$ identically: the diagonal triple is a *spectral*, not a metric, object (the metric role belongs to the non-diagonal multipliers of the truncated operator system on the *true* Camporesi–Higuchi window — the span of complete Dirac eigenspaces, Paper 38 [53] — whose state spaces are proved to converge under the left-translation Lipschitz seminorm; convergence in the Dirac-commutator seminorm is **[OPEN]**, that seminorm being non-degenerate on the window only as measured at $n_{\max} \leq 3$). A real structure $J_{\mathrm{GV}}$ exists on the Dirac sector with $J_{\mathrm{GV}}^2 = -\mathbb{1}$ (the quaternionic/pseudoreal sign for the 2-dimensional Dirac spinor representation of Spin(3) = SU(2)) and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, the signs $(\epsilon, \epsilon') = (-1, +1)$ of the continuum $S^3$ factor (KO-dimension 3 mod 8 [1]). The continuum factor is odd, and so is the true Camporesi–Higuchi window (measured, $n_{\max} \leq 3$: the commutant of its operator system is the scalars), the truncation that converges; only the chirality-doubled Fock/Weyl *model* truncation

is even, via the chirality-doubling grading $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$, with sign triple $(-,+,+)$ and no KO label attached at finite cutoff. **[MEASURED]** On the Fock/Weyl *model* of the operator system (the computational object of this paper, in its Weyl and chirality-doubled forms) the order-zero and order-one conditions fail at finite $n_{\max}$, with residuals that *grow* over the measured cutoffs (order zero at $n_{\max} = 2, 3, 4$, order one at $n_{\max} = 2, 3$; mechanism open). **[MEASURED]** On the true Camporesi–Higuchi window that converges, the charge conjugation $J\psi = (i\sigma_y)\overline{\psi}$ satisfies $J^2 = -\mathbb{1}$, $JD = +DJ$ and $J\mathcal{O}J^{-1} = \mathcal{O}$ exactly, while order zero and order one fail from $n_{\max} = 1$ with residuals that grow toward the sup-norm scale of the generators; there every residual is a Toeplitz boundary term of the compression, confined to the top Dirac levels of the window. **[OBSERVATION]** The four-way $S^3$ coincidence—Fock projection image, Hopf bundle total space $S^3$ (base $S^2$), Camporesi–Higuchi spin carrier, SU(2) gauge manifold—is the statement that the manifold reached by the operator-system truncations in the continuum limit, the spinor bundle of the Camporesi–Higuchi Dirac, and the gauge group of the SU(2) almost-commutative extension (Papers 25/30) all coincide on the unique rank-1 non-abelian compact simply-connected Lie group.

**Scope.** This paper does not derive new physics. It synthesizes existing GeoVac results into the axiomatic spectral-triple language. The synthesis lifts each result from a free-standing observation to a sub-sector of a single named structure, and makes explicit which axioms hold rigorously, which hold up to the standard finite-truncation caveats, and which remain open. **[PANEL-VERIFIED]** The real-structure question—whether the GeoVac Dirac operator $D_{\mathrm{GV}}$ (the diagonal spectral form) carries a real structure matching the signs of $S^3$ exactly at finite $n_{\max}$ or only in the continuum limit—is resolved in favour of finite-$n_{\max}$ exactness *for the three J-relations* $J^2 = -\mathbb{1}$, $JD = +DJ$, $J\mathcal{O}J^{-1} = \mathcal{O}$ only, measured on the Fock/Weyl model objects and, independently, on the true Camporesi–Higuchi window. **[MEASURED]** The order-zero and order-one conditions fail at finite $n_{\max}$ with residuals that grow with the cutoff, on the Fock/Weyl model of the operator system and on the true window alike (on the true window as a boundary term of the compression). The Fock/Weyl model truncation is even (chirality doubling; sign triple $(-,+,+)$, no KO label); the true window is odd (measured, $n_{\max} \leq 3$), like the continuum (its operator system generates all of $B(\mathcal{H})$, so it admits no grading), and there the pair $(\varepsilon, \varepsilon') = (-,+)$ is forced by equivariance; the residual open items are catalogued in Sec. 10.

# 1 Introduction

## 1.1 Why a spectral triple, and why now

Across the thirty-two papers (0–31) preceding this one, the GeoVac framework has carried four distinct strands. Strand 1 is the Fock projection $p_0 = \sqrt{-2E_n}$ that maps the hydrogenic Hamiltonian onto a function of the Laplace–Beltrami operator $\Delta$ on the unit three-sphere, $H = -\frac{Z^2}{2}(1-\Delta)^{-1}$, where $\Delta$ has continuum eigenvalues $\lambda_n = -(n^2-1)$; the graph nodes carry the quantum-number labels $(n, l, m_l)$, not that spectrum (Paper 7). Strand 2 is the Hopf bundle $S^1 \to S^3 \to S^2$ that organizes the discrete edge structure into ladder operators with $U(1)$ phase content (Papers 2, 25). Strand 3 is the Camporesi–Higuchi Dirac spectrum $|\lambda_n^{\mathrm{D}}| = n + 3/2$ with degeneracy $g_n^{\mathrm{D}} = 2(n+1)(n+2)$ that controls QED on $S^3$ (Paper 28). Strand 4 is the Wilson lattice gauge structure with SU(2) link variables on the same graph (Paper 30).

Each strand uses the unit three-sphere in a different role:

(a) $S^3$ as the *conformal image* of the hydrogenic spectrum (Paper 7);

(b) $S^3$ as the *total space* of the Hopf bundle on which Paper 2's $\alpha$ Observation is phrased;

(c) $S^3$ as the *spin carrier* of the Camporesi–Higuchi Dirac operator (Paper 28);

(d) $S^3$ as the *gauge manifold* of SU(2) Wilson links (Paper 30).

[**OBSERVATION**] Working hypothesis WH4 in the project register flags this as the strangest and most consequential coincidence in the framework: $S^3 = \mathrm{SU}(2)$ is the unique rank-1 non-abelian compact simply connected Lie group where (a)–(d) all coincide; in any other dimension or any other rank, the four roles separate. WH4 takes the position, in its current (deflated) form, that the four strands are not four coincidences but one Fock-projection statement plus three forced consequences of $S^3 = \mathrm{SU}(2)$ (its parallelizability and maximal-torus structure).

This paper makes WH4 explicit. We give a concrete construction of a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ (Sec. 3) — a spectral, not metric, object, since its diagonal $D_{\mathrm{GV}}$ commutes with the diagonal algebra, the metric role belonging to the operator-system representative on the true Camporesi–Higuchi window (Paper 38); audit the standard Connes axioms one by one, separating what holds at finite cutoff (the three $J$-relations; oddness on the true window, which the chirality-doubled model replaces by an even grading) from what holds only in the continuum limit (order zero and order one, which fail at finite cutoff on the Fock/Weyl model of the operator system and on the true window) (Sec. 4); read Papers 25, 28, 30, and 31 as structural sub-sectors of the resulting structure (Sec. 5); read the Paper 2 combination formula $K = \pi(B + F - \Delta)$ inside the spectral-action language (Sec. 6); and place the Coulomb/HO asymmetry of Paper 31 in spectral-triple terms (Sec. 7).

### 1.2 What this paper is and is not

**What it is.** A synthesis paper. The construction is explicit, finite, and reproducible from existing GeoVac modules (`geovac/lattice.py`, `geovac/dirac_matrix_elements.py`, `geovac/fock_graph_hodge.py`); the lineage citations are to published mathematical work; the audit of the Connes axioms uses standard textbook definitions [1, 2, 3].

**What it is not.** A derivation of $\alpha$. [**OBSERVATION**] Paper 2 stays where it is: a structural observation with three spectral homes of distinct type for the three pieces $B$, $F$, $\Delta$ but no first-principles derivation of the combination rule. This paper does not change that status; it gives the spectral-action language in which the open question is phrased without promoting it beyond that observation.

**What it is not, second.** A claim of the Connes–Chamseddine Standard Model. The GeoVac construction is a *rank-1* non-abelian almost-commutative extension; the SM spectral triple of [5] is rank $\geq 3$ with specific finite-algebra content $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$. GeoVac is not the Standard Model. It is a finite spectral triple that is an *instance* of Marcolli–van Suijlekom 2014 [6] (with Perez-Sanchez 2025 continuum-limit correction [11]) at the data level and an *extension* at the structural level (the $S^3$ label set singled out by Bertrand × Fock projection, master Mellin engine, explicit Hopf algebra $\mathcal{H}_{\mathrm{GV}}$, Tannakian dual $U^*$), [**OBSERVATION**] with the empirically interesting property that a combination of three invariants drawn from different sectors of the triple, $K = \pi(B+F-\Delta)$, satisfies, after subtraction of the leading Sprint-A correction $\alpha^2$, a residual $K - 1/\alpha - \alpha^2 \approx 1.2 \times 10^{-5}$ (a post-correction relative residual of $8.8 \times 10^{-8}$) whose absolute size matches $\pi^3\alpha^3$ to within 0.25% with CODATA 2018 $\alpha$ — with CODATA 2022 it sits 0.52% below, so the agreement is not stable at this level across editions (Remark 11) (the raw match $|K - 1/\alpha|/(1/\alpha)$ is $4.77 \times 10^{-7}$; CODATA 2022 gives $4.76 \times 10^{-7}$ raw and $8.75 \times 10^{-8}$ for the cubic's root).

**Internal register.** This paper makes explicit the spectral-triple framing already operating informally in the working hypothesis register (kept as §1.7 of the repository's `CLAUDE.md`; the sprint, track and door codenames used throughout this paper are chronicle anchors into that file and the

repository `CHANGELOG.md`; WH1's commutative-factor leg — state-space convergence of the round-$S^3$ truncations on the true window — proved unconditionally as of 2026-06-10, Paper 38 [53]; convergence of the almost-commutative structure is **[OPEN]**). Papers continue under the rhetoric rule of §1.5: dual-description framing, no ontological priority claims, lead with computational results.

## 2 The Connes Data Recalled

A spectral triple is a triple $(\mathcal{A}, \mathcal{H}, D)$ where [1]

- $\mathcal{A}$ is an involutive (unital) algebra,
- $\mathcal{H}$ is a Hilbert space carrying a faithful $*$-representation $\pi : \mathcal{A} \to \mathcal{B}(\mathcal{H})$,
- $D : \mathcal{H} \to \mathcal{H}$ is an unbounded self-adjoint operator with compact resolvent, such that the commutators $[D, \pi(a)]$ are bounded for every $a \in \mathcal{A}$.

The triple is called *even* if there is a self-adjoint involution $\gamma$ commuting with $\pi(\mathcal{A})$ and anticommuting with $D$; otherwise *odd*. A *real structure* on the triple is an antilinear isometry $J : \mathcal{H} \to \mathcal{H}$ with $J^2 = \epsilon \mathbb{1}$, $JD = \epsilon' DJ$, $J\gamma = \epsilon'' \gamma J$ (the signs $(\epsilon, \epsilon', \epsilon'')$ are determined by the KO-dimension mod 8 [1]); for the canonical Riemannian spin manifold of dimension $n$, the KO-dimension is $n$ mod 8 and the signs are tabulated.

The two structural axioms of central interest below are:

(**O1**) *Order one:* $[[D, \pi(a)], \pi(b)^\circ] = 0$ for all $a, b \in \mathcal{A}$, where $b^\circ = JbJ^{-1}$ is the right action induced by the real structure;

(**R**) *Reality:* $J$ exists with the signs above.

For an *almost-commutative* spectral triple $(\mathcal{A}_M \otimes \mathcal{A}_F, \mathcal{H}_M \otimes \mathcal{H}_F, D_M \otimes \mathbb{1} + \gamma_M \otimes D_F)$ [3, 5], the manifold sector $\mathcal{A}_M = C^\infty(M)$ is replaced in our finite setting by the algebra of complex-valued functions on a finite graph, with the convergence to $C^\infty(M)$ carried by the true-window operator-system representative of Theorem 5 (the diagonal algebra itself is the gauge-network convention; see Remark 1). This is exactly the move made by Marcolli and van Suijlekom in their gauge networks construction [6] and continued by Perez-Sanchez [11]: replace the manifold algebra by an algebra on a discrete vertex set, with edges carrying connection data. The GeoVac construction sits in this lineage by design.

## 3 The GeoVac Triple: Explicit Construction

Fix a truncation $n_{\max} \geq 1$. All objects below are finite.

### 3.1 The algebra $\mathcal{A}_{\mathrm{GV}}$

Let $V_{\mathrm{Fock}}$ denote the set of Fock-graph nodes at truncation $n_{\max}$, i.e. quantum-number triples $(n, l, m_l)$ with $1 \leq n \leq n_{\max}$, $0 \leq l \leq n-1$, $-l \leq m_l \leq l$. The total node count is $N_{\mathrm{Fock}}(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2$.

**Definition 1** (Function algebra on the Fock graph)**.** $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\mathrm{Fock}}}$*, the commutative* $*$*-algebra of complex-valued functions on* $V_{\mathrm{Fock}}$ *under pointwise multiplication, with involution* $f^*(v) = \overline{f(v)}$*.*

This is precisely the manifold algebra of the gauge networks of [6] (their $C(V)$, with $V$ the vertex set), with the GeoVac-specific choice that $V$ is the Fock-projected $S^3$ graph of Paper 7 rather than a generic graph. It is finite-dimensional ($\dim_{\mathbb{C}} \mathcal{A}_{\mathrm{GV}} = N_{\mathrm{Fock}}$), unital (1 is the constant function 1), and abelian.

**Remark 1** ($\mathcal{A}_{\mathrm{GV}}$ as the diagonal representative of an operator system)**.** ***[OBSERVATION]*** *In the parallel reading of GeoVac through the spectral-truncation framework of Connes and van Suijlekom [12], Definition 1 describes $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} = \mathbb{C}^{V_{\mathrm{Fock}}}$ as the diagonal (gauge-network) representative attached to the operator system (its $C^*$-envelope is $M_N(\mathbb{C})$, since* prop $= 2$ *gives $\mathcal{O}^2_{n_{\max}} = M_N(\mathbb{C})$)*

$$\mathcal{O}_{n_{\max}} := P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}},$$

*where $P_{n_{\max}}$ is the index-truncation projecting onto the span of the first $n_{\max}$ Fock shells (in the spinor-harmonic basis of $L^2(S^3)$ adapted to the SO(4) decomposition). This is structurally identical to the Connes–van Suijlekom worked example on $S^1$ (their $P_n$ onto* $\mathrm{span}_{\mathbb{C}}\{e_1, e_2, \ldots, e_n\}$*, [12, §3.1]), the spectral projection $\chi_{[1,n]}(D)$ of the circle Dirac operator; $P_{n_{\max}}$ is likewise a spectral projection, onto the first $n_{\max}$ eigenspaces of the Laplace–Beltrami operator. The genuine truncated algebra-image $\mathcal{O}_{n_{\max}}$ is $*$-closed but* not *multiplicatively closed; it carries nontrivial off-diagonal matrix elements*

$$\langle (n,l,m_l)|\, f\, |(n',l',m_l')\rangle = \int_{S^3} \overline{Y_{nlm_l}}\, f\, Y_{n'l'm_l'}\, dV$$

*between distinct n-shells. These overlap integrals are exactly the Gaunt / Wigner-3j angular machinery and the hypergeometric Slater radial integrals already implemented throughout the GeoVac pipeline (Paper 22 angular sparsity; Paper 14 ERI structure). The off-diagonal data therefore already exists in the framework; $\mathcal{A}_{\mathrm{GV}}$ as written in Definition 1 is its diagonal restriction.*

*The two readings agree on diagonal elements and on the spectral data of $D_{\mathrm{GV}}$; they differ in what is taken as the algebraic content surviving the truncation. Connes and van Suijlekom note that taking the $C^*$-envelope of $\mathcal{O}_\Lambda$ typically collapses to a "non-informative full matrix algebra" – the algebraic-structure information lives in the operator-system data thrown away by the envelope construction. Constructing $\mathcal{O}_{n_{\max}}$ explicitly in $\mathcal{B}(\mathcal{H}_{\mathrm{GV}})$ and computing its propagation number and Connes distance on node-evaluation states is the deliverable of the WH1 round-2 sprint (see* `debug/wh1_connes_vs_mapping_memo.md`*, 2026-05-03). The present subsection retains the $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ definition because (a) it matches the Marcolli–van Suijlekom gauge-network convention used throughout Papers 25 and 30, and (b) it carries the spectral data (spectrum, degeneracies, real structure). The two representatives are* not *interchangeable for axiom-level or metric content: on the diagonal representative $[D_{\mathrm{GV}}, \pi(a)] = 0$, so it carries no metric and the order-zero/one conditions hold vacuously, whereas on $\mathcal{O}_{n_{\max}}$ the commutators are nonzero and the order-zero/one conditions fail at finite $n_{\max}$ (Sec. 4).*

Which operator system converges. *The $\mathcal{O}_{n_{\max}}$ defined above — the first $n_{\max}$ Fock shells, and its Weyl and chirality-doubled spinor lifts (*`SpinorTruncatedOperatorSystem`*,* `FullDiracTruncatedOperatorSystem` *— is a* computational model*, and every finite-cutoff axiom measurement in this paper (propagation number, Connes distances, the three J-relations, the order-zero/one residuals) is made on it, except the true-window real-structure audit (item 6 of Sec. 4). The object whose continuum limit is proved (Theorem 5, Paper 38 [53]) is a different one: the truncation to the* true *Camporesi–Higuchi window, the span of the complete Dirac eigenspaces with $|\lambda| \le n_{\max} + \frac{1}{2}$, whose operator system $\mathcal{O}^{\mathrm{CH}}_{n_{\max}}$ has dimension $5, 30, 91$ at $n_{\max} = 1, 2, 3$ (*`tests/test_p38_true_ch_window.py`*). The doubled model has the same labels and Hilbert-space dimensions but is not $\mathrm{SU}(2)_L$-invariant, omits the blocks coupling positive and negative Dirac levels, and reaches only the bands $N \le 2n_{\max} - 1$; its quantities are model quantities, not window quantities (Paper 38, § Truncations). On the true*

*window the Dirac-commutator kernel is exactly the scalars at $n_{\max} \leq 3$ (Paper 38, remark on the Dirac-commutator seminorm), and the real-structure audit is repeated on it in item 6 of Sec. 4; its propagation number is not measured.*

**Proposition 1** (Propagation number of $\mathcal{O}_{n_{\max}}$)**.** ***[PANEL-VERIFIED]*** *Let $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ be the truncated operator system of Remark 1, viewed as a $*$-closed subspace of $M_N(\mathbb{C})$ with $N = N_{\mathrm{Fock}}(n_{\max})$. The propagation number in the sense of Connes & van Suijlekom ([12], Definition 2.39) satisfies*

$$\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2 \qquad \text{for every } n_{\max} \in \{2, 3, 4\}, \tag{1}$$

*matching the Toeplitz operator system $C(S^1)^{(n)}$ ([12], Proposition 4.2:* $\mathrm{prop}(C(S^1)^{(n)}) = 2$ *independently of $n$) verbatim. Equivalently,* $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}^2) = N^2$ *while* $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}) < N^2$ *strictly:*

| $n_{\max}$ | $N$ | $N^2$ | $\dim(\mathcal{O}_{n_{\max}})$ | $\dim(\mathcal{O}_{n_{\max}}^2)$ |
|---|---|---|---|---|
| 2 | 5 | 25 | 14 | 25 |
| 3 | 14 | 196 | 55 | 196 |
| 4 | 30 | 900 | 140 | 900 |

*The dimension fraction* $\dim(\mathcal{O}_{n_{\max}})/N^2$ *falls* $0.560 \to 0.281 \to 0.156$ *across the three truncations: the operator system is a vanishing fraction of its $C^*$-envelope as $n_{\max}$ grows, but its second power fills the envelope completely. (This is the scalar Fock-shell model of Remark 1, not the true Camporesi–Higuchi window $\mathcal{O}_{n_{\max}}^{\mathrm{CH}}$ of the convergence theorem, whose propagation number is not measured here.)*

*Computational verification.* The full construction and verification is documented in `geovac/operator_system.py` (the `TruncatedOperatorSystem` class) and `tests/test_operator_system.py` (24 tests, all passing; CPU time ∼20 s). In summary: a basis of $\mathcal{O}_{n_{\max}}$ is built from the multiplier matrices $M_{NLM}$ corresponding to the hyperspherical harmonics $Y^{(3)}_{NLM}$, with selection rules $|n-n'|+1 \leq N \leq n+n'-1$ (SO(4) triangle), $|l-l'| \leq L \leq l+l'$ with $l+l'+L$ even ($S^2$ Gaunt parity), $M = m-m'$, and the angular $S^2$ part evaluated symbolically via `sympy.physics.wigner.wigner_3j`. Linear independence of $\{M_{NLM}\}$ is verified at machine precision; the propagation number is read off as the smallest $k$ with $\dim(\mathcal{O}_{n_{\max}}^k) = N^2$. The result prop = 2 is robust: it is invariant under the choice of SO(4) radial-overlap placeholder (verified across three placeholders including a pure indicator that retains only the support pattern), so (1) does not depend on the radial value convention (it is the generic value, see the remark below). See `debug/wh1_round2_propagation_memo.md` (2026-05-03) for the full computational record, including the explicit witness pair $a = M_{2,1,0}$ at $n_{\max} = 2$ for which $a^2 \notin \mathcal{O}_{n_{\max}}$ (Frobenius residual $\approx 14.9\%$), with the deficit concentrated on the top-shell diagonal entries. □

**Remark 2** (Significance of prop = 2)**.** *Connes & van Suijlekom prove* $\mathrm{prop}(C(S^1)^{(n)}) = 2$ *for the Toeplitz operator system on the circle by showing that products of two Toeplitz shift generators reach every matrix unit $e_{ij}$ [12]. The companion dual operator system $C(\mathbb{Z})^{(n)}$ has* prop $= \infty$ *(their Proposition 4.3) — so the value* prop $= 2$ *is non-trivial even at this level: it separates the Toeplitz construction from its dual. Eq.* (1) *states that the Fock-projected $S^3$ truncation is in the same propagation class as the Toeplitz $S^1$ at the level of a quantitative Connes–van Suijlekom invariant. **[OBSERVATION]** The value should not be over-read:* prop $= 2$ *is the* generic *value for a $*$-closed unital subspace of $M_N(\mathbb{C})$ of the same complex dimension (random such subspaces of dimension* 14 *in $M_5$ and* 55 *in $M_{14}$ all have* prop $= 2$*), so the invariant certifies that $\mathcal{O}_{n_{\max}}$ is not an abelian or otherwise degenerate comparator, not that it is special among truncations. The strongest*

*quantitative alignment between GeoVac and the published Connes–van Suijlekom framework is the convergence theorem in van Suijlekom's own distance (Theorem 5), not this invariant.*

*The mechanism producing $\mathcal{O}_{n_{\max}} \subsetneq M_N(\mathbb{C})$ is the "raise-then-lower at the top shell escapes the truncation" boundary correction: a multiplier $M_{N\geq 2,L,M}$ acting at the top shell $n = n_{\max}$ would, in the continuum, be balanced by intermediate sums over $n' > n_{\max}$, but the truncation $P_{n_{\max}}$ kills these contributions. This is structurally the same boundary effect as the pendant-edge theorem of Paper 28 [33], $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max}-1)/n_{\max}$ for the graph-native ground-state self-energy, viewed from the operator-system side rather than the QED side.*

**Corollary 1** (prop = 2 separates $\mathcal{O}_{n_{\max}}$ from abelian comparators)**.** ***[PANEL-VERIFIED]*** *The propagation value* $\text{prop}(\mathcal{O}_{n_{\max}}) = 2$ *in Proposition 1 separates the Connes–van Suijlekom spectral truncation of $S^3$ from the abelian (point-evaluation) discretizations of $S^3$; it does* not *single it out among non-abelian truncations, for which* prop = 2 *is The natural circulant-style comparator on $S^3$ at $N$ points*

$$A_{\text{circ},N} := \text{span}\{E_{i,i} : i = 0, \ldots, N-1\} \subset M_N(\mathbb{C}),$$

*i.e., the commutative $C^*$-subalgebra of diagonal matrices in the "point-evaluation" basis of $\mathbb{C}^N$, satisfies*

$$\text{prop}(A_{\text{circ},N}) = \begin{cases} 1 & \textit{(intrinsic envelope: } A_{\text{circ},N} \textit{ is its own } C^*\textit{-algebra),} \\ \infty & \textit{(ambient envelope: } M_N(\mathbb{C}) \textit{ is non-abelian; no power of an abelian subalgebra reaches it).} \end{cases} \tag{2}$$

*Verified at $N \in \{2, 5, 14, 30, 55, 140\}$, matching both dimension-matching conventions (envelope-match $N(\text{circ}) = N(\text{GV})$ and operator-system-match $N(\text{circ}) = \dim(\mathcal{O}_{n_{\max}})$). Either reading of* (2) *is categorically different from the spectral-truncation value* 2*.*

*Computational verification.* The construction lives in `geovac/circulant_s3.py` (the `CirculantS3Truncation` class) with verification in `tests/test_circulant_s3.py` (35 tests, all pass; runtime ∼1.4 s). Algorithmically, $\text{prop}(A_{\text{circ},N})$ is computed via the same vec-stack rank algorithm used in the proof of Proposition 1, applied to the diagonal-matrix basis of $A_{\text{circ},N}$. The intrinsic-envelope reading exits at $k = 1$ since $\dim(A^1_{\text{circ},N}) = N = \dim_{\mathbb{C}}(A_{\text{circ},N})$ already saturates the intrinsic $C^*$-envelope. In the ambient-envelope reading, the $M_N(\mathbb{C})$ envelope has dimension $N^2$ for $N \geq 2$ but $\dim(A^k_{\text{circ},N}) = N$ for every $k$ (commutativity is preserved under self-products), so no finite $k$ achieves saturation; prop = $\infty$ in this ambient-envelope extension of Connes–van Suijlekom's Definition 2.39 (their definition uses the $C^*$-envelope, which here gives the intrinsic value 1). See `debug/wh1_round3_falsification_memo.md` (2026-05-03) for the full record, including the verbatim Connes–van Suijlekom Outlook §6 contrast (lines 2611–2618 of arXiv:2004.14115v2) between spectral and circulant truncations on $S^1$, of which Eq. (2) is the verbatim $S^3$ analog. □

**Remark 3** (Connes distance on $\mathcal{O}_{n_{\max}}$: SDP framework and structural findings)**.** ***[MEASURED]*** *The Connes distance on operator systems ([12], Eq. (2))*

$$d_D(\phi, \psi) = \sup_{x \in \mathcal{O}} \left\{ |\phi(x) - \psi(x)| : \|[D, x]\|_{\text{op}} \leq 1 \right\}$$

*restricted to pure node-evaluation states $\phi_v(x) := \langle v|x|v\rangle$ on the truncated operator system has been computed by semidefinite programming on $\mathcal{O}_{n_{\max}}$ at $n_{\max} = 2, 3$. The implementation lives in `geovac/connes_distance.py` (a `cvxpy` SDP with the operator-norm constraint lifted to the LMI $\left[\begin{smallmatrix} I & [D,x] \\ [D,x]^* & I \end{smallmatrix}\right] \succeq 0$, and a gauge-fixing constraint orthogonalizing $x$ to $\ker([D, \cdot]) \cap \mathcal{O}_h$); 17 tests in `tests/test_connes_distance.py` (all pass; runtime ∼9 s). Three structural findings emerged at the placeholder convention:*

1. ***SO(3) $m$-reflection forces zero distance.*** *For every $|n, l, -m\rangle$ and $|n, l, +m\rangle$ pair, $d_D(\phi_{|n,l,-m\rangle}, \phi_{|n,l,+m\rangle})$ 0 exactly (one such pair at $n_{\max} = 2$, four at $n_{\max} = 3$). This is the SO(3) $m$-reflection symmetry of the operator system showing through, not a pathology: the symmetry identifies the two states as members of the same orbit, and any point-discriminating function must be constant on the orbit.*

2. ***Rational structure at $n_{\max} = 2$ under placeholder multipliers (retracted as a placeholder artifact, Remark 4).*** *Under the placeholder values the five non-zero entries of the $5\times5$ distance matrix at $n_{\max} = 2$ grouped into ratios $1 : 2 : 3$ (specifically $28.87 : 57.73 : 86.60$), with $86.60 \approx 50\sqrt{3} = 86.6025$. This rational/algebraic structure is lost at $n_{\max} = 3$ as the SDP gains degrees of freedom. Under physical values the 1:2:3 grouping does not persist, though the $50\sqrt{3}$ entry does (Remark 4).*

3. ***Non-monotonicity in the Fock-graph distance.*** *Pearson correlation between $d_D(\phi_v, \phi_w)$ and the combinatorial Fock-graph distance $d_{\text{graph}}(v, w)$ (counting $\Delta n = \pm1$ and $\Delta l = \pm1$ ladder steps) is $\rho \approx -0.04$ at $n_{\max} = 2$, growing to $+0.14$ at $n_{\max} = 3$ (Spearman $0.09 \to 0.18$) under placeholder values; under physical values the sign flips (Remark 4), and the absolute level is far from a discretized-geodesic metric at our reachable cutoffs.*

*These findings should be read with three explicit caveats, the first two of which are closed in Remark 4.*

- *The SO(4) radial overlap integral $I^{lLl'}_{nNn'}$ in the construction of $\mathcal{O}_{n_{\max}}$ (Remark 1) is currently a placeholder rather than the genuine Avery–Wen–Avery 3-$Y$ integral on $S^3$. Round 2's propagation-number result (Proposition 1) is invariant under this choice; the Connes distance is not. The absolute distance values reported above are convention-dependent until the placeholder is replaced.*

- *The construction is on the scalar Fock basis $\mathcal{H}_{\text{GV}}$ only; the spinor lift to $\mathcal{H}_{\text{GV}} \otimes \mathbb{C}^4$ with the native Camporesi–Higuchi Dirac in the $(n_{\text{D}}, \kappa, m_j)$ basis is not yet incorporated. The diagonal scalar Dirac on $\mathcal{H}_{\text{GV}}$ has a large $\ker([D, \cdot]) \cap \mathcal{O}$ that makes the SDP unbounded on most state pairs; the metric values reported here use an off-diagonal Dirac proxy as a finite-distance regulator, which is a second convention choice.*

- *No comparison to the round-$S^3$ continuum geodesic distance is attempted. Such a comparison is the actual content of Connes–vS Gromov–Hausdorff convergence. The scalar case is published: Gaudillot-Estrada–van Suijlekom [18] prove spectral-truncation convergence for all compact metric groups, including $\text{SU}(2) \cong S^3$; the spinor case — on the true Camporesi–Higuchi window of complete Dirac eigenspaces, not the chirality-doubled Fock model used in this remark — is the contribution of Theorem 5 below and Paper 38.*

*The SO(3) $m$-reflection forced zeros (and the $50\sqrt{3}$ fingerprint) are robust across the placeholder choice; the rational grouping at $n_{\max} = 2$, the absolute distance values and the non-monotonicity diagnostic are not (Remark 4). See `debug/wh1_round3_connes_distance_memo.md` for the full record. Caveats (1) and (2) above are subsequently closed in Remark 4, where two of the three structural findings (the 1:2:3 grouping and the trend-toward-correlation reading) are retracted as placeholder artifacts.*

**Remark 4** (R3.1+R3.2 closure of round-3 caveats)**.** ***[OBSERVATION]*** *Two of the three caveats above have been closed in subsequent sub-sprints (continuation of the same dispatch on 2026-05-03), and two of the three structural findings (the 1:2:3 grouping and the trend-toward-correlation reading) have been retracted as placeholder artifacts.*

**R3.1 (Avery–Wen–Avery integral, placeholder → physical).** *The placeholder for $I^{lLl'}_{nNn'}$ is replaced with the closed-form Avery–Wen–Avery 3-$Y$ integral on $S^3$, factorized as $I_{\mathrm{rad}} \cdot G_{\mathrm{Gaunt}}$ with $I_{\mathrm{rad}}$ the Gegenbauer triple integral on $\chi$ and $G_{\mathrm{Gaunt}}$ the standard $S^2$ Gaunt integral. Implemented in* `geovac/so4_three_y_integral.py` *(sympy-exact,* `lru_cache`*), verified by hyperspherical orthonormality at $n_{\max} \leq 3$ in exact arithmetic.* $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ *holds at $n_{\max} = 2, 3, 4$ under the physical default with dim sequences $14 \to 25,\ 55 \to 196,\ 140 \to 900$* bit-identical *to the placeholder values—confirming the round-2 invariance claim by replacement, not by sampling.*

*The Connes distance recomputed under physical values retains the $m$-reflection forced zeros and the $50\sqrt{3}$ fingerprint, but* retracts *the $1 : 2 : 3$ rational structure as a placeholder artifact: under Avery the distinct nonzero values at $n_{\max} = 2$ are $\{1.27, 86.60, 87.18, 174.93, 175.53, 262.11\}$ with no clean integer grouping. The $M_{2,0,0}$ multiplier rescales by $\sim 3\times$ between placeholder and physical values, propagating as $\sim 9\times$ on pairs that use it twice. More importantly, the non-monotonicity* flips sign *at $n_{\max} = 3$: Pearson goes from $+0.14$ (placeholder) to $-0.36$ (physical), Spearman from $+0.18$ to $-0.33$. The optimistic "slow trend toward correlation" reading is retracted; under physical values the metric is* anti-correlated *with the Fock-graph distance. Memo:* `debug/wh1_r31_avery_wen_avery_memo.md`.

**R3.2 (spinor lift, scalar Fock → Weyl spinor on $S^3$).** *The construction extends to the Weyl sector of the Camporesi–Higuchi spinor bundle on $S^3$ via Clebsch–Gordan decomposition $|\psi_{l,j=l+1/2,m_j}\rangle = \alpha_+ |Y_{l,m_j-1/2}\rangle \otimes |\uparrow\rangle + \alpha_- |Y_{l,m_j+1/2}\rangle \otimes |\downarrow\rangle$ with $\alpha_\pm = \sqrt{(l \pm m_j + 1/2)/(2l+1)}$. Each scalar Avery multiplier $M_{NLM}$ lifts to a CG-weighted sum of two scalar matrix elements; the resulting $\widetilde{\mathcal{O}}_{n_{\max}}$ is API-compatible with the existing SDP machinery. Implemented in* `geovac/spinor_operator_system.` *$\dim \widetilde{\mathcal{O}} = \dim \mathcal{O}$ at every $n_{\max}$ (the CG decomposition introduces no new linear independencies); identity, $*$-closure, and $m_j$-reflection forced zeros all generalize naturally.*

*Two clean R3.2 findings.* ***(i)*** The truthful Camporesi–Higuchi Dirac on the spinor bundle is too $n$-degenerate *for the Connes distance to be well-defined on cross-shell pure-state pairs: at $n_{\max} = 2$, 24 of 28 cross-pairs give $+\infty$ because the $n$-shell-diagonal multipliers $M_{N,L=\mathrm{even},M=0}$ commute with $D = \mathrm{diag}(n_{\mathrm{Fock}} + 1/2)$, making the SDP unbounded. This is the spinor analog of the round-3* `shell_scalar` *diagnostic. It is a property of the Fock/Weyl spinor* model, *not of the truncated Camporesi–Higuchi triple: on the true window (complete Dirac eigenspaces) the Dirac-commutator kernel is exactly the scalars at $n_{\max} \leq 3$ (Paper 38, remark on the Dirac-commutator seminorm;* `tests/test_p38_true_ch_window.py`*).* ***(ii)*** Under a CH+offdiag perturbation that lifts the $n$-degeneracy, the metric remains anti-correlated *with the Fock-graph distance: Pearson nz $-0.36$ at $n_{\max} = 2$, $-0.26$ at $n_{\max} = 3$. The non-monotonicity is therefore robust under both the placeholder → Avery upgrade (R3.1)* and *the scalar → spinor lift (R3.2). Memo:* `debug/wh1_r32_spinor_lift_memo.md`.

**Disambiguation settled.** *The round-3 §5.4 two-reading question ("placeholder artifact" vs "pure-state-distance pathology") lands on the latter. R3.1 rules out the placeholder reading (anti-correlation* strengthens *under physical values). R3.2 rules out the scalar-Dirac-proxy reading (anti-correlation persists under the spinor lift). The Connes distance on $\mathcal{O}_{n_{\max}}$ or its spinor lift, with pure node-evaluation states, is structurally not a discretization of the round-$S^3$ geodesic distance at any cutoff we can reach. The third caveat above (no GH comparison on the full state space) remains the natural next move; whether the metric becomes monotone in the GH limit via a Kantorovich– Wasserstein lift to mixed states requires Peter–Weyl on* SU(2) *(not addressed by Connes & van Suijlekom 2021, who defer Gromov–Hausdorff questions generically: "will be studied elsewhere", their §3). The $S^3$ GH convergence itself is no longer unpublished: the scalar case is closed by Gaudillot-Estrada–*

*van Suijlekom [18] for all compact metric groups including* SU(2)*, and the spinor case, on the true Camporesi–Higuchi window (not the chirality-doubled Fock/Weyl model of this remark), by Theorem 5 and Paper 38; the Kantorovich–Wasserstein monotonicity question remains open.*

**Remark 5** (Two-sided structural alignment)**.** ***[OBSERVATION]*** *Connes & van Suijlekom's worked $S^1$ example exhibits the spectral-vs-circulant distinction along two axes simultaneously: the Toeplitz operator system $C(S^1)^{(n)}$ has* prop $= 2$ *(their Proposition 4.2), while the circulant matrix algebra $C(C_m)$ has* prop $= 1$ *trivially (it is its own $C^*$-algebra). Eqs.* (1) *(Proposition 1) and* (2) *(Corollary 1) reproduce both axes verbatim on $S^3$:*

| | *spectral truncation* | *circulant analog* |
|---|---|---|
| $S^1$ *(Connes–vS)* | $C(S^1)^{(n)},\ \text{prop} = 2$ | $C(C_m),\ \text{prop} = 1$ |
| $S^3$ *(this work)* | $\mathcal{O}_{n_{\max}},\ \text{prop} = 2$ | $A_{\text{circ},N},\ \text{prop} = 1$ *or* $\infty$ |

*The two-sided match separates spectral truncation from abelian discretization on both manifolds; it does not foreclose the generic non-abelian value (Remark above). Together with the SO(4)-invariance of the truncation $P_{n_{\max}}$ (whole Fock shells; the isometry-preservation property Connes & van Suijlekom themselves identify as distinguishing spectral truncation from circulant discretization), it is a consistency check on the construction; the load-bearing alignment is Theorem 5.*

### 3.2 The Hilbert space $\mathcal{H}_{\text{GV}}$

Let $V_{\text{Dirac}}$ denote the set of Dirac-graph nodes at the matching truncation, labeled by Camporesi–Higuchi triples $(n_{\text{D}}, \kappa, m_j)$ with $0 \le n_{\text{D}} \le n_{\max} - 1$ (the standard convention $n_{\text{Fock}} = n_{\text{CH}} + 1$ from `geovac/dirac_matrix_elements.py`), Dirac–Coulomb relativistic angular momentum $\kappa \in \{-(l+1), +(l+1)\}$ for each $0 \le l \le n_{\text{D}}$ — equivalently the fixed $j = l + \frac{1}{2}$ carrying a $\pm 1$ chirality label — and total angular momentum projection $m_j \in \{-j, \dots, j\}$ where $j = |\kappa| - 1/2$. The total node count is

$$N_{\text{Dirac}}(n_{\max}) = \sum_{n=0}^{n_{\max}-1} g_n^{\text{D}} = \sum_{n=0}^{n_{\max}-1} 2(n+1)(n+2) = \tfrac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2).$$

**Definition 2** (Spinor space on the Dirac graph)**.** $\mathcal{H}_{\text{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\text{Dirac}}}$*, with the standard Hermitian inner product on* $\mathbb{C}^{N_{\text{Dirac}}}$*.*

The faithful $*$-representation of $\mathcal{A}_{\text{GV}}$ on $\mathcal{H}_{\text{GV}}$ is by the natural pullback: for $f \in \mathcal{A}_{\text{GV}}$ and $\psi \in \mathcal{H}_{\text{GV}}$,

$$(\pi(f)\psi)(n_{\text{D}}, \kappa, m_j) := f(n_{\text{Fock}}(n_{\text{D}}), l(\kappa), m_l(\kappa, m_j)) \cdot \psi(n_{\text{D}}, \kappa, m_j), \tag{3}$$

where the maps $n_{\text{Fock}}, l, m_l$ are the standard Dirac-to-Fock quantum-number bridges: $n_{\text{Fock}}(n_{\text{D}}) = n_{\text{D}} + 1$, $l(\kappa) = |\kappa| - 1$ for both signs, and $m_l = m_j - \frac{1}{2}$ clipped into $[-l, l]$ (a fixed choice of Fock node: the Clebsch–Gordan decomposition of $|j, m_j\rangle$ superposes $m_l = m_j \mp \frac{1}{2}$ and defines no map). Each $\pi(f)$ acts diagonally; in the spinor basis it is the diagonal matrix $\text{diag}(f(v_{\text{Fock}}(\cdot)))$. The representation is faithful: $\pi(f) = 0 \Leftrightarrow f \equiv 0$ (every Fock node has at least one Dirac node above it for $n_{\max} \ge 1$).

### 3.3 The Dirac operator $D_{\text{GV}}$

The continuum Camporesi–Higuchi Dirac operator on the unit three-sphere has spectrum $|\lambda_n^{\text{D}}| = n + 3/2$ with degeneracy $g_n^{\text{D}} = 2(n+1)(n+2)$ [23]. We promote this to a finite operator on $\mathcal{H}_{\text{GV}}$ by truncation of its spectral decomposition.

**Definition 3** ($D_{\mathrm{GV}}$)**.** *$D_{\mathrm{GV}}$ is the diagonal operator on $\mathcal{H}_{\mathrm{GV}}$ with eigenvalues $\lambda_n^{\mathrm{D},\pm} = \pm(n_{\mathrm{D}} + 3/2)$ on the eigenstate $|n_{\mathrm{D}}, \kappa, m_j\rangle$ with sign + for $\kappa < 0$ and − for $\kappa > 0$ (a GeoVac Dirac–Coulomb labelling convention for the two signs of the Camporesi–Higuchi spectrum; Camporesi–Higuchi [23] supply the spinor bundle and the spectrum, not a $\kappa$ labelling).*

This is the operator used everywhere in the paper. (The relocated Q5′ cochain record, §VIII, used the hopping operator $D = \Lambda + \kappa A_{\mathrm{graph}}$; on $D_{\mathrm{GV}}$ every degree-$\geq 1$ JLO cochain vanishes, since the algebra's idempotents commute with it.) Every axiom verification below, the real structure $J_{\mathrm{GV}}$ of Proposition 4, and every finite-truncation spectral-action computation are carried out on this diagonal $D_{\mathrm{GV}}$ (`geovac/full_dirac_operator_system.py`, `geovac/real_structure.py`; backing tests `tests/test_full_dirac_operator_system.py`, `tests/test_real_structure.py`). The $\pi$-free certificate of the Dirac sector (Paper 28 §graph_native_qed.6) is a statement about this spectrum: every eigenvalue is a half-integer rational, every degeneracy is a positive integer.

**Remark 6** (No separate "graph form" of $D_{\mathrm{GV}}$)**.** *The paper carries one Dirac operator for its theorems, the diagonal one above. A hopping operator with weights chosen to reproduce its spectrum would be unitarily equivalent to it by the spectral theorem alone;* ***`geovac/dirac_matrix_elements.py`*** *holds closed-form angular (Szmytkowski) and radial (hydrogenic) matrix elements in the $(\kappa, m_j)$ basis, not a hopping operator, and the hopping operator of the relocated Q5′ record is a separate computational object. The Dirac-sector* graph *that appears elsewhere in the corpus (*`geovac/dirac_lattice.py`*, the E1-dipole adjacency of Paper 28's graph-native QED) is a different operator, $H = D + W$, whose adjacency encodes the photon-vertex selection rules rather than $D_{\mathrm{GV}}$ itself. The module contents asserted here are pinned by* `tests/test_paper32_dirac_module_contents.py`*.*

### 3.4 Bounded commutators

**Proposition 2** (Bounded commutators)**.** *For every $a \in \mathcal{A}_{\mathrm{GV}}$, the commutator $[D_{\mathrm{GV}}, \pi(a)]$ is a bounded operator on $\mathcal{H}_{\mathrm{GV}}$.*

*Proof.* $D_{\mathrm{GV}}$ has finite-dimensional domain $\mathcal{H}_{\mathrm{GV}}$ at every $n_{\max}$, hence bounded. $\pi(a)$ is a diagonal multiplication operator, also bounded. Any commutator of bounded operators is bounded. In fact more is true: $D_{\mathrm{GV}}$ and $\pi(a)$ are both diagonal in the basis $|n_{\mathrm{D}}, \kappa, m_j\rangle$, so $[D_{\mathrm{GV}}, \pi(a)] = 0$ identically. □

**[SYMBOLIC PROOF]** The check is therefore trivial in a strong sense: the commutator seminorm $a \mapsto \|[D_{\mathrm{GV}}, \pi(a)]\|$ vanishes on all of $\mathcal{A}_{\mathrm{GV}}$, the Connes distance it would induce is degenerate, the first-order differential forms $\Omega^1_{D_{\mathrm{GV}}}$ are zero, and $D_{\mathrm{GV}}$ admits no inner fluctuations. The diagonal triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ is a *spectral*, not a metric, object: it carries the Camporesi–Higuchi spectrum, the degeneracies and the real structure, but no Lipschitz geometry. The metric role belongs instead to the *non-diagonal* multipliers of the truncated operator system on the true Camporesi–Higuchi window (Remark 1; Paper 38 [53]), whose state spaces converge under the left-translation Lipschitz seminorm (Theorem 5); the theorem uses no Dirac commutator, and convergence in the Dirac-commutator seminorm is **[OPEN]** (that seminorm is non-degenerate on the window as measured at $n_{\max} \leq 3$, `tests/test_p38_true_ch_window.py`). Both halves of the contrast — $[D_{\mathrm{GV}}, \pi(f)] = 0$ identically on the diagonal representative, and nonzero commutators on the non-diagonal multipliers of the chirality-doubled model $\mathcal{O}_{n_{\max}}$ — are regression-pinned in `tests/test_paper32_diagonal_triple.py`.

### 3.5 The triple, summarized

Putting Definitions 1, 2, 3 and Proposition 2 together:

**Theorem 1** (The GeoVac spectral triple)**.** *For every $n_{\max} \geq 1$, the data $(\mathcal{A}_{\mathrm{GV}}^{(n_{\max})}, \mathcal{H}_{\mathrm{GV}}^{(n_{\max})}, D_{\mathrm{GV}}^{(n_{\max})})$ is a finite spectral triple in the sense of [1]: $\mathcal{A}_{\mathrm{GV}}$ is unital and involutive, $\pi : \mathcal{A}_{\mathrm{GV}} \to \mathcal{B}(\mathcal{H}_{\mathrm{GV}})$ is a faithful $*$-representation, $D_{\mathrm{GV}}$ is self-adjoint with bounded commutators.* ***[INTERNAL THEOREM]*** *As $n_{\max} \to \infty$ the state spaces converge to that of the canonical spectral triple of the unit Riemannian spin three-sphere $S^3$ with the Camporesi–Higuchi Dirac operator* for the operator-system representative on the true Camporesi–Higuchi window $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}^{\mathrm{CH}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ *of Theorem 5 (van Suijlekom's state-space Gromov–Hausdorff distance, translation-seminorm metrization, limit identified up to isometry; the target's spectral-triple properties are inputs, not outputs, of the convergence); neither the Fock/Weyl model $\mathcal{O}_{n_{\max}}$ of the axiom audit nor the diagonal (gauge-network) representative $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ of Remark 1, used for the axiom audit above, is the object that converges (Corollary 1 places the diagonal representative on the comparator side of the Connes–van Suijlekom distinction).*

The continuum-limit clause is van Suijlekom's state-space Gromov–Hausdorff convergence of spectral truncations [13] (in the operator-system framework of [12]), proved for the true window $\mathcal{O}_{n_{\max}}^{\mathrm{CH}}$ in Paper 38; no Cauchy-sequence statement for the diagonal algebra is made.

## 4 Axiom Audit

We check each Connes axiom in turn, distinguishing what holds rigorously at finite $n_{\max}$ from what holds only in the continuum limit.

### 4.1 Order zero (involutive faithful representation)

**[SYMBOLIC PROOF]** Holds by construction from Definitions 1–2 and Eq. (3): $\pi(f)$ acts diagonally in the $(n_{\mathrm{D}}, \kappa, m_j)$ basis, so $\pi$ is multiplicative and $\pi(f^*) = \pi(f)^\dagger$, and it is faithful because the quantum-number bridges of Eq. (3) map $V_{\mathrm{Dirac}}$ onto $V_{\mathrm{Fock}}$. No numerical certificate is involved.

### 4.2 Bounded commutators (regularity)

Holds rigorously at every $n_{\max}$ by Proposition 2.

### 4.3 Compact resolvent / finite summability

In the finite-$n_{\max}$ truncation $\mathcal{H}_{\mathrm{GV}}$ is finite-dimensional so this is automatic. The continuum statement is that $D_{\mathrm{GV}}$ has compact resolvent and is $3^+$-summable (Dixmier-trace summable in dimension $3 + \epsilon$); this is the Camporesi–Higuchi summability of the continuum operator on $S^3$ and is a standard result [23, 1].

### 4.4 Order one

**Proposition 3** (Order one, diagonal representative)**.** *For every $a, b \in \mathcal{A}_{\mathrm{GV}}$ (the diagonal representative), $[[D_{\mathrm{GV}}, \pi(a)], \pi(b)^\circ] = 0$ holds trivially, because already $[D_{\mathrm{GV}}, \pi(a)] = 0$ (Proposition 2).*

This is vacuous rather than informative: the diagonal triple has no nonzero commutators to constrain. It is *not* the statement for the Fock/Weyl model $\mathcal{O}_{n_{\max}}$ of the operator-system representative, whose multipliers do not commute with $D$ and for which the order-zero and order-one

conditions (with the real structure) *fail* at finite $n_{\max} \geq 2$ (item 5 of the finite-$n_{\max}$ audit below; Table 1 labels the two representatives separately). Order-one becomes non-trivial only when one extends to an almost-commutative triple $\mathcal{A}_{\mathrm{GV}} \otimes M_n(\mathbb{C})$ with non-abelian fiber, at which point it constrains the form of fluctuations of $D$. This is exactly the setting where Papers 25 ($n = 1$, $U(1)$ fiber) and 30 ($n = 2$, SU(2) fiber) live; we treat them in Sec. 5.

### 4.5 Real structure

$S^3$ is odd-dimensional; in dimension $n$ the canonical KO-dimension is $n$ mod 8 for a Riemannian spin manifold, and the sign table [1, 3] for $n = 3$ gives

$$\epsilon = -1, \quad \epsilon' = +1, \quad \epsilon'' = \text{ (none)} \quad \text{(no } \gamma \text{ in odd dimension)}.$$

The sign $\epsilon = -1$ reflects the fact that the 2-dimensional spinor representation of $\mathrm{Spin}(3) = \mathrm{SU}(2)$ is quaternionic (pseudoreal): the standard charge-conjugation operator on this representation squares to $-\mathbb{1}$, equivalently $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$ acts on a complex 2-dimensional space with antiunitary $J$ satisfying $J^2 = -\mathbb{1}$. This is the same sign that appears in the canonical Lorentzian Dirac formalism in $3 + 1$ dimensions for the charge-conjugation matrix $C$.

**Proposition 4** (Real structure on $\mathcal{H}_{\mathrm{GV}}$)**.** *There exists an antilinear isometry $J_{\mathrm{GV}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ with $J_{\mathrm{GV}}^2 = -\mathbb{1}$ and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, namely the charge-conjugation operator on the Camporesi–Higuchi spinor bundle restricted to the truncation.*

*Construction.* The continuum charge conjugation $J_{S^3}$ on the spinor bundle of $S^3$ satisfies the listed signs by construction [1] (the KO-dimension-3 row; Camporesi–Higuchi [23] supply the spinor bundle and spectrum, not a charge conjugation); the finite-cutoff signs are verified directly by `tests/test_real_structure.py`. Restriction to the truncation $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})}$ commutes with the spectral truncation because $D_{\mathrm{GV}}$ acts diagonally in the spectral basis and $J_{S^3}$ permutes the spectral basis (sending $|n_{\mathrm{D}}, \kappa, m_j\rangle \mapsto |n_{\mathrm{D}}, \kappa, -m_j\rangle$ up to a phase — *$m_j$ only, with the chirality label held fixed* — whose antilinear-squaring gives $-\mathbb{1}$ on the quaternionic 2-spinor representation; a chirality flip would give $\varepsilon' = -1$, since $D_{\mathrm{GV}} = \chi\,(n + \frac{1}{2})$). The implemented $J$ (`geovac/real_structure.py`) maps $m_j$ alone, as the real-structure tests verify. The restriction therefore inherits the signs. □

**Finite-$n_{\max}$ audit (Sprint WH1-Connes Step 2, 2026-05).** The qualitative argument above is now backed by a quantitative computational audit at $n_{\max} \in \{1, 2, 3\}$ on both the Weyl sector (`geovac/spinor_operator_system.py`, $\dim_{\mathcal{H}} = 2, 8, 20$) and the full-Dirac sector (`geovac/full_dirac_operator_s` $\dim_{\mathcal{H}} = 4, 16, 40$), with $J_{\mathrm{GV}}$ realized as an antilinear permutation-phase operator $J = U\mathcal{K}$ (with $\mathcal{K}$ complex conjugation) and phase convention $\sigma(l, m_j) = i^{2m_j}(-1)^l$ giving $\sigma_a \overline{\sigma_{J(a)}} = -1$ on every basis label. Implementation in `geovac/real_structure.py`; verification in `tests/test_real_structure.py` (44 tests passing); raw audit data `debug/data/wh1_real_structure_audit.json`. The audit verdict for the four Connes-axiom conditions involving $J$ at finite $n_{\max}$ on the truthful Camporesi–Higuchi Dirac is as follows; items 1–5 are measured on the Fock/Weyl and chirality-doubled *model* objects of Remark 1, and item 6 repeats the audit on the true window whose continuum limit Theorem 5 proves:

1. $J^2 = -\mathbb{1}$ **HOLDS EXACTLY** at every $n_{\max} \in \{1, 2, 3\}$ on both sectors (bit-exact zero residual). The phase formula $\sigma_a \overline{\sigma_{\pi(a)}} = -1$ is verified pair-by-pair across the spinor and full-Dirac bases.

2. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{truthful}} = +D_{\mathrm{GV}}^{\mathrm{truthful}} J_{\mathrm{GV}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual). Mechanism: the truthful CH Dirac is diagonal in $(n_{\mathrm{Fock}}, l, m_j, \text{chirality})$ with eigenvalue chi $\cdot$ $(n_{\mathrm{Fock}} + 1/2)$ depending only on $n$ and chi; $J_{\mathrm{GV}}$ permutes only $m_j$, leaving $n$ and chi fixed, so the matrix relation $U\overline{D} = DU$ is diagonal-by-diagonal.

3. $J_{\mathrm{GV}} \mathcal{O}_{n_{\max}} J_{\mathrm{GV}}^{-1} = \mathcal{O}_{n_{\max}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual on each generator $a$, with 0 of $1, 14, 55$ generators failing the $\mathcal{O}$-membership test). Mechanism: each multiplier matrix $\widetilde{M}_{NLM}$ has $J$-conjugate $\widetilde{M}_{NL(-M)}$ via the $m_l \to -m_l$ symmetry of the Wigner-3j coupling coefficients (Edmonds Eq. 3.7.5 / Avery–Wen–Avery 3-Y integral); the operator system is closed under this involution.

4. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{offdiag}} = +D_{\mathrm{GV}}^{\mathrm{offdiag}} J_{\mathrm{GV}}$ **FAILS** structurally (max residual $2 \times 10^0$ at $n_{\max} \in \{2, 3\}$; $10^{-2}$ at $n_{\max} = 1$ from the m-linear lifter alone). This is the natural counter-example: the round-3 `offdiag` Dirac is a hand-crafted Hermitian perturbation with E1-type hopping $D[a, b] = \alpha$ on the rule $|\Delta n| = |\Delta l| = 1, |\Delta m_j| \leq 1$, and the permutation-phase $\sigma$ with $(-1)^l$ parity does not commute with this perturbation because the $l$-parity sign flips under bra/ket $l \to l \pm 1$. The genuine 4-component Camporesi–Higuchi spinor Dirac (which couples large/small components via the proper $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ angular gradient with Clifford-algebra $\gamma$ matrices) would resolve this, but lies outside the present construction; the offdiag CH is best read as an SDP-bounding device (round-3 R3.5 Connes distance), not a spectral-action-foundational object.

5. **Order-zero** $[a, JbJ^{-1}] = 0$ **and order-one** $[[D, a], JbJ^{-1}] = 0$ **FAIL at finite** $n_{\max} \geq 2$ on the Weyl-sector model $\widetilde{\mathcal{O}}_{n_{\max}}$ of the operator system (not the true window of Theorem 5), with residuals of order 0.05–0.2 (largest matrix entry in absolute value, not operator norm). At $n_{\max} = 2$: 124 of 196 pairs fail order-zero (max residual $5.5 \times 10^{-2}$); 43 of 196 pairs fail order-one (max residual $1.0 \times 10^{-1}$); at $n_{\max} = 3$: 2,594 of 3,025 and 1,337 of 3,025 respectively (max residuals $7.9 \times 10^{-2}$ and $2.0 \times 10^{-1}$). The conditions hold trivially at $n_{\max} = 1$ (single-generator operator system). **[MEASURED]** The order-zero failure *grows* with the cutoff (Weyl sector): max residual $5.5 \times 10^{-2} \to 7.9 \times 10^{-2} \to 9.6 \times 10^{-2}$ and failing fraction $63\% \to 86\% \to 92\%$ (17,970 of 19,600 pairs) at $n_{\max} = 2, 3, 4$; on a *fixed* set of generators (the degree-1 harmonics, $N = 2$) the operator-norm residual runs $0.051, 0.068, 0.076, 0.081$ at $n_{\max} = 2$–$5$, and its ratio to the (growing) generator norm runs $0.225, 0.212, 0.209, 0.208$ — still drifting down, so whether it tends to a nonzero limit is not established (measured through $n_{\max} = 5$). The order-zero counts/residuals, the model order-one counts at $n_{\max} = 2, 3$ and the fixed-generator growth are regression-pinned in `tests/test_paper32_order_condition_growth.py`.

6. **True Camporesi–Higuchi window.** **[MEASURED]** The audit repeated on the object of Theorem 5: spinors $L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2$ in the left-invariant frame, basis $e^j_{mm's} = \sqrt{2j+1}\, D^j_{mm'} \otimes e_s$, $D = 2\sigma \cdot J_R + \frac{3}{2}$, window $|\lambda| \leq n_{\max} + \frac{1}{2}$, $\mathcal{O}^{\mathrm{CH}}_{n_{\max}} = P\, C(S^3)\, P$ of dimension $5, 30, 91, 204$ at $n_{\max} = 1$–$4$, spanned by the compressed matrix coefficients $D^J_{MM'}$ of bands $N = 2J + 1 \leq 2n_{\max}$ (sup norm 1). The charge conjugation is $J = UK$, $U\, e^j_{mm's} = (-1)^{m-m'}\, (i\sigma_y)_{s's}\, e^j_{-m,-m',s'}$: pointwise complex conjugation of the spinor ($\overline{D^j_{mm'}} = (-1)^{m-m'} D^j_{-m,-m'}$) twisted by $i\sigma_y$, which reverses $J_R$ and $\sigma$ and so fixes $\sigma \cdot J_R$. It commutes with $D$, hence preserves the window, and $J^2 = -\mathbb{1}$, $JD = +DJ$ hold exactly (test-pinned at $n_{\max} = 1$–$4$). The signs are not a phase convention: the antiunitaries intertwining $D$ and the isometry generators ($J_L$, $J_R + \sigma/2$) form a family of one phase per irreducible block (dimension $2n_{\max}$), every member has $J^2 = -\mathbb{1}$, and no equivariant antiunitary anticommutes with $D$. $J\mathcal{O}^{\mathrm{CH}}J^{-1} = \mathcal{O}^{\mathrm{CH}}$ holds exactly; indeed $J(PfP)^*J^{-1} = PfP$. Order zero $[a, Jb^*J^{-1}]$ and order one $[[D, a], Jb^*J^{-1}]$

*fail*, already at $n_{\max} = 1$ (where the model holds trivially), over all basis pairs:

| $n_{\max}$ | order 0 failing | max | order 1 failing | max entry | op. norm |
|---|---|---|---|---|---|
| 1 | 12/25 (48%) | 1/2 | 4/25 (16%) | 3/4 | 0.750 |
| 2 | 572/900 (64%) | 5/9 | 513/900 (57%) | 5/3 | 1.863 |
| 3 | 5842/8281 (71%) | 11/16 | 5856/8281 (71%) | 21/8 | 2.902 |
| 4 | 30462/41616 (73%) | 19/25 | 31021/41616 (75%) | 18/5 | 3.923 |

(max entry in the Peter–Weyl basis; at order zero it equals the operator norm). On the fixed degree-1 generators the order-zero operator-norm residual runs $1/2, 5/9, 11/16, 19/25, 29/36, 41/49$ and the order-one residual $3/4, 7/9, 13/16, 21/25, 31/36, 43/49$ at $n_{\max} = 1$–6, increasing toward 1, the product of the generators' sup norms (the circle analogue: on $\mathrm{span}\{e^{ik\theta}\}_{|k|\le n}$ the commutator of the compressions of $e^{\pm i\theta}$ is a difference of two rank-one projections, of norm 1 at every $n$). Relative to the generator norm the true-window residual (0.72 at $n_{\max} = 2$, 0.86 at $n_{\max} = 5$) is *larger* than the model's ($\approx 0.21$). *Mechanism (exact).* Since $fg = gf$, $[PfP, PgP] = Pg(1-P)fP - Pf(1-P)gP$; at order one $[D, f] = c(df)$ commutes with $g$ pointwise, so the same identity holds with $f$ replaced by $c(df)$. A band-$N$ multiplier moves the Dirac level by at most $N - 1$ (Paper 38's shell-difference rule), so every residual vanishes on all levels more than $\max(N_a, N_b) - 1$ below the top (measured: largest entry outside that strip $\le 3 \times 10^{-15}$, $n_{\max} = 2$–5). The degree-1 residual lives on the top Dirac level alone, whose share of the window is $3/(n_{\max} + 2) \to 0$, and its normalised Hilbert–Schmidt norm $\|C\|_2/\sqrt{\dim \mathcal{H}}$ falls $0.30, 0.29, 0.28, 0.26$ at $n_{\max} = 2$–5. The failure is a boundary term of the compression — full operator norm, shrinking weight — not a bulk non-commutativity. Two independent constructions (Clebsch–Gordan multipliers with the closed-form $J$; sampled Wigner-$D$ functions with $J$ obtained from pointwise conjugation of the sampled basis) agree pair by pair to $10^{-10}$ at $n_{\max} = 1, 2$ (`tests/test_true_window_real_structure.py`). Test-pinned there: the $J$-relations (with $\dim \mathcal{O} = 5, 30, 91, 204$) at $n_{\max} = 1$–4, the order tables at $n_{\max} = 1$–3, the degree-1 residuals at $n_{\max} = 1$–4, and the boundary strip at $n_{\max} = 2, 3$; the $n_{\max} = 4$ order-zero/one table row, the degree-1 values at $n_{\max} = 5, 6$, the Hilbert–Schmidt weights and the strip check at $n_{\max} = 4, 5$ are measured in the audit driver, not test-pinned.

**Reading of the order-condition failures.** The order-zero condition $[a, JbJ^{-1}] = 0$ is the abelian-classical statement *multiplications by classical functions commute*, automatic for $f, g \in C^\infty(S^3)$ in the continuum. At finite $n_{\max}$, the truncated objects $P_{n_{\max}} f P_{n_{\max}}$ and $J P_{n_{\max}} g P_{n_{\max}} J^{-1}$ are no longer multiplications by classical functions; they are operator-system elements ([12, Sec. 2]), and their commutator is generically nonzero. That explains why the residual is nonzero; it does *not* explain its size or its trend. The residuals do not decay at the cutoffs reached: they grow, and on fixed low-degree generators they do not decay through $n_{\max} = 5$. The same growth is measured on the true window that converges (item 6), where it is identified exactly as a Toeplitz boundary term of the compression. No contradiction with Theorem 5 arises: GH convergence of state spaces does not by itself control the commutator $[a, JbJ^{-1}]$ in norm, and on the true window the residual, though of full operator norm, is confined to a boundary strip of vanishing weight. **[MEASURED]** status, regression-pinned in `tests/test_paper32_order_condition_growth.py` (model) and `tests/test_true_window_real_structure.py` (true window); the mechanism on the model (the boundary-term identity of item 6 is the plausible reading there too), and whether the continuum identity is recovered in any weaker (e.g. weak-operator or state-space) sense, are open.

**Net status of the real structure at finite $n_{\max}$.** The introductory abstract flagged as open "whether the GeoVac Dirac operator $D_{\mathrm{GV}}$ (the diagonal spectral form) carries a real structure matching the signs of $S^3$ exactly at finite $n_{\max}$ or only in the continuum limit." The Sprint WH1-Connes Step 2 audit closes this question *positively for the three $J$-relations on the truthful CH Dirac*: $J^2 = -\mathbb{1}$, $JD = +DJ$, and $J\mathcal{O}J^{-1} = \mathcal{O}$ all hold with bit-exact zero residual at every finite $n_{\max} \in \{1, 2, 3\}$ on both the Weyl and full-Dirac (chirality-doubled) model sectors, and on the true Camporesi–Higuchi window (test-pinned at $n_{\max} = 1$–$4$), where the signs are forced by equivariance (item 6). It does *not* close it for the order-zero and order-one conditions (involving the abelian-multiplication identity that defines the classical limit of the algebra): on the model those fail at finite $n_{\max} \geq 2$ with residuals that grow over the measured cutoffs (item 5 above; **[MEASURED]**, regression-pinned in `tests/test_paper32_order_condition_growth.py`, mechanism open), and on the true window they fail from $n_{\max} = 1$ with residuals growing toward the generators' sup-norm scale, as a Toeplitz boundary term of the compression (item 6; **[MEASURED]**, `tests/test_true_window_real_structure.py`). The offdiag CH Dirac fails $JD = +DJ$ structurally and is therefore not a spectral-action-foundational object; the truthful CH is the natural domain for the present audit. (On the diagonal representative $\mathcal{A}_{\mathrm{GV}}$ the truthful CH Dirac has no nonzero commutators and so no inner fluctuations, Proposition 2; an inner-fluctuation calculus for the Track 3 almost-commutative extension needs the operator-system representative or a non-diagonal Dirac, Sec. 6 discussion.)

### 4.6 $\mathbb{Z}_2$-grading

**Proposition 5** (Odd continuum; even model truncation; odd true window)**.** *(a) The continuum spectral triple of the unit Riemannian spin $S^3$ with the Camporesi–Higuchi Dirac operator is* odd*: it carries no $\mathbb{Z}_2$-grading commuting with $C^\infty(S^3)$ and anticommuting with the Dirac operator. (b) The* chirality-doubled model *truncation $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ is* even*: the chirality-doubling operator $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$ (exchanging the two chirality copies of each $(n, l, m_j)$ label) is a self-adjoint involution that commutes with $\pi(\mathcal{A}_{\mathrm{GV}})$ and anticommutes with $D_{\mathrm{GV}}$. (c) The* true *Camporesi–Higuchi window of Theorem 5 (complete Dirac eigenspaces, the object that converges) is* odd *(measured, $n_{\max} \leq 3$), like the continuum: at $n_{\max} = 1, 2, 3$ the commutant of its operator system $\mathcal{O}^{\mathrm{CH}}_{n_{\max}}$ is the scalars, so no $\mathbb{Z}_2$-grading commutes with $\mathcal{O}^{\mathrm{CH}}$ and anticommutes with $D$. The evenness in (b) is a property of the chirality-doubled model, not of the converging truncation.*

*Proof.* (a) $S^3$ is odd-dimensional; this is the standard statement for odd Riemannian spin manifolds [1]. (b) $\pi(f)$ does not depend on the chirality label (Eq. (3)), so it commutes with any operator acting on the chirality index alone, in particular with $\sigma_x \otimes \mathbb{1}$; $D_{\mathrm{GV}} = \chi\,(n + \frac{1}{2})$ is $\sigma_z \otimes (n + \frac{1}{2})$ in the chirality index, and $\sigma_x$ anticommutes with $\sigma_z$. Measured bit-exactly at $n_{\max} \leq 3$ together with $J\gamma = +\gamma J$ (`geovac/chirality_grading.py`; `tests/test_almost_commutative.py`, §8.2; `tests/test_paper32_diagonal_triple.py`, which also contrasts the diagonal $\sigma_z \otimes \mathbb{1}$ candidate, which commutes with $D_{\mathrm{GV}}$ instead of anticommuting). **[SYMBOLIC PROOF + MEASURED]** (c) **[MEASURED]** At $n_{\max} = 1, 2, 3$ ($\dim \mathcal{H} = 4, 16, 40$) the commutant of $\mathcal{O}^{\mathrm{CH}}_{n_{\max}}$ (the null space of $X \mapsto [X, a]$ over a basis of $\mathcal{O}^{\mathrm{CH}}$) has dimension 1 (Peter–Weyl construction, `tests/test_true_window_real_structure.py`). Two further routes in the same test file corroborate it at $n_{\max} = 1, 2$: the sampled-function construction of $\mathcal{O}^{\mathrm{CH}}$, whose commutant is again the scalars and whose generated unital algebra is all of $B(\mathcal{H})$ (dimension $16,256$, so by Burnside's theorem its commutant is the scalars; dimension 1600 at $n_{\max} = 3$ is computed outside the test), and an independent polynomial construction on $S^3 \subset \mathbb{R}^4$ (no Wigner-$D$, Clebsch–Gordan or sampling code). At $n_{\max} = 3$ the oddness rests on the Peter–Weyl route. A scalar cannot anticommute with

the invertible $D$, so the grading space is $\{0\}$. On the true window only the pair $(\varepsilon, \varepsilon') = (-, +)$ is defined, and it is forced by equivariance (audit item 6). □

The chirality-doubled model triple of (b) therefore carries the sign triple $(\varepsilon, \varepsilon', \varepsilon'') = (-, +, +)$, and, per the rule stated in §8.2, *no* KO-dimension label is attached at finite cutoff: the continuum factor is KO-3 (odd), the model's grading reads as the KO-4 column, and any single label at finite cutoff is a convention. The grading is a doubling artefact of the chirality-labelled model basis, not a chirality of $S^3$; it is absent on the true window (c) (measured, $n_{\max} \leq 3$) and in the continuum limit, where the odd description is the correct one. Odd-dimensional spectral triples construct their spectral action from $|D|$ rather than from $D^2$ projected onto chirality eigenspaces, which is the structure that Paper 28's QED-on-$S^3$ computations use.

### 4.7 Summary

**Theorem 2** (State-space limit identification)**.** *The state spaces of the true-window truncations (Theorem 5) converge, in van Suijlekom's state-space Gromov–Hausdorff distance under the translation seminorm, to the state space $(P(S^3), W_d)$, with $d$ the dual-Coxeter geodesic distance (a homothety of the unit-sphere metric), of the canonical spectral triple of the round Riemannian spin three-sphere with the Camporesi–Higuchi Dirac operator, up to isometry (Paper 38, limit-identification proposition). That target triple is real, odd, and of* KO*-dimension* 3*; these are classical properties of the target, not outputs of the convergence.*

**[INTERNAL THEOREM]** This is a structural theorem; no fitting parameters or empirical inputs. It is a statement about the continuum limit only: at finite cutoff the true-window truncation is odd like its limit (measured, $n_{\max} \leq 3$; Proposition 5(c)), only the chirality-doubled model being even (Proposition 5(b)), and the Fock/Weyl model of the operator system fails order zero and order one, as does the true window, by a boundary term of the compression (Table 1, audit items 5–6). The audit in Table 1 establishes that the spectral-triple framing is not metaphorical at the level of spectral data and the three $J$-relations.

## 5 Sub-sector Identification: Papers 25, 28, 30, 31

We now read the major framework papers as projections of the GeoVac triple.

### 5.1 Paper 28 (QED on $S^3$): the spectral action

Paper 28 [33] computes Tr $f(D^2/\Lambda^2)$ and related spectral sums on the Dirac operator $D_{\mathrm{GV}}$ in the continuum limit, with SO(4) vertex selection rules from the Hopf-bundle decomposition. In the language of Sec. 2:

- Theorem T9 (Paper 28 Sec. T9): the spectral zeta of $D^2$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s$. In spectral-triple language: the heat-kernel coefficients $a_k(D^2)$ on $S^3$ saturate at $k = 2$ (Paper 28's "SD two-term exactness theorem").
- Self-energy structural zero (Paper 28 Theorem 4): $\Sigma(n_{\mathrm{ext}} = 0) = 0$ on the continuum spectral sum, identically by vertex parity. This is the standard heat-kernel bosonic sector cancellation specialized to the Camporesi–Higuchi spectrum.
- Vector-photon QED on $S^3$ (Paper 28 §vector_photon_qed): adding photon $(q, m_q)$ labels with Wigner $3j$ vertex coupling recovers 7/8 selection rules for scalar electrons and 8/8 for Dirac electrons. The per-loop calibration is the $S^2$ Weyl constant $1/(4\pi)$ (M1 volume slot).

In the spectral-triple language Paper 28 reads as the spectral-action computation $\mathrm{Tr}\, f(D_{\mathrm{GV}}^2/\Lambda^2)$ at one loop; the Camporesi–Higuchi spectrum is the input data of $D_{\mathrm{GV}}$; the SO(4) selection rules are the harmonic decomposition of the spinor bundle. Read this way, Paper 28 is not a free-standing observation about $S^3$ but the spectral action of the present triple (a reading, **[OBSERVATION]**).

### 5.2 Paper 25 (Hopf graph as lattice gauge): the Cartan torus

Paper 25 [32] introduces the signed node-edge incidence matrix $B$ of the Fock graph, with $L_0 = BB^\top$ the node Laplacian (spectrally inert on $S^3$ and $S^5$) and $L_1 = B^\top B$ the discrete Hodge-1 Laplacian. **[OBSERVATION]** The spectral-triple reading of $L_1$ is *not* an inner fluctuation of $D_{\mathrm{GV}}$: $[D_{\mathrm{GV}}, \pi(f)] = 0$ (Proposition 2), so $\Omega^1_{D_{\mathrm{GV}}} = 0$ and $D_{\mathrm{GV}}$ has no fluctuations at all. The reading goes through the gauge-network (edge-hopping) Dirac of Papers 25/30 — the Marcolli–van Suijlekom setting of Sec. 9 — whose $U(1)$ link variables fluctuate the edge data and whose quadratic kinetic term is $L_1$. In that setting $L_1$ is the Cartan-subalgebra (maximal-torus) sector, and its assignment as "photon propagator" in Paper 25 is the gauge-network spectral-action reading. A statement of the same identification on the non-diagonal multipliers of $\mathcal{O}_{n_{\max}}$, with a backing test, is open.

### 5.3 Paper 30 (SU(2) Wilson gauge): the full non-abelian sector

Paper 30 [34] builds the SU(2) Wilson lattice gauge theory on the same graph and proves: (a) the maximal-torus reduction reproduces Paper 25's $U(1)$ action exactly; (b) the weak-coupling kinetic term equals $L_1$ acting on $\mathfrak{su}(2)$-valued 1-cochains. **[OBSERVATION]** In the spectral-triple language the natural setting is the almost-commutative algebra $\mathcal{A}_{\mathrm{GV}} \otimes M_2(\mathbb{C})$ (a rank-1 non-abelian $M_2$ fiber). With the diagonal $D_{\mathrm{GV}} \otimes \mathbb{1}$ this extension still has no inner fluctuations ($[D_{\mathrm{GV}} \otimes \mathbb{1}, a \otimes m] = [D_{\mathrm{GV}}, a] \otimes m = 0$); the identification of Paper 30's $\mathfrak{su}(2)$-valued link variables as the fluctuations, and of the Wilson plaquette action as the lattice-gauge implementation of the spectral-action curvature term, is made in the Marcolli–van Suijlekom gauge-network setting with its edge-hopping Dirac (Sec. 9), where the order-one condition is non-trivial.

Read that way, Paper 30 is the rank-1 non-abelian almost-commutative gauge-network extension over the GeoVac label set; Paper 25 is the abelian projection of Paper 30; together they exhibit the standard Cartan torus / full group structure of that gauge sector. They are not fluctuations of the diagonal $D_{\mathrm{GV}}$.

### 5.4 Paper 31 (universal/Coulomb partition): the $\mathcal{A}/D$ split

Paper 31 [35] partitions GeoVac results into a universal sector that depends only on the algebra $\mathcal{A}$ (angular sparsity, composed scaling, selection rules) and a potential-specific sector that depends on the Dirac operator $D$ (spectrum, calibration $\kappa$, transcendental content, rigidity, $\alpha$ observation). In the spectral-triple language this is exactly the $\mathcal{A}/D$ split: properties of $\mathcal{A}_{\mathrm{GV}}$ alone (rank, dimension, representation content) transfer to any spectral triple with the same algebra; properties of $D_{\mathrm{GV}}$ specifically depend on the choice of Dirac operator. The $D$-dependent side is a statement about *spectral* data (spectrum, degeneracies, the transcendentals of spectral sums): on the diagonal representative $D_{\mathrm{GV}}$ commutes with $\mathcal{A}_{\mathrm{GV}}$ (Proposition 2), so the split carries no metric content of its own.

Paper 31 is the structural reading of Paper 22 (angular sparsity theorem, an $\mathcal{A}$-statement) and Paper 23 (nuclear shell model, a different $D$ on the harmonic-oscillator label set, hence a different $\mathcal{A}$ sharing the angular $(l, m)$ factor): when one keeps $\mathcal{A}$ fixed and varies $D$, the universal results

transfer; when one varies $\mathcal{A}$ (say, replace $S^3$ with $S^5$, Paper 24), the entire structural picture reorganizes.

### 5.5 Summary of sub-sector identification

**Track NI as an explicit composed real-space spectral triple.** The "Different $\mathcal{A}$ and $D$" row of Table 2 (Paper 23 [30], nuclear shell model) covers a second class of construction in Paper 23 §VI: the composed nuclear–electronic deuterium Hamiltonian, a 26-qubit Hilbert-space tensor product of a 16-qubit deuteron register (harmonic-oscillator basis) with a 10-qubit hydrogenic electronic register (Fock-projected $S^3$ lattice), coupled through a cross-register hyperfine $\mathbf{I} \cdot \mathbf{S}$ operator. This is structurally distinct from both the SM almost-commutative geometry (which couples a Riemannian factor with a finite *internal* factor) and from the standard spectral-truncation convergence machinery (which handles single triples or one-infinite-plus-one-finite products). **[OBSERVATION]** It is read as a Connes-style real-space multi-particle spectral triple; no axiom audit of the composed triple is carried out. **[PANEL-VERIFIED]** Its Hamiltonian reproduces the singlet–triplet hyperfine pattern $(-3A/4, +A/4)$ for the input coupling $A_{\mathrm{hf}}$ set to the 21 cm value ($1.62 \times 10^{-7}$ Ha for the $3A/4$ shift) — a structural check of the Clebsch–Gordan pattern, not a measurement against an external reference, since the coupling is the input. **[OBSERVATION]** The positioning observation (that the canonical NCG / spectral-triple literature does not contain a directly comparable construction) is formalized in the standalone Track NI Zenodo memo (`papers/group4_quantum_computing/track_ni_spectral_triple_zenodo.md`). The W2b cross-manifold structural blocker of Sec. 8.5 is exactly the convergence-theorem gap that surrounds this construction.

**Paper 39 treats the same-manifold tensor-product case at the state-space GH level.** The same-manifold sub-case of W2b — the tensor product $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ of two truncated Camporesi–Higuchi spectral triples on $S^3$ at distinct focal lengths $\lambda_a \neq \lambda_b$ — is treated in a companion standalone paper (Paper 39 [39]) at the level of van Suijlekom's state-space Gromov–Hausdorff convergence; the Latrémolière-propinquity question that defines W2b stays open. The result extends Paper 38's convergence to the KO-dim-6 graded tensor product via the *lifted-state* construction (leading constant 1): Paper 38's unconditional-theorem facts, lifted factor-wise and metrized by the translation seminorm, with the limit the graded (Clifford-doubled) tensor product of the two odd Camporesi–Higuchi real spectral triples — the naive $\gamma_a \otimes D_b$ being unrealisable ($\gamma_a = 0$ on $S^3$). This is the mathematical foundation for the GeoVac multi-focal-composition program, where two electrons in distinct hydrogenic shells live on $S^3$ at distinct focal lengths $\lambda = n/Z$; the convergence theorem of Paper 39 says that the cross-register joint operator system on $\mathcal{O}^{\mathrm{CH}}_{n_{\max,a}} \otimes \mathcal{O}^{\mathrm{CH}}_{n_{\max,b}}$ (the true Camporesi–Higuchi windows, not the Fock/Weyl label-set systems) converges qualitatively to the continuum tensor algebra $C^\infty(S^3 \times S^3)$ in van Suijlekom's state-space Gromov–Hausdorff distance (the Latrémolière propinquity strengthening is the same open gap as for the single $S^3$ factor; Remark 19). The two remaining cross-manifold variants ($\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{Bargmann}}_{S^5}$ and $\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{Riem}}_{S^5}$) remain the W2b content of Sec. 8.5.

**Remark 7** (Multi-center composition: the reducible–irreducible transition)**.** *The multi-focal program above concerns two electrons at distinct focal lengths on a single $S^3$; the complementary* multi-center *composition concerns the orthogonal projections $P_A$ onto the single-center orbital subspaces of a polyatomic molecule in the shared Hilbert space with overlap metric $G$. The natural setting is the* $*$-representation *theory of orthogonal projections. For a* diatomic *the pair $\{P_A, P_B\}$ is* tame *(Type I): by the two-projection theorem (Halmos [59], with the principal-angle norm*

*and block calculus catalogued in Böttcher–Spitkovsky [60]) the generated $*$-algebra decomposes into blocks of dimension $\leq 2$, completely classified by the principal angles between the two subspaces—**[MEASURED]** for LiH at $R_{\mathrm{eq}}$ the three angles $7.6°/44.7°/67.3°$, the middle one within $0.3°$ of $45°$, so that $\|[P_A, P_B]\| = 0.49997$ sits essentially at the ceiling $\frac{1}{2}$ (the ceiling itself, and the reducibility of any pair, are forced by the two-projection theorem **[SYMBOLIC]**; the molecular data supply only the angles). The $*$-algebra generated by* three or more *projections is, by contrast, $*$-wild in general; the passage from two to three centers is precisely this tame-to-wild threshold. **[PANEL-VERIFIED]** What the molecular data adds is that the three-center configuration lands on an* irreducible *(Schurian) point of that wild family: for linear $\mathrm{BeH_2}$ (the $\sigma$ pair $\{1s, 2p_0\}$ per center) the three center-projections generate the* full *matrix algebra $M_6$ and act irreducibly (commutant $= \mathbb{C}\mathbb{1}$), welding the whole space into a single indecomposable block that admits no decomposition; every pair remains reducible, as it must by the two-projection theorem.*

*Tameness is not composability, and conflating the two misreads the obstruction. The diatomic pair is completely classified, but it emphatically does not commute—$\|[P_A, P_B]\| = 0.49997$ is essentially the maximal possible value $\frac{1}{2}$—and it is that* two*-projection non-commutativity, not the three-projection wildness, which constitutes the multi-focal composition wall recorded above. The obstruction accordingly has two distinct rungs. At two centers the projections fail to commute while remaining completely classified by their principal angles: the obstruction is quantitative, and the cost of removing it is known in closed form. At three centers the classification itself fails, and no complete list of invariants exists. Forcing commutation at the first rung—Löwdin orthogonalization—is the framework's sparsity-destroying move; at the second rung there is no corresponding normal form available to force.*

*The wildness is specific to the subspace configuration and is* not *a generic consequence of non-commutativity. The framework's other non-commutative angular structure—$SO(3)/SU(2)$ coupling, the Gaunt and $6j$ machinery—is the representation theory of a* compact *group, hence type I by Peter–Weyl, hence tame however non-abelian it may be. **[MEASURED]** The contrast is quantitative: on the menu $l = 0, 1, 2$ the $\mathfrak{su}(2)$ generators satisfy $\|[J_a, J_b]\| = 2$ yet their commutant is three-dimensional (one dimension per irreducible constituent, and $\sum_\lambda m_\lambda^2$ in general—four for $l=1$ at multiplicity two, five for $l=1 \oplus l=2 \oplus l=2$), whereas the three center-projections, at the* smaller *non-commutativity $\|[P_A, P_B]\| \approx 0.49$, have commutant $\mathbb{C}\mathbb{1}$. Non-abelianness and wildness are independent properties. This is the structural reason the framework's angular sparsity is graded, finite and potential-independent (Paper 22), and the reason no Gaunt-style graded reduction should be expected to rescue multi-center composition: arbitrary configurations of subspaces are not group representations and escape Peter–Weyl.*

*Two invariants must be kept separate. **[PANEL-VERIFIED]** (i) The* categorical *statement—pairs reducible, triple irreducible—is robust: it survives enriching the per-center $\sigma$ menu $\{1s, 2p_0\} \to \{1s, 2s, 2p_0\}$ ($M_6 \to M_9$, irreducible at the physical $Z = 2$ charge, regression-pinned; the structural $Z = 1$ case is likewise regression-pinned, `tests/test_paper32_composition_pins.py`) and holds at real molecular geometries. It is, however, a statement about center* count*: three coupled projections in general position are irreducible at essentially any bonding-relevant separation, the commutant reverting to reducible only once residual overlaps fall below resolution. **[MEASURED]** (ii) The bonding-dependent content is instead the* magnitude *of the irreducible three-body coupling, measured by the nested commutator $\|[[P_A, P_B], P_C]\|$ (which cannot even be formed for a diatomic):* 0.32 *at bonding (regression-pinned, `tests/test_paper32_composition_pins.py`), decaying smoothly to zero as the centers separate.*

***[PANEL-VERIFIED]** Geometry sharpens the picture. At* exactly *linear geometry the in-plane basis $\{s, p_x, p_z\}$ per center decouples by axial symmetry into a $\sigma$ and a $\pi$ sector and the nine-dimensional algebra is reducible;* any *bend mixes the sectors and welds the full space into one*

*irreducible $M_9$ block (regression-pinned at the* $H_2O$ *geometry* $104.5°$*; the sweep across* $160° \to 90°$ *and apex charges* $Z = 1, 2, 4$ *is likewise regression-pinned,* `tests/test_paper32_composition_pins.py`*)—the operator-algebra face of the Renner–Teller mechanism.* ***[OBSERVATION]*** *As an internal observation the configuration operator* $F = \sum_i P_i$ *carries, over molecular geometry, a lattice of conical intersections each with a* $\pi$ *(*$\mathbb{Z}_2$*) Berry phase and, under bending, a structural Renner–Teller promotion; this is the* generic *topology of a symmetric-triatomic degeneracy, distinct from the molecule's physical electronic conical intersection, not an identity with it. (Computed and regression-pinned in* `tests/test_paper32_composition.py` *on validated two-center overlaps, with the commutant routine self-validated on cases of known answer; chronicle in CHANGELOG v5.1.0.)*

**Remark 8** (The configuration operator's Berry curvature: a flat $\mathbb{Z}_2$ bundle over nuclear geometry)**.** *The configuration operator* $F = \sum_i P_i$ *of the previous remark varies over the nuclear-geometry manifold* $\mathcal{M}$*, and its geometric-phase structure is the finished form of the "lattice of conical intersections" noted there.* ***[SYMBOLIC PROOF]*** *Because the* $\sigma$ *orbitals are real,* $F$ *is real-symmetric at every geometry, so its Berry connection is* flat *with only a* $\mathbb{Z}_2$ $(\pm)$ *holonomy: the curvature is not a smooth field but a sum of* $\pi$*-flux Dirac deltas at the conical intersections,*

$$\Omega \;=\; \sum_{CI} \pi\, \delta^2(m - m_{CI}),$$

*the Longuet–Higgins sign-change / molecular Aharonov–Bohm structure (Longuet-Higgins; Berry; Mead–Truhlar) realized here for* $F$ *rather than the electronic Hamiltonian.* ***[SYMBOLIC PROOF]*** *On the linear-stretch plane* $(d_1, d_2)$ *the Fukui–Hatsugai–Suzuki lattice flux is exactly* $0$ *or* $\pi$ *on every plaquette (forced by the reality of* $F$*; the machine-exact regression check is a consistency check, not independent evidence).* ***[MEASURED]*** *There are three* $\pi$*-sources—the central intersection at* $(2.45, 2.45)$ *and an antisymmetric-stretch mirror pair— carried by the two crossing bands and absent on the other four (the central source and the mirror pair are both regression-pinned,* `tests/test_paper32_composition_pins.py`*). The holonomy is the* $\mathbb{Z}_2$ *representation of* $\pi_1(\mathcal{M} \setminus \mathrm{CI})$ *sending a loop to* $\pi \cdot (\#\, \textit{enclosed} \bmod 2)$*, cross-checked by direct parallel transport (*$\pi$ *around one intersection and around all three).*

***[MEASURED]*** *Each intersection is the real (equatorial) section of a charge-*$\frac{1}{2}$ *Berry monopole: in a fixed real frame the local* $2 \times 2$ *model is* $F_0 + a_z\sigma_z + a_x\sigma_x$ *with the* $\sigma_y$ *coefficient* identically zero *(****[SYMBOLIC]****: forced by* $F$ *real-symmetric)—the real family lives on the sphere's equator—and branching vectors* $g = \nabla a_z$ *along the symmetric stretch (the tuning mode) and* $h = \nabla a_x$ *along the antisymmetric stretch (the coupling mode), with nonzero winding.* ***[SYMBOLIC PROOF]*** *The* $\mathbb{Z}_2$ *(rather than* $U(1)$*) character is* forced by time-reversal symmetry *—the reality of the Coulomb overlaps—not by planarity: real orbitals give real overlaps in any geometry, so a non-coplanar real center leaves* $F$ *real-symmetric and does not lift the phase. Breaking time reversal does.* ***[PANEL-VERIFIED]*** *A magnetic (Peierls) phase* $\phi$ *threading the* H–Be–H *loop makes* $F$ *complex-Hermitian, opens the empty* $\sigma_y$ *axis, and turns each intersection into a genuine* $U(1)$ *Berry monopole of integer charge: the first Chern number over a small sphere enclosing* $(d^*, d^*, 0)$ *in* $(d_1, d_2, \phi)$ *is* $\pm 1$ *(opposite on the two crossing bands), and the real* $\phi = 0$ *plane cuts the monopole at its equator, recovering the* $\pi$ *holonomy.* ***[OBSERVATION]*** *This is an internal observation about the geometry of the composition, distinct from the molecule's physical electronic conical intersection. (Regression-pinned in* `tests/test_paper32_berry_curvature.py`*; chronicle in CHANGELOG v5.1.0.)*

**Remark 9** (Four centers: a continuous four-body modulus (the $\tilde{D}_4$ analog))**.** *The three-center geometric phase of Remark 8 is discrete (*$\mathbb{Z}_2$ *conical intersections). A* fourth *center adds a gen-*

*uinely new,* continuous *four-body invariant. With* $S_{ij} = Q_i^\top Q_j$ *the metric-orthonormal overlaps between center subspaces, the loop holonomy of this overlap connection around the* 4*-cycle,* $W = \mathrm{Tr}(S_{12}S_{23}S_{34}S_{41}) = \mathrm{Tr}(P_1P_2P_3P_4)$*, is invariant under the per-center gauge* $Q_i \mapsto Q_iO_i$ *(*$O_i \in O(r)$*) and is therefore a physical configuration invariant. Two facts make it the four-center analogue of an elliptic modulus.* ***[PANEL-VERIFIED]*** *First, a* threshold*: for four rank-one (single-orbital) subspaces* $W$ *is a function of the pairwise and triple traces—the* $4 \times 4$ *Gram is the entire configuration, so there is no four-body degree of freedom—whereas for rank* $\geq 2$ *it is not (a local Jacobian-nullspace test gives residual* $\gtrsim 0.3$*). A genuine four-body invariant appears exactly when each center carries a* $\sigma$*–*$\pi$ *pair, mirroring the rank-one→rank-two step in the tame four-subspace (*$\tilde{D}_4$*) problem.* ***[MEASURED]*** *Second, over molecular shape* $W$ *is a* smooth continuous modulus*: for a planar rhombus with* $\{1s, 2p\}$ *per center it runs monotonically (from* $\approx 0.8$ *to* $\approx 0.04$ *over the sprint's diagonal-ratio range; the monotonicity and the sprint-range endpoints (*$0.79 \to 0.042$*) are both regression-pinned,* `tests/test_paper32_composition_pins.py`*) as the diagonal ratio varies —a one-parameter family of inequivalent four-center configurations, the continuous freedom the three-center case lacks. For real orbitals* $W$ *is real (its phase is* $\mathbb{Z}_2$*); a magnetic (Peierls) flux through the* 4*-cycle makes* $\arg W$ *grow continuously from zero—the four-body* $U(1)$ *lift.* ***[OBSERVATION]*** *The object is the* $*$*-representation (non-abelian holonomy) analogue of the* $\tilde{D}_4$ *cross-ratio: four* orthogonal *projections are* $*$*-wild rather than tame, so the modulus is a matrix loop holonomy, not a single scalar cross-ratio. It is a geometry-axis invariant—a function of nuclear shape—distinct from the density-axis cross-ratio of the three-center integral in Paper 59 [58]. (Regression-pinned in* `tests/test_paper32_four_center_modulus.py`*; chronicle in CHANGELOG v5.1.0.)*

**Remark 10** (The configuration operator is the composed basis's overlap metric)**.** *The three preceding remarks treat* $F = \sum_i P_i$ *as an abstract configuration operator. It is not abstract. Let* $G$ *be the Gram matrix of the union of the per-center bases, with blocks* $G_{ij} = \langle \chi_i | \chi_j \rangle$*, and suppose each* intra*-center block is the identity—automatic when each center's own functions are orthonormal, and exact for the Shibuya–Wulfman metric of Paper 60 [50]. Put* $X = \mathrm{chol}(G)^\mathsf{T}$*, so that* $X^\mathsf{T}X = G$ *and the columns of* $X$ *are the basis vectors expressed in an orthonormal ambient frame; let* $W_k$ *collect the columns belonging to center* $k$*. Then* $W_k^\mathsf{T}W_k = G_{kk} = I$*, so* $W_k$ *has orthonormal columns, the pseudoinverse in* $P_k = W_kW_k^+$ *collapses to* $W_k^\mathsf{T}$*, and*

$$F \;=\; \sum_k W_kW_k^\mathsf{T} \;=\; XX^\mathsf{T} \;=\; X\,(X^\mathsf{T}X)\,X^{-1} \;=\; X\,G\,X^{-1}.$$

***[SYMBOLIC PROOF]*** *Hence* $F \simeq G$*—*similar*, not merely cospectral. Every spectral statement in the preceding remarks (the conical intersections, the band gaps, the* $\mathbb{Z}_2$ *holonomy of the eigenvectors) is therefore a statement about the* overlap metric of the composed basis *over nuclear geometry.* ***[MEASURED]*** *Verified across the whole construction: linear* $BeH_2$*, bent* $BeH_2$ *(including the* $H_2O$ *angle* $104.5°$*), and the four-center configuration, with* $\max|\operatorname{spec} F - \operatorname{spec} G| \leq 3 \times 10^{-15}$ *and* $\|FX - XG\|_{\max} \leq 9 \times 10^{-16}$*. The intra-center hypothesis is load-bearing rather than decorative: perturbing one intra-center block off the identity by* 0.3 *breaks the identity outright (spectral discrepancy* $6.7 \times 10^{-2}$*). The four-center holonomy of Remark 9 is the same statement read blockwise,* $\mathrm{Tr}(P_1P_2P_3P_4) = \mathrm{Tr}(G_{12}G_{23}G_{34}G_{41})$ *(residual* $3 \times 10^{-16}$*).*

*Two consequences follow, one positive and one negative.* ***[MEASURED]*** (i) An orthogonalization criterion. *Since the intersections are degeneracies of the metric, the eigenvectors of* $G$ *are double-valued around one: parallel transport of the near-degenerate eigenvector around a small loop enclosing the central intersection returns it with overlap* $-1.0000$*, against* $+1.0000$ *on a CI-free control. Any orthogonalization that* selects *metric eigenvectors—canonical orthogonalization with a drop threshold, the standard remedy for near-linear-dependence—is therefore*

*path-dependent along a geometry scan through such a point, whereas symmetric (Löwdin) orthogonalization $G^{-1/2}$, being a function of the matrix rather than of its eigenvectors, returns bit-exactly (**[SYMBOLIC]**: single-valuedness of any matrix function; the numerical return is a consistency check). **[MEASURED]*** (ii) It is not a conditioning statement. *The intersections are* mid-*spectrum degeneracies (crossing eigenvalue* $\approx 0.71$ *against* $\lambda_{\min} \approx 0.13$*), and* $\mathrm{cond}(G)$ *passes through* $d^*$ *monotonically (*$19.9 \to 16.2 \to 15.0$*); the metric-conditioning growth of Paper 60 is a different functional of the same object, and no kink at* $d^*$ *should be expected or claimed. The distinction drawn in the preceding remarks between* $F$ *and the molecule's* physical electronic *conical intersection is unchanged; what changes is that* $F$ *is an invariant of the framework's own encoding rather than a generic operator introduced for the analysis. (Regression-pinned in* `tests/test_paper32_config_operator_metric.py`*; chronicle in CHANGELOG v5.1.2.)*

## 6 The $K = \pi(B + F - \Delta)$ Formula in Spectral-Action Language

Paper 2 [25] observed the numerical coincidence $K = \pi(B+F-\Delta) \approx 1/\alpha$ at $4.77 \times 10^{-7}$ (raw; the $8.8 \times 10^{-8}$ figure is the match of the root of Paper 2's cubic $1/\alpha = K - \alpha^2$, not of $K$ itself), with

$$B = 42, \quad F = \tfrac{\pi^2}{6}, \quad \Delta = \tfrac{1}{40},$$

and the Phase 4B–4I sprint series identified spectral homes of distinct type for the three pieces (project development-frontier log):

- $B$ is the finite Casimir trace at $m = 3$: $B = \sum_{n=1}^{3}(2l+1)l(l+1)|_{n,l} = 42$ (Paper 2 Eq. 17).
- $F = D_{n^2}(d_{\max} = 4) = \zeta_R(2)$, the Fock-degeneracy Dirichlet series at the packing exponent (Phase 4F $\alpha$-J).
- $\Delta^{-1} = g_3^{\mathrm{D}} = 2(3+1)(3+2) = 40$, the total (both-sign) Camporesi–Higuchi Dirac degeneracy at level $k = 3$, $|\lambda| = k + 3/2 = 9/2$ (Paper 7's $g_3^{\mathrm{Dirac}}$; Phase 4H SM-D).

In spectral-triple language, $B$, $F$, $\Delta$ live in three different sectors of the same triple:

- $B$ is a *finite scalar Casimir trace* on the truncated algebra: in the language of Sec. 4, it is $\mathrm{tr}_{\mathcal{A}_{\mathrm{GV}}}(C \cdot \not{K})$ at $n_{\max} = 3$, either for $C = L^2$ the SO(3) Casimir $l(l+1)$ (the sum printed above) or for $C = (n^2-1)/2$, half the SO(4) Casimir $L^2 + A^2 = n^2 - 1$; with the unhalved SO(4) Casimir the same trace is $84 = 2B$ (both routes' agreement at 42, and the un-halved trace at 84, are regression-pinned in `tests/test_paper32_B_normalisation.py`).
- $F$ is an *infinite scalar spectral zeta*: it equals $\zeta_{1-\Delta_{S^3}}(2) = \sum_{n\geq 1} n^2 \cdot n^{-4} = \zeta(2)$ in the continuum limit, where $\Delta_{S^3}$ is the scalar Laplace–Beltrami operator on $S^3$ and $1 - \Delta_{S^3}$ has eigenvalue $n^2$ with the Fock degeneracy $n^2$. It is *not* $\zeta_{-\Delta_{S^3}}(2) = \sum_{n\geq 2} n^2/(n^2-1)^2 \approx 0.885$.
- $\Delta^{-1}$ is a *single-level Dirac degeneracy*: it equals $g_3^{\mathrm{D}}$, the total multiplicity (both signs) of the Camporesi–Higuchi level $k = 3$, $|\lambda_3^{\mathrm{D}}| = 9/2$.

These three are categorically different objects: a finite combinatorial trace, an infinite analytic Dirichlet series, and a single-level multiplicity. They sit in different sectors of the spectral triple (scalar/spinor; finite/infinite; trace/zeta/dimension) and **[OBSERVATION]** Phases 4B–4I made eight unsuccessful attempts to find a common spectral generator that produces all three simultaneously, without success (twelve counting Sprint A and Sprint K-CC; see the E6 no-single-mechanism observation, Observation 4).

**Observation 1** (Three-sector composition)**.** *The Paper 2 combination rule $K = \pi(B + F - \Delta)$ is a sum across three structurally distinct sectors of the GeoVac spectral triple. Each piece has been individually identified as a clean spectral invariant; the combination rule itself remains a numerical observation.*

This rephrasing is what working hypothesis WH5 in the project register is built on: $\alpha$ is the conversion factor between three projection regimes that do not share a generator. The spectral-triple framing does not derive $\alpha$; it makes precise where the obstruction is.

**Remark 11** (Sprint A, $\pi^3\alpha^3$ residual)**.** ***[MEASURED]*** *Sprint A (April 2026, working-hypothesis register) verified at 80 dps that the post-cubic residual $K - 1/\alpha - \alpha^2 = 1.2079 \times 10^{-5}$ matches $\pi^3\alpha^3 = 1.2049 \times 10^{-5}$ to within 0.25% (CODATA 2018 $\alpha$; with the CODATA 2022 value the residual is $1.1986 \times 10^{-5}$, 0.52% below $\pi^3\alpha^3$, so the agreement is not stable at this level across editions). The structural reading is $\pi^3 = \mathrm{Vol}(S^5)$, hinting at a contribution from the Bargmann–Segal $S^5$ HO sector (Paper 24 [31]) at higher order. This is a structural hint, not a derivation; the cross-check (Paper 24's HO triple is a distinct spectral triple, not a sub-sector of the GeoVac triple) shows that the $\pi^3\alpha^3$ identification, if real, is a* cross-triple *relation and not internal to the GeoVac spectral-action expansion.*

# 7 The Coulomb/HO Asymmetry in Spectral-Triple Language

Paper 31 [35] formalized a layered asymmetry (Paper 24 §5 catalogues seven layers, six standing; the table below tabulates structural rows, including the spectrum-computing role of $L_0$, withdrawn and marked; the CM Hodge structure is Paper 24 §5) between the Coulomb spectral triple of the present paper and the harmonic oscillator spectral triple of Paper 24 [31], summarized here:

**[OBSERVATION]** In spectral-triple language the asymmetry is structural: the Coulomb case is (in its continuum limit) a Riemannian-spin spectral triple of KO-dimension 3, the HO case is a spectral triple of complex Euler type on the holomorphic Hardy space of $S^5$. These are different objects even though both are "almost-commutative" in the broad sense. Calibration $\pi$ enters in the Coulomb case through the $k = 0$ volume slot $\mathrm{Vol}(S^2)/4$ in the spectral-action coefficients; it does not enter the HO case because the HO spectral action is built from a linear (first-order) operator whose heat kernel does not produce $\pi$ at the rational level of the truncation [31].

**Observation 2** (Coulomb-specificity of $\pi$)**.** *The presence of $\pi$ in the spectral-action coefficients of QED is specific to the Coulomb-projected $S^3$ triple of this paper, not a universal feature of GeoVac-style discretizations. The HO triple is $\pi$-free at the discrete level. On this reading QED's $\pi$ enters through the Coulomb-projection $S^3$ triple: through the $k = 0$ volume slot $\mathrm{Vol}(S^2)/4$ (M1) and the spectral action's $\pi^{2k}$ (M2).*

**Theorem 3** (G3 closure: Bertrand-rigidity category obstruction)**.** ***[PROOF-BY-ARGUMENT; scoped to published constructions]*** *Within standard Connes spectral triples and Berezin–Toeplitz quantization, there is no tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}^{\mathrm{BL}}_{S^5}$ that simultaneously preserves:*

*(a) the Riemannian-Dirac character of $\mathcal{T}_{S^3}$ (second-order conformal projection and spectral action built from $D^2$, spinor bundle, first-order Camporesi–Higuchi Dirac with spectrum $|\lambda_n| = n + 3/2$); and*

*(b) the Bargmann character of $\mathcal{T}^{\mathrm{BL}}_{S^5}$ ($\pi$-free certificate, Hardy-sector tower structure, first-order Euler spectrum $E_N = N + 3/2$, Paper 24).*

*The obstruction is categorical and rests on three premises:*

- Bertrand's theorem (1873): *the only central potentials with all bounded orbits closed are $V = -Z/r$ (Coulomb) and $V = \omega^2 r^2$ (harmonic oscillator).*
- Fock rigidity (Paper 23 [30], Thm. 4): *the $S^3$ conformal projection is unique to $-Z/r$; the projection is second-order (stereographic / conformal), yielding a Riemannian spectral triple with the Laplace–Beltrami operator.*
- HO rigidity (Paper 24 [31], Thm. 3): *the $S^5$ Bargmann–Segal projection is unique to $\omega^2 r^2$; the projection is first-order (complex-analytic / holomorphic), yielding a Berezin–Toeplitz construction with the Euler number operator on the Hardy space.*

*Proof sketch. Step 1 (category identification).* Fock rigidity and HO rigidity place the two potentials in different mathematical categories: Coulomb → second-order conformal → Riemannian spectral triple; HO → first-order holomorphic → Berezin–Toeplitz quantization.

*Step 2 (exhaustion of tensor-product options).* Three constructions are available: (i) the standard graded tensor product with the round-$S^5$ Riemannian Dirac—this preserves (a) but violates (b) ($\pi$ re-enters via the Seeley–DeWitt coefficients on $S^5$); (ii) the tensor product with the Bargmann–Euler operator—this preserves (b) but violates (a) ($E_{\mathrm{BL}}$ is not a Dirac-type operator on $S^5$, being the radial part of the Laplacian restricted to the holomorphic sector, not an elliptic differential operator on the total space); (iii) a hypothetical framework-extending construction beyond standard Connes spectral triples and Berezin–Toeplitz quantization—we know of no such framework in the published mathematics (a literature-absence premise, which is why the theorem is scoped to published constructions).

*Step 3 (irreducibility).* Bertrand exhausts the potentials; the two rigidity theorems fix the projection type for each. The category mismatch is a structural feature of central-force quantum mechanics, not an artifact of the GeoVac construction. The asymmetry layers of Paper 24 § V (seven catalogued, six standing; five are listed here, the remaining two, including the CM Hodge structure, are in Paper 24 § V) — spectrum-computing role of $L_0$ (*withdrawn*: $L_0$ does not converge to $\Delta_{S^3}$ and is spectrally inert on both spheres); calibration $\pi$; Wilson gauge with natural matter coupling (a representation-theoretic layer, not an $L_0$ one); modular-Hamiltonian Pythagorean orthogonality; spectral-action gravity termination — are, the withdrawn one excepted, read here as downstream of this single categorical obstruction; the proof sketch argues the obstruction itself, not each layer's derivation from it. □

**Remark 12** (G3 scope)**.** *Theorem 3 does not assert that $S^3$ and $S^5$ are unrelated — they are the two Bertrand geometries, deeply connected through the angular sparsity theorem (Paper 22). It asserts that their* spectral-triple-type structures *live in different mathematical categories, and no published framework bridges the gap while preserving both structural contents. Cross-manifold unification, if achievable, requires framework-extending mathematics beyond the scope of this paper and the published NCG literature.*

## 8 The $\pi$-Source Case-Exhaustion Theorem

**[PANEL-VERIFIED]** The empirical principle of Paper 35 [37] (Prediction 1, graduated to load-bearing status on a cumulative 208/208 confirmation across sprints KG and TX-B, May 2026) states that a GeoVac observable contains $\pi$ if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete label/Dirac spectrum. Read

in the spectral-triple language of §3–§4, this principle has a structural underpinning: in chains of the first fifteen Paper 34 projections, the places where such continuous integrations occur are anchored to three named mechanisms of the present triple. **[CONDITIONAL]** We state the case-exhaustion below and give its proof sketch; the (i)⇔(ii) leg rests on the empirical Paper 35 verification rather than on a proof, and the converse (iii)⇒(i) is not established: no mechanism is known to put $\pi$ into every one of its outputs (see the proof sketch).

**Theorem 4** ($\pi$-Source Case-Exhaustion)**.** *Let $\mathcal{C} = (P_{i_k} \circ \cdots \circ P_{i_1})$ be any finite composition of projections drawn from the Paper 34 list [36] § III.1–§ III.15, applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. The following hold: (i)⇔(ii) (the empirical leg) and (ii)⇒(iii). The converse (iii)⇒(i) is not established (see the proof sketch):*

*(i) The converged value $\mathcal{C}(\mathrm{datum})$ contains a transcendental of the form $\pi$, $\sqrt{\pi}$, $\pi^k$ for $k \in \mathbb{Q}$, or $\pi$ in a denominator.*

*(ii) The chain $\mathcal{C}$ contains at least one projection $P_{i_j}$ whose evaluation pipeline includes a continuous integration over a parameter promoted from the discrete label/Dirac spectrum (heat-kernel time, Matsubara frequency, Mellin contour, $S^2$ angular (volume-slot) measure, $\mathrm{SU}(N)$ Haar measure, or analytic-continuation parameter).*

*(iii) The chain $\mathcal{C}$ engages at least one of three spectral-triple mechanisms:*

- ***M1*** *(the $k = 0$ volume slot): introduces $\pi = \mathrm{Vol}(S^2)/4$ via the Hopf fibration $S^1 \to S^3 \to S^2$, or its higher-rank analogues [32, 34].*
- ***M2*** *(Seeley–DeWitt heat-kernel coefficient): introduces $\sqrt{\pi}$ via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on unit $S^3$ (Paper 28 Theorem T9 [33]: $\zeta_{D^2}(s)$ is a two-term polynomial in $\pi^2$ with rational coefficients at every integer $s \geq 1$; at half-integer $s \geq 5/2$ it carries odd zeta values instead). The sub-mechanism for the spinor sector is the half-integer Hurwitz $\zeta(s, 3/2)$.*
- ***M3*** *(vertex-topology Dirichlet L-content): introduces Catalan $G = \beta(2)$ and Dirichlet $\beta(s)$ via quarter-integer Hurwitz shifts $\zeta(s, 3/4)$, $\zeta(s, 5/4)$ at two-loop sunset vertices (Paper 28 Theorem 3: $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$).*

*Proof sketch.* (i) ⇔ (ii) is Paper 35 Prediction 1, verified across 208 individual checks (sprints KG, TX-B, May 2026 [37]).

(ii) ⇒ (iii): the proof body that follows enumerates the fifteen Paper 34 projections in §III.1–§III.15 (the corpus state at this theorem's drafting; Paper 34 has since grown to twenty-eight projections [36], and the theorem's claim is here restricted to those first fifteen pending an extension of the case-by-case enumeration to projections 16–28). Each of the fifteen Paper 34 projections is either $\pi$-free at the projection step (projections 1, 3, 4, 5, 8, 9, 12, 13, 14: Fock conformal, Bargmann–Segal, Stereographic, Sturmian, Wigner $3j$, Wigner $D$, Molecule-frame, Drake–Swainson, Rest-mass) or $\pi$-bearing through a continuous integration that reduces to one of M1, M2, M3 by direct identification (Hopf bundle, Spectral action, Camporesi–Higuchi spinor lift, Wilson plaquette, Vector-photon promotion, Observation/temporal-window). The nine $\pi$-free projections live in either Paper 31's universal sector (algebra-only, transfers to any spherical-fermion system) or Paper 22's potential-independent angular sparsity tier; neither sector contains $\pi$. The six $\pi$-bearing projections are tagged, one by one, to the mechanism of their $\pi$-bearing integration: Hopf bundle, Wilson plaquette (Haar volume) and vector-photon promotion to M1 (Paper 25); spectral action (also M1), Camporesi–Higuchi spinor lift (half-integer Hurwitz) and observation/temporal window (Matsubara $\zeta_R(2k)$) to M2 (Paper 28 T9); M3 arises only in the composition of the spinor lift with a

vertex-parity rule (Paper 28 Theorem 3). This leg is a tagging, made case by case and recorded in the sprint chronicle (CHANGELOG); it has no regression test, which is one reason the theorem is **[CONDITIONAL]**.

The converse (iii) ⇒ (i) is not established: engaging a mechanism is not known to force $\pi$ into the converged value. Paper 55 [56] tags the rational Casimir trace $B = 42$ as M1's "rational core" (a tagging convention, not a proof that $B$ is the converged value of a chain engaging M1); the spinor spectral zeta of M2's sub-mechanism at half-integer $s$ carries no known $\pi$, e.g. $\zeta_{D^2}(5/2) = 14\,\zeta(3) - \frac{31}{2}\,\zeta(5)$ (Paper 28 T9; this lies outside the pure-Tate M2 ring, which is scoped to Seeley–DeWitt coefficients and integer $s$); M3's convergent closed form at $s = 4$ gives $2^3(\beta(4) - \beta(2)) = 8(\beta(4) - G)$, with no known $\pi$ (none of $\zeta(3)$, $\zeta(5)$, $G$, $\beta(4)$ is known to lie in $\mathbb{Q}[\pi]$, nor proved not to). What remains closed-form is the forward identification per mechanism: M1's $\mathrm{Vol}(S^2)/4 = \pi$ and M2's heat-kernel $a_0(S^3) = \sqrt{\pi}$ carry $\pi$ where those specific quantities enter. □

**Corollary 2** (M1 sub-mechanism lies in the localised pure-Tate sub-ring generated by $\pi$)**.** *The M1 sub-mechanism (the $k = 0$ volume slot of the master Mellin engine, carrying the factor $\mathrm{Vol}(S^2)/4 = \pi$ of Paper 25's $S^3 \to S^2$ bundle; Paper 38's $4/\pi$ is* twice *the unit-radius sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ and not a base-to-total ratio) produces period values in the localised pure-Tate sub-ring $\mathbb{Q}[\pi, \pi^{-1}]$ of mixed Tate periods over $\mathbb{Q}$, at depth 0 and Tate weight $\in \mathbb{Z}$. **[PANEL-VERIFIED]** Every closed form placed in the M1 slot across the corpus (the Fejér-constant entries are placed there as a reading, not a derivation; Paper 38) (Paper 25 unit-sphere volume slot $\mathrm{Vol}(S^2)/4 = \pi$; Paper 38 §L2 Fejér-moment constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$, a kernel constant and not the convergence rate; Paper 40 [54] Thm 1 universal Fejér-moment constant $4/\pi$ across compact connected simple Lie groups (derived at rank 1, with an analytical universality mechanism at proof-sketch tier and numerical support at rank 2 plus a noisier rank-3 datapoint; a circle factor carries $2/\pi$); and the M1 factor $\pi^{-2}$ of the $\zeta(3)$ term $-3\zeta(3)/(16\pi^2)$ of the free-scalar F-value in Paper 50 [55] §3, an M1×M3 cross-product whose $\zeta(3)$ is M3 content — a continuum Klebanov–Pufu–Safdi value re-derived there, not a finite-cutoff output) is a $\mathbb{Q}$-linear combination of $\pi^n$ for $n \in \mathbb{Z}$, with $|n| \le 2$ on the empirical witnesses logged so far. Higher integer powers of $\pi$ enter only through M2 (Seeley–DeWitt) or M3 (vertex-parity sums). The M1 ring does not engage M2's $\sqrt{\pi}$ vs $\pi^2$ volume-normalisation choice, because the volume slot is itself a volume integral and does not engage the heat-kernel asymptotic; it is* not *free of the Killing-form normalisation, however: under Kac's basic form the Paper 38 constant reads $2\sqrt{2}/\pi$ (discussion of Theorem 5; Paper 38), which lies in $\mathbb{Q}(\sqrt{2})[\pi, \pi^{-1}]$ rather than $\mathbb{Q}[\pi, \pi^{-1}]$, so the ring statement holds in the corpus's dual-Coxeter convention. Sprint M1 Pure-Tate (June 2026, memo `debug/sprint_m1_pure_tate_memo.md`).*

**Corollary 3** (M2 sub-mechanism inherits Fathizadeh–Marcolli mixed-Tate classification, with pure-Tate refinement)**.** *At the standard volume-normalized Connes–Chamseddine convention, the M2 sub-mechanism Seeley–DeWitt coefficients on $S^3$ are mixed-Tate periods over $\mathbb{Q}$ by direct inheritance from Fathizadeh–Marcolli [7] (the static-R–W sub-case $a(t) \equiv 1$ on $\mathbb{R} \times S^3$). **[PANEL-VERIFIED]** The GeoVac $S^3$ M2 sector is in fact the strictly smaller pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ of even-weight Tate periods, with no $\zeta(3), \zeta(5)$, or MZV content appearing at the SD level. This refines F–M's continuum classification in two structural ways: (i) the Dirac SD expansion is two-term exact ($a_k^{D^2} = 0$ for $k \ge 2$, Paper 51 [47] Cor. 2.1), not generically infinite; (ii) the scalar Laplacian SD coefficients have the closed-form geometric-series collapse $a_k^{\Delta} = 2\pi^2/k!$. The M3 sub-mechanism (vertex-parity Hurwitz, Catalan $G$, $\beta(s)$) lives in cyclotomic mixed Tate at level 4 (mixed Tate over $\mathbb{Z}[i]$, not over $\mathbb{Q}$) and does* not *contaminate the SD coefficients, by the operator-order partition of Paper 18 [28] § III.7 ($k = 2$ for SD vs $k = 1$ for vertex parity). Sprint Mixed-Tate Test (June 2026, memo `debug/sprint_mixed_tate_test_memo.md`).*

**Corollary 4** (M3 sub-mechanism inherits Deligne–Glanois cyclotomic mixed-Tate classification at level $\leq 4$)**.** *The M3 sub-mechanism (vertex-parity Hurwitz / Dirichlet L / depth-k loop tower) on the discrete Camporesi–Higuchi $S^3$ spectral triple lies in the cyclotomic mixed-Tate period ring at level $\leq 4$ over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$ at depth at most the loop order, by direct application of Deligne [8] (proven for $N \in \{2, 3, 4, 6, 8\}$) and the explicit basis of Glanois [9]. Three sub-rings are realised:*

$$\mathcal{MT}(\mathbb{Z}) \ \subset \ \mathcal{MT}(\mathbb{Z}[1/2]) \ \subset \ \mathcal{MT}(\mathbb{Z}[i, 1/2]),$$

*selected by (i) un-restricted vs vertex-restricted summation, and (ii) loop order bounding the motivic depth. Specifically:*

- *Un-restricted Dirac Dirichlet $D(s)$ at integer $s \geq 2$ (Paper 28 [33] Thm 2) lies in $\mathcal{MT}(\mathbb{Z})$ at depth 1;*
- *Vertex-restricted $D_{\text{even}} - D_{\text{odd}} = 2^{s-1}(\beta(s) - \beta(s-2))$ at integer $s \geq 2$ (Paper 28 [33] Thm 3) lies in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level 4, depth 1, via Hurwitz at quarter-integer shifts; Catalan $G = \beta(2)$ and $\beta(4)$ are level-4 Dirichlet L-value periods in this ring;*
- *The two-loop $S_{\min}$ (Paper 28 §$S_{\min}$; resolved in closed form in $\mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$) lies in $\mathcal{MT}(\mathbb{Z}[1/2])$; the depth-k tower (Paper 28 Prop. depth_k) lies in $\mathcal{MT}$ at level $\leq 4$ at depth at most the loop order (realized depths can drop, Paper 55).*

*The vertex-parity-as-$\chi_{-4}$ Dirichlet character mechanism is read as an instance of the Deligne–Glanois Galois descent from level 4 to level 2 (the $N' = 2 \mid N = 4$ instance of Glanois [9] Cor. 1.2) realised on the framework's natural QED observables. Combined with Corollary 3, the master Mellin engine of Theorem 4 now has a period-theoretic stratification over the cases it treats (the theorem is **[CONDITIONAL]** and scoped to the first fifteen projections): M1 in $\mathbb{Q}[\pi, 1/\pi]$, M2 (Seeley–DeWitt coefficients and integer-s values) in the pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, and M3 in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level $\leq 4$. Specific motivic identification of constants at depth $\geq 3$ within the Deligne–Glanois basis is an open question at each depth ($S_{\min}$ is resolved in closed form, above). Sprint M3 Cyclotomic Mixed-Tate (June 2026, memo `debug/sprint_m3_cyclotomic_mixed_tate_memo.md`).*

**Remark 13** (Joint engagement)**.** *M1, M2, M3 are not set-theoretically disjoint as constructions: the spectral action invokes both M1 (the $k = 0$ volume slot on the spinor bundle) and M2 (Seeley–DeWitt coefficients) simultaneously. The theorem requires only that* at least one *of the three be engaged; multiple engagement is the typical case at higher loop order. The mechanisms are role-disjoint in the Paper 18 § III.7 Mellin-engine sense (M1 = packing-tier $\omega_d$, M2 = heat-kernel via Mellin/Gaussian, M3 = vertex-parity quarter-integer Hurwitz character mechanism). The Track TS-E3 sprint (May 2026) reads M1, M2, M3 as sub-cases of a single master Mellin-engine mechanism $\mathrm{Tr}(D^k \cdot e^{-tD^2})$ on a compact Riemannian manifold, with $k \in \{0, 1, 2\}$ selecting the sub-mechanism.*

**Remark 14** (Status of $K = \pi(B+F-\Delta)$)**.** ***[OBSERVATION]** Paper 2's combination $K = \pi(B+F-\Delta) \approx 1/\alpha$ matches its target at $4.77 \times 10^{-7}$ (raw; $8.8 \times 10^{-8}$ through the cubic's root after the $\alpha^2$ correction). Under Theorem 4, the chains of K's ingredients (Fock ∘ Hopf∘ spinor) engage M1 (the explicit prefactor $\pi$, identified by Sprint A as $\mathrm{Vol}(S^2)/4$ [25]) and M2 (in $F = \pi^2/6 = D_{n^2}(d_{\max})$ via the Mellin transform of the scalar Fock-degeneracy). The $\pi$-content is therefore consistent with case-exhaustion. The remaining anomaly is at the* depth-prediction *layer of Paper 34 § VIII.3 (depth predicts $\sim 3\%$ residual at three projections; K matches at $4.8 \times 10^{-7}$, the cubic's root at $8.8 \times 10^{-8}$): a quantitative-error puzzle, distinct from the qualitative case-exhaustion of the present theorem, and unaffected by it.*

**Remark 15** (Falsification target)**.** ***[OPEN]*** *A counter-example to Theorem 4 would be a finite chain of the first fifteen projections introducing $\pi$ via a fourth mechanism not in {M1, M2, M3}; a sixteenth projection doing so would fall outside the theorem's scope and would instead block its extension. Three natural candidates were investigated in Track TS-E3 (May 2026) and each REDUCED to combinations of the three existing mechanisms: continuum-gravity coupling (M1+M2), anomaly-class projections (M1+M2+M3 via the eta-invariant's first-order Mellin), and non-trivial principal-bundle projections (M1 normalisations against integer integrands). The strongest remaining computational falsification target is the discrete first-Chern-class invariant on the Hopf $S^3$ graph at $n_{\max} = 3$, expected $\pi$-free by the case-exhaustion reading; if a $\pi$ appears unexpectedly, a fourth mechanism M4 would need to be named.*

**Remark 16** (Mechanism-as-domain reading (Sprint MR-A/B/C, May 2026))**.** *A three-track follow-up sprint (MR-A: Dirac state-space GH rate; MR-B: spectral-action modular residual; MR-C: L2 next-order constant) tested whether the master Mellin engine of Paper 18 § III.7 is also read (a reading, not a derivation) as organizing convergence rates, not only tagging $\pi$-sources. The combined verdict is a mechanism-as-domain* reading *alongside the case-exhaustion theorem.*

- *M1 ($k = 0$) is the slot of volume-slot normalizations; the first-moment constant of the SU(2) scalar central Fejér kernel, $4/\pi$ (dual-Coxeter normalisation; $2/\pi$ on the unit $S^3$), shares its value (a reading, not a derivation, Paper 38). It is a kernel constant, not the state-space GH rate: Paper 38 measures $O(1/n_{\max})$ convergence with a tapered amplitude. The constant is the sphere-volume quotient $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$ for unit-radius spheres, twice that in the dual-Coxeter normalisation; it is not the Hopf fibration's base-to-total ratio, whose base carries half the radius (Paper 38, volume-ratio note).*

- *M2 ($k = 2$) is the natural domain of heat-kernel and spectral-action convergence. Sprint MR-B derived a closed form for the Camporesi–Higuchi modular residual:*

$$\varepsilon(t) \;=\; \sum_{m\geq 1} (-1)^m \sqrt{\pi}\, e^{-m^2\pi^2/t} \left[ t^{-3/2} - 2m^2\pi^2\, t^{-5/2} - \tfrac{1}{2}\, t^{-1/2} \right].$$

  *Verified at $> 100$ digits at every tested $t$; every coefficient is $\sqrt{\pi}$ times an element of $\bigoplus_k \pi^{2k}\,\mathbb{Q}$; modular exponent exactly $\pi^2$ via Jacobi $\vartheta_2$.*

- *M3 ($k = 1$) is the natural domain of vertex-restricted parity-character observables (Paper 28 § QED-vertex: Catalan $G$, $\beta(4)$). Sprint MR-A demonstrated that asking for M3 from a half-integer-only Peter–Weyl truncation (the state-space GH machinery of Theorem 5) is a* category error*: the half-integer-spin truncation cannot span the constant function $\chi_0$ on* $\mathrm{SU}(2)$*, so the kernel does not localize and the moment is identically $\gamma_n^{\mathrm{Dirac}} = \pi$ for every $n_{\max} \geq 1$. The scalar Fejér moment's $(4/\pi)\log(n)/n$ decay (dual-Coxeter normalisation; $2/\pi$ on the unit $S^3$) is an integer×half-integer interference effect; removing the integer sector kills it. This is consistent with the M1 reading when the sub-bundle lacks the constant function.*

*The reading is therefore: the index $k \in \{0, 1, 2\}$ is read as indexing* both *the sub-mechanism and the natural domain of observable from which its transcendental signature can be extracted. The case-exhaustion theorem 4 (**[CONDITIONAL]**, first fifteen projections) tags* which *transcendentals appear in the cases it covers; the domain partition refines to* which observables *can produce them. Sprint MR-C extracted the L2 next-order constant $c = 4.10932146748987562042\ldots$ to* 22 *verified decimal digits but found no PSLQ identification in $M_1 \cup M_2 \cup M_3 \cup \{\gamma_E, \log 2, G, \zeta(3)\}$ at coefficient ceiling $10^4$. This is consistent with $c$ being a downstream Stein–Weiss IBP derivative*

*of the M1-read Fejér constant rather than a direct Mellin output, and is a partial negative against extending the engine's classification to all asymptotic-expansion coefficients along subsequent analytical pipelines. Within each domain the engine organizes the observed values (for $k = 0$ the domain is the Fejér kernel's moment, read as M1, not the GH convergence rate; Paper 38). Source:* `debug/mr_a_dirac_propinquity_*`, `debug/mr_b_spectral_action_rate_*`, `debug/mr_c_l2_subleading_*`.

**The Q5′ cochain-level record.** The Stage-1/Stage-2 Q5′ sprint remarks (JLO and Connes–Moscovici cochains, substrate candidates, bridge identities, path counts and the pro-system and Tannakian follow-ons) are kept verbatim in `docs/chronicles/paper_32_chronicle.tex`. Their degree-$\geq 1$ content was computed on the hopping operator $D = \Lambda + \kappa A_{\text{graph}}$, not on $D_{\text{GV}}$, on which every degree-$\geq 1$ JLO cochain vanishes; their degree-0 content, the continuum Mellin-residue identification of $M_1$, $M_2$, $M_3$, is owned by Paper 18 §III.7, and the Tannakian programme by Paper 56.

**Remark 17** (Rational-residue extension of the case-exhaustion theorem: the Bernoulli ladder and the functional-equation reading of the two-layer split (Sprint GB, 2026-05-29))**.** *The case-exhaustion theorem 4 classifies (conditionally, for chains of the first fifteen Paper 34 projections) the $\pi$-content as $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$. The companion question is where the framework's* rational *coefficients come from. A diagnostic on the gravity sector's small rationals places them on a single Bernoulli ladder, with each rung $B_{2n}$ glued across two windows by the Riemann functional equation:*

$$\zeta(2n) = (-1)^{n+1}\frac{B_{2n}(2\pi)^{2n}}{2(2n)!} \in \pi^{2n}\,\mathbb{Q} \qquad \Longleftrightarrow \qquad \zeta(1-2n) = -\frac{B_{2n}}{2n} \in \mathbb{Q}.$$

*The left (positive-even) window is the M2 transcendental ring of Layer 2 observables; the right (negative-integer) window is the $\pi$-free rational residue carried by Layer 1 skeleton coefficients. Verified placements (all exact in* `sympy`*):*

- *Conical-defect tip coefficient $\frac{1}{12} = -\zeta(-1) = B_2/2$ (§4.15); per-$t$ UV target $\frac{1}{24} = -\zeta(-1)/2 = B_2/4$ (the string $c/24$); scalar $S^3$ Casimir $\frac{1}{240} = +\zeta(-3)/2$ ($B_4$ rung); the $\alpha$-observation's $F = \pi^2/6 = \pi^2 B_2 = \zeta(2)$ (Paper 2; the M2 window of the same $B_2$ whose skeleton window is the conical tip).*
- Non-trivial internal check*: the replica-entropy derivative is forced, not free — $\frac{d}{d\alpha}\big[\frac{1}{12}(\frac{1}{\alpha} - \alpha)\big]\big|_{\alpha=1} = -\frac{1}{6} = -B_2 = 2\zeta(-1)$, the $\alpha$-derivative of the tip's $B_2/2$.*
- Controls behave*: $\kappa = -1/16$ (numerically coincident with $1/\Omega^4(0) = 1/16$, the fourth power of the inverse Fock conformal factor; an Observation, Paper 0 [24]) has no clean $\zeta$ form; the Dirac Casimir $17/480$ misses the scalar $B_4$ rung by exactly $1/32$ and requires the half-integer Hurwitz shift $\zeta(-3, \frac{1}{2})$, i.e. the scalar/spinor distinction is the integer/half-integer Hurwitz split (the half-integer shift is M2's spinor sub-mechanism).*

*The reading is the rational-residue extension of Theorem 4: the rational coefficients are the negative-integer analytic continuation of the* same *Mellin object whose positive-even values are the Layer-2 transcendentals, and the empirical two-layer split (Paper 34/35) is the Riemann functional-equation reflection $\zeta(s) \leftrightarrow \zeta(1-s)$ instantiated per Bernoulli order.*

Honest scope. *This is a structural correspondence at the same epistemic level as Theorem 4, not a forcing proof; it concerns $\zeta$ at* integer *arguments (special values), not the non-trivial zeros, so it makes no contact with the critical-line structure (cf. WH6 and the three classical-RH walls RH-N/O/M, which remain in force). The $B_2$, $B_4$,* and *$B_6$ rungs are now populated (Sprint GB-3):*

*the $B_6$ rung's falsifiable target was met by the conformal-scalar Casimir on $S^5$ (a GeoVac manifold via the Bargmann–Segal lattice, Paper 24), which evaluates to $-31/60480 = \frac{1}{24}(\zeta(-5) - \zeta(-3))$ — the $B_6$ rung $\zeta(-5) = -1/252$ enters with clean coefficient $+\frac{1}{24}$, alongside $B_4$ at $-\frac{1}{24}$, with $B_2$* absent. *The $\zeta(-5) - \zeta(-3)$ spreading (rather than a pure $\zeta(-5)$) is the* same *quadratic-multiplicity mechanism that obstructs the functional equation in Paper 28's functional-equation obstruction remark (the $S^5$ harmonic degeneracy is quartic), now seen on the Casimir side from an independent observable; the method reproduces the $S^3$ control $1/240 = \frac{1}{2}\zeta(-3)$ (Paper 35). The integer/half-integer Hurwitz split recurs: the conformal scalar gives the clean ladder value, the half-integer HO/Bargmann $S^5$ Casimir gives $-17/3840$ (the "17" echoing the Dirac-$S^3$ 17/480). Next rung: $S^7$ conformal Casimir is predicted to spread across $\zeta(-7), \zeta(-5), \zeta(-3)$. The ladder is thus supported across three rungs and two observable types, but remains a structural correspondence, not a forcing theorem. Source:* `debug/sprint_gb_bernoulli_ladder_memo.md`, `debug/sprint_gb3_b6_rung_memo.md`.

**Remark 18** (Bisognano–Wichmann reading of the M1 sub-mechanism on temporal compactifications (Track D, May 2026))**.** *The M1 sub-mechanism of the master Mellin engine, $\mathcal{M}[\mathrm{Tr}(D^0\, e^{-tD^2})]$, produces a* Vol/Mellin-measure *factor on the spatial $S^3$ ($4/\pi = 2\,\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3)$ at unit radius, numerically $\mathrm{Vol}(S^2)/\pi^2$, the L2 Fejér-moment constant in the dual-Coxeter normalisation, read as this factor (a reading, not a derivation; Paper 38) —* not *a quantity on the Hopf base, whose radius is half the total space's; Remark 16) and on a temporal $S^1_\beta$ factor (the bare circumference $\beta$ itself; Sprint TD Track 1, $T_{S^3} \otimes T_{S^1_\beta}$). Sprint TD Track 4 (`debug/sprint_td_track4_memo.md`, May 2026) extends the temporal-side instantiation to the Euclidean Schwarzschild cigar, where smoothness at $r = 2M$ forces the $\tau$-circle period $\beta_{\mathrm{cigar}} = 8\pi M$, consistent with the Hawking temperature $T_H = 1/(8\pi M)$ (sympy residual zero): the M1 volume slot re-expresses the continuum value, it does not derive it—$M$ is external (it comes from the Einstein equations) and $\beta$ is posited at finite cutoff. Track D (`debug/bisognano_wichmann_track_d_memo.md`, May 2026) makes explicit the dual physical reading of M1 on temporal compactifications: under the standard Wick-rotation chain (Hartle–Hawking [43]; Sewell [42]; Bisognano–Wichmann [40, 41]; reviewed in [45]), the framework's M1 $2\pi$ on $S^1_\tau$ matches the period of the modular-flow / Lorentz-boost orbit of a stationary observer outside the Schwarzschild horizon. Equivalently, the M1 sub-mechanism on temporal compactifications corresponds, by analogy, to the Wick-rotated Bisognano–Wichmann modular-flow mechanism; identifying the framework's generator with a boost is open (Paper 42's O0). This is a* structural correspondence *(an analogy), not a literal identification: two mathematical objects (a volume factor on the Riemannian side; a modular-automorphism period on the Lorentzian side) are matched by the published Wick-rotation prescription (Bisognano–Wichmann 1976; Sewell 1982; Hartle–Hawking 1976), not by an operator-level proof inside the truncated spectral triple $\mathcal{T}_{n_{\max}}$. The natural follow-up that would turn the correspondence into an operator-level identification — compute the modular Hamiltonian $K$ on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi} =$ identity at finite $n_{\max}$, then take the GH limit using the lemmas of Theorem 5 — is named as the principal falsifier in Track D §3.1; estimated beyond sprint scale; priority MEDIUM-HIGH. The case-exhaustion theorem 4 itself is unchanged: M1's $2\pi$ carries* two readings of one number *(the $k = 0$ volume slot, the framework's own reading; and the boost-orbit period via Bisognano–Wichmann, an analogy under Wick rotation), not two independent constraints on the master Mellin engine partition. The corollary on Rindler wedges (the Unruh temperature $T_U = a/(2\pi)$, with $\beta_{\mathrm{Rindler}} = 2\pi/a$ posited and re-expressed through the same M1 volume slot — a consistency check, not a derivation) is carried out in the Unruh paragraph below.*

***Unruh pendant (May 2026).*** *The Track D §5.4 follow-up was executed as a pendant sprint (`debug/unruh_pe`s May 2026). Wick-rotating Rindler coordinates $(\eta, \rho)$ for an observer at proper acceleration $a$ produces Euclidean polar $(\eta_E, \rho)$ with conical regularity at $\rho = 0$ forcing $\eta_E$-period $2\pi$, hence*

*proper-time period $\beta_U = 2\pi/a$ and Unruh $T_U = a/(2\pi)$ (Unruh [44]). Sprint TD Track 1's* `matsubara_spectrum()` *applied verbatim with $\beta = 2\pi/a$ reproduces the Unruh-thermal Matsubara spectrum bit-identically (sympy residual zero on $T_U = 1/\beta_U$, lowest bosonic mode $= a$, lowest fermionic mode $= a/2 = \pi T_U$). The single $2\pi$ in $\beta_U$ is the same M1 ($k = 0$ volume slot) / $\mathrm{Vol}(S^1)$ signature that drove the cigar correspondence, instantiated on the Rindler $\eta_E$-circle instead of the cigar $\tau$-circle. The reading is identical: under the Wick-rotation chain, the framework's M1 $2\pi$ on $S^1_{\eta_E}$ matches the period of the modular-flow / boost-orbit of the accelerated observer (an analogy; identifying the framework's generator with a boost is open, Paper 42's O0). Hawking, Bisognano–Wichmann, and Unruh are three faces of one continuum Wick-rotation theorem, whose shared $2\pi$ the framework's M1 ($k = 0$) volume slot matches by analogy (it re-expresses the continuum value and does not derive it); the published Wick-rotation prescription supplies the bridge to the Lorentzian-side modular-flow vocabulary in each case. The* operator-level falsifier *(modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted state has analytic period $2\pi$) named in Track D §3.1 covers all three faces simultaneously, since the bridge runs through the same KMS-Tomita-Takesaki algebra in every case; lifting the period-closure check on any one face lifts the structural correspondence on Hawking, Bisognano–Wichmann, and Unruh together. Same scope (structural correspondence, not an operator-level identification); same falsifier; same MEDIUM-HIGH priority and scale.*

***The finite-cutoff period closure (Sprints L1, L1-tighten). [SYMBOLIC (trivial) + MEASURED]*** *Taking the rotation generator $K = 2J_z$ as modular generator on $\mathcal{T}_{n_{\max}}$ gives $\sigma_{2\pi} = \mathrm{id}$ exactly, but for a reason that carries no Bisognano–Wichmann content: $K$ has integer spectrum, so any operator is returned after period $2\pi$, and any other integer-spectrum generator passes the same check (*`tests/test_modular_period_closure_automatic.py`*, which also shows a non-integer-spectrum generator fails it). $K$ anticommutes with the reflection $R_{\mathrm{polar}}$ that defines the wedge projector, so $P_W K P_W = 0$, and the wedge operator is a separately defined diagonal operator, not a restriction of $K$ (*`tests/test_p32_polar_generator_wedge.py`*). The Tomita–Takesaki cross-check of Sprint L1-tighten is circular, since its state is built as $e^{-K}/Z$. The falsifier named above —* derive *$K$ as the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state, rather than choose it, and check its period — remains* ***OPEN****. The sprint record is kept verbatim in* `docs/chronicles/paper_32_chronicle.tex`*.*

**Theorem 5** (Gromov–Hausdorff convergence on $S^3$)**.** *Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\mathrm{CH}})$ be the round-$S^3$ metric spectral triple with the Camporesi–Higuchi Dirac, and let $\mathcal{T}_{n_{\max}} = (\mathcal{O}^{\mathrm{CH}}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ be the Connes–van Suijlekom truncation at cutoff $n_{\max}$ on the* true *Camporesi–Higuchi window: $\mathcal{H}_{n_{\max}}$ is the span of the complete Dirac eigenspaces with $|\lambda| \le n_{\max} + \frac{1}{2}$, $\mathcal{O}^{\mathrm{CH}}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ ($\dim = 5, 30, 91$ at $n_{\max} = 1, 2, 3$), and the state spaces are metrized by the left-translation Lipschitz seminorm (Paper 38; this is* not *the Fock-shell operator system of sprint R2.1, a model object, Remark 1). Then the truncated triples converge to $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance (the strengthening to the Latrémolière quantum propinquity remains an open named gap; see Remark 19):*

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}} \;\xrightarrow[n_{\max}\to\infty]{}\; 0,$$

*where $\gamma_{n_{\max}}$ is the L2 mass-concentration moment of the central spectral Fejér kernel on SU(2). Its asymptotics $\gamma_{n_{\max}} = (4/\pi + o(1)) \log n_{\max}/n_{\max}$ (its $O(1/n_{\max})$ remainder is the measured part) (dual-Coxeter normalisation, $\mathrm{Cas}(\mathrm{ad}) = h^\vee$; equivalently $2/\pi$ on the unit $S^3$ this paper declares, and $2\sqrt{2}/\pi$ under the field-standard Kac form — the constant scales with the metric) is* ***[INTERNAL THEOREM + MEASURED]*** *in Paper 38 [53]; the constant is the Fejér kernel's, and with a tapered amplitude Paper 38 measures $d_{\mathrm{GH}} = O(1/n_{\max})$* ***[MEASURED]****. What is proved unconditionally is the bound $d_{\mathrm{GH}} \le \gamma_{n_{\max}}$ and $\gamma_{n_{\max}} \to 0$, while the uniform bound $\gamma_{n_{\max}} \le 6 \log n_{\max}/n_{\max}$*

*($n_{\max} \geq 2$) is **[MEASURED]** (a finite-range check, not proved for all $n_{\max}$); the small-n closed forms are consistent with it. Convergence in the Dirac-commutator seminorm, and of the almost-commutative (gauge-network, $\mathcal{A} \otimes M_n$) structure, is **[OPEN]**.*

*Proof sketch.* The theorem is established by Paper 38's [53] lifted-state unconditional route in the van Suijlekom state-space Gromov–Hausdorff distance (*not* a Latrémolière propinquity — that strengthening is the open named gap of Remark 19). The five-lemma UCP-pair assembly of WH1 sprint R2.5, whose lemma L5 fails ($B_{n_{\max}}$ is finite-band, so $\text{height}_B \equiv 1$ at every cutoff; Paper 38), and the module panel that measured it (`geovac/gh_convergence.py`) are kept verbatim in `docs/chronicles/paper_32_chronicle.tex`. □

**Remark 19** (Status and named gaps)**.** *One named gap stands for Theorem 5, item (iii) below; items (i), (ii) and (iv) concern the relocated UCP-pair assembly (its lemma L5 fails) and do not affect the theorem. (i) The L2 cb-norm statement conflated the kernel's Plancherel* mass distribution *with its convolution-*multiplier symbol*: the smoothing map is unital completely positive with cb-norm exactly* 1*, and the quantity* $2/(n_{\max}+1)$ *is the mass-distribution maximum, not a cb-norm; the moment* $\gamma_{n_{\max}}$*, its closed forms, and its* $4/\pi$ *asymptote are unaffected. (ii) The* $\text{reach}_P$ *dual estimate in L5 is a* named gap*: a forward cb-norm controls no partial inverse; the correct mechanism is an antiderivative-transference lemma in the style of Leimbach–van Suijlekom [14]; for scalar sectors on compact metric groups the conclusion is covered by Gaudillot-Estrada–van Suijlekom [18]. (iii) The distance proved is van Suijlekom's state-space Gromov–Hausdorff distance [13]; the Latrémolière propinquity label is an open named gap — the Latrémolière framework [51] builds tunnels from quantum isometries, not UCP pairs. (iv) The kernel condition* $L(a) = 0 \iff a \in \mathbb{C}\mathbf{1}$ *for the Dirac-commutator seminorm. The counts* $10/14$ *(*$n_{\max} = 2$*) and* $26/55$ *(*$n_{\max} = 3$*) for the chirality-diagonal CH Dirac, and* $1/14$*,* $1/55$ *for the engineered offdiag Dirac, are counts on the* doubled Fock model *of Remark 1, not on the truncated Camporesi–Higuchi triple: on the true window the Dirac-commutator kernel is exactly the scalars at* $n_{\max} \leq 3$ *(**[MEASURED]**, Paper 38, remark on the Dirac-commutator seminorm;* `tests/test_p38_true_ch_window.py`*). Item (ii) is bypassed by the translation-seminorm reframing of the companion* $S^3$ *paper, whose lifted-state route uses no dual reach; item (iii) stands (the propinquity label remains a named gap). The reframing was adopted because its compression and lifted-state estimates close, not because the Dirac seminorm fails; whether the convergence can be proved in the Dirac seminorm is **[OPEN]**. The truncated state space is metrized by the left-translation Lipschitz seminorm (equal to the Lipschitz constant on the continuum), whose kernel is exactly the scalars on the true window at every cutoff (Schur multiplicity-one plus per-band injectivity of band-limited compression); the dual direction is supplied by an exact-fit spinor lifted state (the Fejér vector* $h \otimes \chi$*, lying in the window by the* $V_j \otimes V_{j\pm 1/2}$ *block decomposition of the Camporesi–Higuchi eigenspaces), and both almost-inverse defects are Fejér smoothings at the common moment* $\gamma_{n_{\max}}$ *— no partial inverse or transference estimate remains. **[INTERNAL THEOREM]** The convergence theorem is unconditional in van Suijlekom's state-space framework, bounded by* $\gamma_{n_{\max}} \to 0$*; **[INTERNAL THEOREM + MEASURED]** the Fejér moment is* $(4/\pi + o(1)) \log n_{\max}/n_{\max}$ *in the dual-Coxeter normalisation (*$2/\pi$ *on the unit* $S^3$*), and a tapered amplitude gives **[MEASURED]*** $O(1/n_{\max})$ *(Paper 38).*

**Remark 20** (Limit identification)**.** ***[INTERNAL THEOREM]** Paper 38 (its limit-identification proposition) identifies the state-space limit,* up to isometry*, as* $(P(S^3), W_d)$*: the unconditional theorem gives GH convergence of the true-window state spaces, metrized by the Monge–Kantorovich metric of the left-translation seminorm* $L$*, to* $(S(C(S^3)), \text{MK}_L)$*; a GH limit is unique up to isometry, and since* $L = \text{Lip}_d$*, Kantorovich–Rubinstein duality gives* $\text{MK}_L = W_d$*, the Wasserstein distance*

*for the (dual-Coxeter) geodesic distance $d$. The identification is for the translation-seminorm metric, not for the Dirac Connes distance $d_{D_{n_{\max}}}$: convergence in the Dirac-commutator seminorm is **[OPEN]**.*

**Remark 21** (WH1 keystone: the commutative factor)**.** *Theorem 5 closes the commutative-factor leg of the WH1 keystone (working-hypothesis register). **[INTERNAL THEOREM]** The Connes–van Suijlekom truncations of the commutative factor — the round-$S^3$ Camporesi–Higuchi triple — are proved to converge in state-space Gromov–Hausdorff distance ($d_{\mathrm{GH}} \le \gamma_{n_{\max}} \to 0$). **[OPEN]** The structural claim of WH1 as a whole — that GeoVac is an almost-commutative spectral triple whose truncations, including the gauge-network $\mathcal{A} \otimes M_n(\mathbb{C})$ structure, converge — is* not *proved: convergence of the almost-commutative structure is open, as is convergence in the Dirac-commutator seminorm. **[INTERNAL THEOREM + MEASURED]** The Fejér-kernel bound is $(4/\pi + o(1)) \log n_{\max}/n_{\max}$ (its $O(1/n_{\max})$ remainder is the measured part; a tapered amplitude gives **[MEASURED]** $O(1/n_{\max})$) (dual-Coxeter normalisation,* $\mathrm{Cas}(\mathrm{ad}) = h^\vee$*; equivalently $2/\pi$ on the unit $S^3$ this paper declares, and $2\sqrt{2}/\pi$ under the field-standard Kac form — the constant scales with the metric) (Paper 38), on the true Camporesi–Higuchi window under the translation-seminorm metrization. The L3 constant $C_3 = 1$ holds for every $f$ and every $n_{\max}$ by compression (Paper 38, Lemma L3, **[INTERNAL THEOREM]**). The remaining open questions (Higgs-gap analysis on the four-way $S^3$ coincidence; whether the order-zero and order-one conditions, which fail at finite $n_{\max}$ **[MEASURED]**, are recovered in any weaker sense in the limit — the KO sign pair itself is settled, Sec. 4) are downstream of this theorem.*

**Remark 22** (Pythagorean orthogonality of the L2-E residual; a normalisation-dependent $1/\pi^2$)**.** *The case-exhaustion theorem 4 classifies (conditionally, for chains of the first fifteen Paper 34 projections) the transcendental ring in which the $\pi$-content must sit. A post-Sprint-L2-E PSLQ probe (`debug/h_local_residual_pslq_memo.md`, 2026-05-17) records the following operator-level identity; its $\pi$-content is fixed by a posited normalisation (below), so it is* not *an instantiation of that theorem. On the truncated Krein-carrier triple (Paper 42 [46] §11.7), the local Hamiltonian $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$ (the diagonal wedge relabelling of Paper 42 Def. 5.1 of the posited generator $2J_z$; BW identification open, Paper 42's O0) and the wedge-restricted Lorentzian Dirac $D_W^L$ are Hilbert–Schmidt orthogonal,*

$$\langle H_{\mathrm{local}}, D_W^L \rangle_{\mathrm{HS}} \;=\; 0 \qquad \textit{(bit-exact at every tested } (n_{\max}, N_t)\textit{)},$$

*and the squared residual decomposes as a Pythagorean sum:*

$$\left\| H_{\mathrm{local}} - D_W^L \right\|_F^2 \;=\; \frac{\kappa_g^2 \cdot S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

*with $S, D \in \mathbb{Q}[n_{\max}]$ given by cumulative wedge Casimir traces (Paper 42 §11.7, Cor. 11.15 therein). The sole transcendental in the residual is $1/\pi^2$, and it enters* only *through the posited normalisation $H_{\mathrm{local}} = K/\beta$ with $\beta = 2\pi/\kappa_g$: it is the continuum KMS period imported as a normalisation and squared (a temporal-compactification $\pi$ in the sense of Paper 35),* not *an M1 Mellin output. The orthogonality itself is a chirality-parity identity that holds for any rescaling of $K$ and is independent of $\beta$ (Paper 42, Thm. 11.14).*

*The residual's ring $\mathbb{Q}[n_{\max}] \oplus \pi^{-2}\,\mathbb{Q}[n_{\max}]$ is therefore a statement about the chosen normalisation, and it does not add an instantiation of the mechanism-as-domain partition of Remark 16 (M1 volume-slot normalizations, the Fejér-moment constant sharing the value as a reading; M2 heat-kernel residuals, M3 vertex-restricted parity observables).*

*The orthogonality holds for all six modular witnesses (Paper 42 [46], the corollary on the single operator-system construction; Paper 42 §11.7), via $\kappa_g$-linearity of $H_{\text{local}}$ and $\kappa_g$-independence of $D_W^L$ — by construction, since the witnesses share one posited $K$. The mechanism is a chirality-parity identity, proved in general (Paper 42, Thm. 11.14 and Cor. 11.15): in the unfolded wedge spinor basis $H_{\text{local}}$ is real, diagonal and independent of the chirality $\chi$ (eigenvalues* $\text{two_m}_j$*), while the wedge Dirac is chirality-diagonal with $\chi$-paired eigenvalues of opposite sign, so the pairs cancel in the trace. The regression test (`tests/test_paper43_pythagorean_hs_orthogonality.py`) checks 7 cells — Riemannian $n_{\max} = 2, 3, 4$ and Lorentzian $(n_{\max}, N_t) = (2,1), (3,1), (2,3), (3,3)$.*

Cross-references. *Paper 42 §11.7 Cor. 11.15 (closed form and witness-universality); Paper 42 §8, the Hilbert–Schmidt orthogonality remark (which states that the orthogonality does* not *underlie the six-witness agreement, which holds by construction); Paper 24 §5.3 Coulomb/HO asymmetry (seven catalogued, six standing; the structural finding that the Pythagorean orthogonality is $S^3$-specific to the construction — the $S^5$ Hardy sector has no clean analog, adding the Pythagorean orthogonality for the posited wedge generator as a further Coulomb/HO asymmetry layer alongside calibration $\pi$ and Wilson matter coupling).*

**Remark 23** (BH Phase 0 — clean negative on wedge KMS area-law)**.** *The modular-Hamiltonian construction of Sprint L1 (a finite-cutoff period closure for a chosen integer-spectrum generator, this section; it identifies no Bisognano–Wichmann flow) and Sprint L2-E (its signature-blind Krein-carrier counterpart) is compatible with the Hartle–Hawking and Unruh* temperatures *via the M1 mechanism (§ 18). A natural follow-up question is whether the same construction also reproduces the Bekenstein–Hawking* entropy $S_{BH} = A/(4G)$ *at finite cutoff via the von Neumann entropy of the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ (the unit-$\beta$ Gibbs state of the diagonal wedge relabelling $K_\alpha^W$ (values $2m_j$ on the $m_j > 0$ states; Paper 42 Def. 5.1) of the posited generator $2J_z$, not BW-normalised: that would be $e^{-\pi K_\alpha^W}/Z$, Paper 50 [55]). Sprint BH-Phase0 (`debug/bh_phase0_diagnostic_memo.md`, 2026-05-22) tests this directly at $n_{\max} \in \{2, 3, 4, 5, 6, 7\}$.*

Result. *$S(\rho_W) \approx 1.944 \log(n_{\max}) + 0.565$ at $R^2 = 0.99991$ across the panel. No polynomial scaling appears in any of the four candidate fits (log-linear, power-law, poly-2, poly-3). An s-deformation $\rho(s) = e^{-sK_\alpha^W}/Z$ across $s \in \{0.01, 0.05, 0.1, 0.5, 1, 2, 2\pi\}$ shows power-law exponents $\alpha \in [0.64, 0.72]$, never close to the $\alpha = 2$ that a naive Bekenstein–Hawking area-law $S \sim n_{\max}^2$ would predict.*

Mechanism. *The structural reading is*

$$S(\rho_W) \;\approx\; \log\bigl(n_{\text{equator}}\bigr), \qquad n_{\text{equator}} = n_{\max}\,(n_{\max}+1),$$

*where $n_{\text{equator}}$ is the count of wedge states with smallest $|m_j| = 1/2$ (the equator analog). The unit-$\beta$ state ($s = 1$) is effectively the maximally mixed state on the lowest-$K_\alpha$ shell, with exponentially suppressed contributions $\sim e^{-(2k+1)}$ from shells $k = 1, 2, \ldots$. The two asymptotic limits saturate bit-exactly: $S(s \to 0) = \log(\dim W)$ (hot, maximally mixed) and $S(s \to \infty) = \log(n_{\text{equator}})$ (cold, lowest shell only). The unit-$\beta$ state ($s = 1$) sits between, closer to the cold limit.*

*The slope $\approx 2$ is the exponent of the $n_{\max}(n_{\max}+1)$ degeneracy of the lowest shell of the posited generator **[MEASURED]**. This is structurally weaker than the QFT Bekenstein–Hawking area-law $S \sim A \cdot \Lambda^2/4$, which would scale as a power of the UV cutoff times the boundary dimension.*

Structural reading. *The operator-system truncation on which the finite-cutoff period closure holds (§ VIII.F, Sprint L1) does* not *preserve the continuum QFT vacuum entanglement that would give the Bekenstein–Hawking area-law. The modular-flow period closure (a posited $2\pi$ period, tagged M1) and the Bekenstein–Hawking area-law are structurally distinct features of the gravitational thermodynamics chain; at finite cutoff the GeoVac framework exhibits the former only in the automatic integer-spectrum sense, and not the latter. Consistent with the structural-skeleton-scope pattern*

*(project development-frontier log, sprints H1, LS-8a, HF-3/4/5, multi-focal-composition wall, W3): the framework supplies an integer-spectrum candidate for the algebraic skeleton of modular flow ($K = 2J_z$ posited; its Bisognano–Wichmann identification is open, Paper 42's O0) but does not autonomously generate the calibration data (here: Newton's constant) that links wedge-entropy to physical area.*

*Cross-references: Paper 34 §V.B Bekenstein–Hawking probe row; Sprint TD Track 4 (Hawking temperature, machinery-witness, M1 closure); the May-16 BH scoping memo (`debug/bekenstein_hawking_sprint`; full Connes–Chamseddine cigar route) remains the longer-route alternative if a literal $S_{BH} = A/(4G)$ closure is pursued, but with the honest scope statement that Newton's constant is calibration input not derivation.*

## 8.1 The unified $U(1) \times SU(2) \times SU(3)$ gauge content and its limits

The case-exhaustion theorem of Sec. 8 (**[CONDITIONAL]**, first fifteen projections) tags M1 (the $k = 0$ volume slot) as the mechanism by which the Wilson plaquette-Haar projection injects $\pi$ into spectral-triple observables. This subsection records a complementary structural fact about the family of GeoVac sub-triples that engage M1: they realise all three Standard-Model gauge groups as Wilson lattice constructions, on three structurally distinct sub-manifolds, with a weak-coupling coefficient $1/(4\,N_c)$ shared by the two non-abelian constructions (the $U(1)$ value $\frac{1}{4}$ coincides with it at $N_c = 1$). The construction is honest about what it is and is not: a unified *catalogue* of three Wilson-Yang–Mills instances of the Marcolli–van Suijlekom gauge-network framework, not a unified spectral triple in the Connes–Chamseddine sense [5].

### Three Wilson constructions, a shared non-abelian kinetic coefficient

**[PANEL-VERIFIED]** Three independent sprints have produced gauge-invariant, finite, Haar-normalisable Wilson lattice gauge theories on GeoVac sub-graphs (one abelian, two non-abelian):

- *Paper 25* [32] constructs the abelian $U(1)$ Wilson–Hodge structure on the Fock-projected $S^3$ Coulomb graph (Paper 7), with phases $\chi_v$ on nodes and $A_e$ on edges covariant under the standard incidence operator.
- *Paper 30* [34] constructs the non-abelian SU(2) Wilson lattice gauge theory on the same $S^3 = \mathrm{SU}(2)$ graph, with link variables $U_e \in \mathrm{SU}(2)$, plaquettes from primitive non-backtracking walks, and Haar measure $dU$. Paper 30 §5.2 verifies the weak-coupling kinetic coefficient symbolically and numerically.
- *Sprint ST-SU3* (May 2026, modules `geovac/su3_wilson_s5.py`, `tests/test_su3_wilson_s5.py`, 47 pass and 2 skipped of 49 as of 2026-09-02) constructs the non-abelian SU(3) Wilson lattice gauge theory on the Bargmann–Segal $S^5$ graph of Paper 24 [31], with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (independent of the shell index $N$), plaquettes from primitive non-backtracking walks, and SU(3) Haar measure. Sprint ST-SU3 §4 verifies Theorem 2 (kinetic coefficient 1/12) symbolically and numerically at $N_{\max} = 2, 3$ (7 765 plaquettes), with maximal Cartan reduction to a coupled $U(1) \times U(1)$ theory verified at machine precision.

The three constructions are summarised in Table 4. The columns are not independent discoveries: each is a specialisation of the gauge-networks framework of Marcolli–van Suijlekom [6], with the Perez-Sanchez correction [11] that the continuum limit of such gauge networks (whose lattice action is of Wilson type) is the Yang–Mills action *without* a Higgs sector.

In each construction the edge Laplacian $L_1 = B^\top B$, where $B$ is the signed node–edge incidence matrix, plays the role of weak-coupling kinetic operator on $\mathfrak{g}$-valued 1-cochains. Paper 25's "discrete Hodge–1 Laplacian" is therefore the same combinatorial construction across all three; the host graph is the $S^3$ Coulomb graph for the first two and the $S^5$ Bargmann graph for the third, and the target Lie algebra changes ($\mathfrak{u}(1)$, $\mathfrak{su}(2)$, $\mathfrak{su}(3)$). **[OBSERVATION]** This is the single construction unifying the three sectors at the kinetic level.

**The two-step SU(3) story**

The SU(3) row of Table 4 requires a specific clarification, because an earlier Sprint 5 Track S5 (April 2026, memo `debug/s5_gauge_structure_memo.md`) returned a clean negative when asking whether the $S^5$ Bargmann–Segal lattice carries a non-abelian gauge structure analogous to Paper 25's $U(1)$. The two findings are not in tension: they answer two different questions, and read together they give a two-step honest picture.

- *Step 1 (Sprint 5 Track S5, negative).* An adjacency-preserving SU(3) action on the $(N, 0)$ tower of the Bargmann graph fails Wilson's fixed-group-on-every-link requirement, because the transitions $(N, 0) \to (N + 1, 0)$ are Clebsch–Gordan intertwiners between distinct SU(3) irreducible representations of growing dimension $\dim(N, 0) = (N + 1)(N + 2)/2$, not group elements of a single SU(3). The natural shell-respecting gauge group of the Bargmann graph is $U(1)$, and this is recorded in Paper 25 §VII.A as a structural asymmetry between the $S^3$ and $S^5$ sectors.
- *Step 2 (Sprint ST-SU3, positive at the gauge level).* A *shell-independent* Wilson lattice gauge theory with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (not in any of the $(N, 0)$ irreps) is a structurally different object that bypasses the Step 1 obstruction. The links live as external gauge degrees of freedom on edges; they do not act on shell labels. Sprint ST-SU3 verified gauge-invariance of the Wilson action under node-local $U_e \mapsto g_{\mathrm{src}(e)} U_e g^\dagger_{\mathrm{tgt}(e)}$ to machine precision and computed the kinetic coefficient $1/12$ both symbolically and numerically.
- *Matter coupling (Sprint ST-SU3 Theorem 3, mixed).* When one tries to couple this Wilson SU(3) to the natural shell matter on the $(N, 0)$ tower, the original Step 1 obstruction is rediscovered: a fundamental-representation link cannot uniformly couple to representations of growing dimension, and the matter coupling reverts to the Clebsch–Gordan intertwiner structure. Wilson SU(3) on the Bargmann graph is therefore self-consistent as a *pure-gauge* theory but does not admit a natural matter coupling in the sense of Paper 30's spin-1/2 Dirac fermions on the Coulomb graph.

The two-step picture is consistent with Paper 24's Coulomb/HO asymmetry (seven layers catalogued, six standing, Papers 24 and 31) and sharpens its Wilson gauge-with-natural-matter layer: on the $S^3$ Coulomb side the gauge group acts in the same representation as the matter, so matter–gauge coupling is structurally available (argued, not constructed). On the $S^5$ Bargmann side the combinatorial Wilson–Hodge vocabulary is universal but the gauge–matter coupling is not.

**Marcolli–vS lineage of the unified content**

Each construction in Table 4 is a specific gauge-networks instance in the sense of Marcolli and van Suijlekom [6], with the Perez-Sanchez correction [11] that the continuum limit of such a gauge network is the continuum Yang–Mills action (the lattice action being of Wilson type) *without* a

Higgs. The vertex algebra is $\mathcal{A}_v = \mathbb{C}$ (or $C(V)$ globally) in all three rows; the vertex Hilbert space and the gauge group differ row-by-row, as listed in the table. In each row the gauge-network spectral action reduces to the standard Wilson action plus a quadratic kinetic term in the weak-coupling expansion, with coefficient $1/(4N_c)$ in the two non-abelian rows (the $U(1)$ value $\frac{1}{4}$ coincides with it at $N_c = 1$).

The lineage placement is identical to that of Sec. 9, and each Wilson construction inherits the same status: each is a published-machinery instance with a specific manifold and gauge group; what is not duplicated in the literature is the *simultaneous* existence of all three SM gauge groups as siblings across GeoVac sub-manifolds.

**What this is not: structural distance to a Standard Model**

Table 4 is a catalogue, not a unified theory, and the project's rhetoric rule forbids treating it as the latter. The structural distance from this catalogue to Connes' SM construction lives in four named gaps.

**G1** (Three generations of matter.) Each construction in Table 4 carries a single matter species: Paper 30 has one generation of Dirac fermions (Paper 14 Tier 2 [27]); Sprint ST-SU3 has one tower of $(N, 0)$ irreps. Sprint 4H Track SM-B (April 2026) tested the naive shell-to-generation map $\sum_f N_c Q_f^2 = 8 = |\lambda_3|$ and returned a clean negative: the per-generation contribution is charge-universal across the three SM generations whereas every per-shell $S^3$ invariant varies with $n$, so no shell→generation map is consistent with charge-universal per-generation structure. The $8 = 8$ equality was a numerical coincidence with no representation-theoretic content. There is no GeoVac mechanism currently selecting a generation tripling of the Hilbert space.

**G2** (Higgs as inner fluctuation of $D$.) In Connes' SM, the Higgs field arises as the off-diagonal block of an inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ ($\omega = \sum_i a_i[D, b_i]$) for $a_i, b_i \in \mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ in an almost-commutative algebra. The Marcolli–vS gauge-network lattice action [6] does carry a finite vertex Dirac and hence vertex-Higgs terms; Perez-Sanchez 2025 [11] shows that the Higgs does not survive the *continuum limit*, which is Yang–Mills without a Higgs. To keep a Higgs in the limit, an almost-commutative finite factor with an off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block is needed. The natural extension for GeoVac is $\mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F$ with the electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (deferring color), and with a finite Dirac $D_F$ on the four-dimensional electroweak fiber whose off-diagonal block encodes the Yukawa coupling matrix. Inner fluctuations on this extension formally split into a gauge-1-form sector (which must reproduce the unified $U(1) \times \mathrm{SU}(2)$ content of Papers 25–30) and a candidate Higgs scalar sector living in the off-diagonal $(\mathbb{C}, \mathbb{H})$ component. Whether the required $D_F$ can be selected from GeoVac structure rather than imposed is the load-bearing open question; we report the result of this test (Sprint H1) below in Sec. 8.2. Note that the operator-system non-multiplicative-closure of Remark 1 (witness pair $M_{2,1,0}^2$, 14.9% residual against $\mathcal{O}_{n_{\max}=2}$) is a separate, finite-resolution feature of the truncated operator system itself; inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, which uses ALGEBRA elements (the multiplier matrices, which are an algebra over $\mathbb{C}$ at the level of $M_{\dim H}(\mathbb{C})$ even though their span $\mathcal{O}_{n_{\max}}$ is not closed under multiplication), so the $a \cdot b \notin \mathcal{O}$ feature does not block the fluctuation calculus. Sprint H1 (Sec. 8.2) sharpens this gap to the Yukawa-undetermined verdict; Sprint G3 (Sec. 8.3) offers a structural reading of why $Y$ is undetermined — $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings, the Yukawa lives on the $\gamma_F$ flip, and no GeoVac-side data couples to that flip.

**G3** (Hypercharge / electroweak unification.) The SM relation $Q = T_3 + Y_W/2$ requires co-located $U(1)_Y$ and $\mathrm{SU}(2)_L$ on a common Hilbert space with a left–right chiral asymmetry. Paper 25's $U(1)$ and Paper 30's SU(2) are co-located on the same $S^3$ graph (the co-location Sprint G3 tested and closed in the negative, §8.3), but the Camporesi–Higuchi Dirac (Sec. 3 Definition 3) is non-chiral by default. The chirality grading introduced in Sprint TS Track E2 (R3.5, full Dirac with both chiralities) is a $\mathbb{Z}_2$-grading of the spinor bundle, not the SM weak-isospin chirality, and it does not distinguish left- and right-handed sectors of a charged matter field. Sprint G3 (Sec. 8.3) closed this in the negative on $S^3$ alone: $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, with operator-norm residual exactly 2 at every $n_{\max} \in \{1, 2, 3\}$. Identification is structurally impossible by tensor-product factorisation, and the electroweak co-location route is therefore pushed to G4.

**G4** (Cross-manifold obstruction, the dominant gap.) Connes' SM uses a *single* spectral triple $(\mathcal{A}_{\mathrm{SM}}, \mathcal{H}_{\mathrm{SM}}, D_{\mathrm{SM}})$ on a single base manifold, with all three SM gauge groups emerging from inner automorphisms of one almost-commutative algebra. The GeoVac construction has, at present, three separate spectral triples on two different graphs (the $S^3$ Coulomb graph for $U(1)$ and SU(2); the $S^5$ Bargmann graph for SU(3)). There is no GeoVac operator that geometrically unifies the $S^3$ and $S^5$ graphs into a single spectral triple containing both as commuting factors: at best the Bargmann and Coulomb graphs can be encoded as orthogonal factors of a tensor-product Hilbert space, but the resulting object is a direct sum of two Wilson-YM instances rather than a unification. This is the most consequential gap. It is also the most directly tied to the Coulomb/HO asymmetry of Sec. 7 (seven layers catalogued, six standing): the absence of cross-manifold unification is a structural symptom of the same asymmetry that forbids a calibration-$\pi$ analog and a natural Wilson matter coupling on $S^5$. Sprint G4 scoping (Sec. 8.4) sharpens this gap into G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$; since built, closed positive-thin: it defines the construction but selects no Yukawa, hypercharge or generation data) and G4b (genuine cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$, blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry). With G3 negative and G4a built, no sprint-scale electroweak-unification path remains open.

A footnote on the GUT-embedding direction: Sprint 4H Track SM-C (April 2026) tested whether $1/40 = \Delta$ in Paper 2's $K = \pi(B + F - \Delta)$ formula appears naturally in SU(5), $E_8$, or Spin(10) as a Chern number, $\eta$-invariant, or Chern–Simons level, and returned a clean negative: the dual-Coxeter rewrite $1/40 = (3/8)/15 = \sin^2\theta_W^{\mathrm{GUT}}/\dim(\bar{\mathbf{5}} \oplus \mathbf{10})$ is tautological, recycling the integers 5 and 8 already present in Paper 2. The naive shell-to-generation embeddings $\mathrm{shell}_n \to \bar{\mathbf{5}}$ fail at the $\mathrm{SU}(3) \times \mathrm{SU}(2)$ branching level. GeoVac therefore has no current contact with the GUT-embedding lineage of the SM, beyond the tautological numerical coincidences.

### Unification targets after G3: what remains

Of the four gaps, G4 (cross-manifold) is the most consequential. G3 is closed in the negative (§8.3: identification is structurally impossible by tensor-product factorisation); G4a is built and closed positive-thin (§8.4); G4b is blocked at the NCG-framework level; G1 has no current selecting mechanism (§8.1). Paper 25's $U(1)$ and Paper 30's SU(2) already live on the same underlying graph (the Fock-projected $S^3$ Coulomb graph) and share the same incidence structure $B$ and the same edge Laplacian $L_1 = B^\top B$ (neither uses the continuum Camporesi–Higuchi Dirac operator; their Dirac operators are the graph-side gauge-network operators of Papers 25 and 30). The two constructions differ only at the gauge-group level, and Paper 30 §5.1 (Result 1) establishes that Paper 25 is exactly the maximal-torus reduction of Paper 30. The natural unified construction is therefore an

$\mathrm{SU}(2) \times U(1)$ Wilson lattice gauge theory on the same graph, with link variables $U_e \in \mathrm{SU}(2) \times U(1)$ and a chosen embedding of hypercharge as the $U(1)$ phase. Whether such a construction admits a Higgs-style inner fluctuation through an almost-commutative extension (closing G2 within the electroweak sector) is an open structural question. Whether the Camporesi–Higuchi Dirac admits a chirality grading compatible with weak-isospin is *not* open: Sprint G3 closed it in the negative on $S^3$ alone (§8.3). *The open question* — the almost-commutative lift of Paper 30's SU(2) Wilson instance, *not* the chirality-grading question Sprint G3 closed — is a clean $S^3$-only target that does not require any progress on the $S^5$ side. It is not by itself an electroweak construction: what it would deliver is a non-chiral $\mathrm{SU}(2) \times \mathrm{U}(1)$ Wilson product on $S^3$; chiral electroweak co-location on $S^3$ is what Sprint G3 closed (§8.3), and G4a (Connes SM on the $S^3$ triple alone, below) is built and closed positive-thin.

We do not claim such a unification has been achieved, or even that its component pieces have been verified. The status of this subsection is: a recorded structural fact (the shared non-abelian $1/(4N_c)$ coefficient, with which the $U(1)$ value coincides at $N_c = 1$, and the lineage placement are real); a recorded honest gap analysis (G1–G4 are real); and a recorded open direction (the almost-commutative lift of Paper 30's SU(2) Wilson instance is the most concrete $S^3$-only target; G4a is built and closed positive-thin, and the chirality-grading route is closed in the negative, §8.3). Per the project's rhetoric rule, no claim is made that GeoVac contains the Standard Model; the claim is only that it contains the three SM gauge groups as separate Wilson lattice gauge theories on its own sub-manifolds, with a weak-coupling kinetic coefficient shared by the non-abelian constructions and a unified gauge-network lineage.

## 8.2 Sprint H1: Higgs as inner fluctuation on the GeoVac electroweak almost-commutative extension

The G2 gap of Sec. 8.1 (Higgs as inner fluctuation) admits a constructive sprint-scale test: build the minimal electroweak almost-commutative extension $\mathcal{T}_{\mathrm{AC}} = \mathcal{T}_{\mathrm{GV}} \otimes \mathcal{T}_F$ with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ on the truthful Camporesi–Higuchi GeoVac triple, compute the inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ explicitly at finite $n_{\max}$, and ask whether the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block of the finite Dirac $D_F$ that turns Marcolli–vS-without-Higgs into a Higgs construction can be *selected from GeoVac structure* rather than imposed. This is the H1 sprint reported in `debug/h1_ac_extension_memo.md`; we summarize its content here.

**Construction.** Following Connes–Marcolli 2008 Ch. 1 §13 (the finite noncommutative geometry) [2], the finite electroweak Hilbert space is doubled into matter and antimatter sectors: $\mathcal{H}_F = \mathcal{H}_F^{\mathrm{matter}} \oplus \mathcal{H}_F^{\mathrm{antimatter}} = \mathbb{C}^4 \oplus \mathbb{C}^4$, with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ acting on the matter sector only (the opposite-algebra action on antimatter is realized via $J$). The 8 by 8 finite Dirac is $D_F = \mathrm{block_diag}(M, \bar{M})$ where the matter-sector $M$ is the 4 by 4 Yukawa-Dirac with off-diagonal $L \leftrightarrow R$ block $Y = \mathrm{diag}(y_\nu, y_e)$. The combined Dirac is $D = D_{\mathrm{GV}} \otimes \mathbb{1}_F + s_{\mathrm{GV}} \otimes D_F$ with $s_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ (Track 2 closure): this commutes with $D_{\mathrm{GV}}$ on the truthful CH Dirac, and is *not* the chirality-doubling grading $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$ of Proposition 5(b) used below. **[SYMBOLIC + MEASURED]** The combined real structure is $J = J_{\mathrm{GV}} \otimes J_F$ with $J_F$ swapping matter and antimatter via complex conjugation; matter-antimatter doubling makes $J_F^2 = +\mathbb{1}$ and $J_F D_F = +D_F J_F$ (the KO-dim 6 signs of $\mathcal{A}_F$ in the Connes–Marcolli convention). The signs of the combined $J$ follow factor by factor: $J^2 = J_{\mathrm{GV}}^2 \otimes J_F^2 = -\mathbb{1}$, and since $s_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ is a function of $D_{\mathrm{GV}}$, $JDJ^{-1} = D$, i.e. $JD = +DJ$. The combined sign pair $(\varepsilon, \varepsilon') = (-, +)$ is the sign pair of the GV factor, which $J_F$ leaves unchanged; in Connes' sign table [1] (in the three-sign convention of [3]) it is shared by the KO-2, KO-3 and KO-4 columns, KO-3 being the odd continuum factor's own label. It is *not* the

KO-dimension $3+6 \equiv 1 \pmod 8$ that the additive rule for products of real spectral triples would assign: KO-1 has $(\varepsilon, \varepsilon') = (+,-)$, and the additive rule is a theorem about the product whose Dirac operator carries the grading of the even factor, $D_{\mathrm{GV}} \otimes \gamma_F + \not{1} \otimes D_F$ (the $\widetilde{\mathcal{D}}$ of Dabrowski–Dossena [4], eq. (18), which their §4.2 prescribes when the first factor is odd; their eq. (10) is the classical Cartesian-product analogue); for that product their Table 5 (row $3_+$) shows that Connes' KO-6 real structure on the finite factor (column $6_+$) admits no consistent $\varepsilon'$ at all, and the alternative real structure $J_F\gamma_F$ (column $6_-$) gives KO-1 (entry $1_-$) with $J^2 = +\not{1}$, $JD = -DJ$. Our $D$ is the ungraded sum above, so the additive rule does not apply to it, and we label it by its signs (derived factor by factor above and confirmed numerically). Both signs are verified to machine precision at $n_{\max} \le 3$ (`tests/test_almost_commutative.py`, `test_J_combined_sign_pair`). **[PANEL-VERIFIED]** The pair $(-,+)$ alone does not fix a KO-dimension: the KO-2, KO-3 and KO-4 columns of Connes' table [1, 3] share it; KO-3 is the label of the odd continuum GV factor ($S^3$ carries no chirality). At finite cutoff the chirality-doubling grading $\gamma_{\mathrm{GV}} = \sigma_x \otimes \not{1}$ (`geovac/chirality_grading.py`), extended as $\gamma_{\mathrm{GV}} \otimes \not{1}_F$, squares to one, anticommutes with the combined $D$ and commutes with $J$ bit-exactly at $n_{\max} \le 3$, so the finite combined triple built on the chirality-doubled model is *even* with $(\varepsilon, \varepsilon', \varepsilon'') = (-,+,+)$, the KO-4 column, i.e. the sign triple of that column (pinned in the same test). We therefore attach *no* KO-dimension label to the finite model truncation (the true window carries no grading, Proposition 5(c)) and record the measured sign triple $(-,+,+)$ instead: the continuum factor is KO-3 and the finite grading reads as KO-4, so any single label at finite cutoff is a convention, not a result.

**Inner fluctuations split cleanly into gauge and Higgs sectors.** For $\omega = \sum_i a_i[D, b_i]$ with $a_i, b_i \in \mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F$, we have

$$\omega = \underbrace{\sum_i a_i^{\mathrm{GV}}[D_{\mathrm{GV}}, b_i^{\mathrm{GV}}] \otimes a_i^F b_i^F}_{\text{gauge piece}} + \underbrace{\sum_i a_i^{\mathrm{GV}} b_i^{\mathrm{GV}} s_{\mathrm{GV}} \otimes a_i^F [D_F, b_i^F]}_{\text{Higgs piece}}. \tag{4}$$

For the diagonal $\mathcal{A}_{\mathrm{GV}}$ the gauge piece vanishes identically ($[D_{\mathrm{GV}}, b^{\mathrm{GV}}] = 0$, Proposition 2); the module (`geovac/almost_commutative.py`) takes $b^{\mathrm{GV}}$ from the operator-system multipliers of $\mathcal{O}_{n_{\max}}$ (`FullDiracTruncatedOperatorSystem`), which do not commute with $D_{\mathrm{GV}}$, and the nonzero $\|\omega_{\mathrm{gauge}}\|$ values below are on that representative. The gauge piece is read as reproducing Papers 25 ($U(1)$ from the $\mathbb{C}$-summand) and 30 (SU(2) from the $\mathbb{H}$-summand) **[OBSERVATION]**; an operator-level identification with those papers' link variables is not established here. The Higgs piece is non-zero exactly when $[D_F, b_i^F]$ has off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ matrix elements — which happens *iff* the imposed Yukawa $Y$ is non-zero. At $n_{\max} = 1, 2, 3$, with three test choices of $Y$:

| $D_F$ choice | $\|\Phi\|_{\max}$ | $\|\omega_{\mathrm{gauge}}\|_{\max}$ | **Falsifier** |
|---|---|---|---|
| $Y = 0$ | 0.000 | 0.000–0.580 | **HOLDS** |
| $y_e = 0.1$, $y_\nu = 0$ | 0.046–0.080 | 0.000–0.580 | FAILS |
| $y_e = 0.3$, $y_\nu = 0.2$ | 0.171–0.271 | 0.000–0.580 | FAILS |

(50 random generators per cell; data `debug/data/h1_falsifier.json`).

**Verdict: positive-thin.** The Higgs construction is *well-defined* on the GeoVac AC extension at finite $n_{\max}$: the algebra, Hilbert space, Dirac, and real structure satisfy the relevant Connes axioms (order zero/one on the GV factor excepted, as in Table 1), and inner fluctuations, taken over the operator-system multipliers, decompose into a gauge sector read as recovering Papers 25/30 plus a Higgs sector $\Phi$ that is non-trivial whenever the imposed Yukawa $Y$ is non-zero.

*However*, GeoVac structure does **not autonomously select** a non-trivial $Y$. Two candidate sources for $D_F$ off-diagonal data were flagged in the H1 scoping memo (`debug/almost_commutative_scoping_memo.`
candidate A used the offdiag CH chirality bridging variant of the GeoVac Dirac, but Track 2 (`debug/real_structure_finite_nmax_memo.md` §3.D) ruled this out by showing $JD = +DJ$ fails at residual $\sim 2.0$ on offdiag CH; candidate B was a Sturmian / DUCC inter-shell coupling whose physical interpretation as a Yukawa is itself speculative and was not pursued.

The strong falsifier of the scoping memo §5 ("every Hermitian $D_F$ derivable from GeoVac structure forces the Higgs sector to vanish") therefore does not fire as a clean negative. The construction admits a Higgs given any $Y$; the question is which $Y$, and GeoVac data is silent. This is exactly the positive-thin reading of the scoping memo §0: GeoVac sits on the *Marcolli–vS-without-Higgs* side of the 2014/2024 distinction at the structural level — not because inner fluctuations cannot define a Higgs (they can), but because no GeoVac mechanism selects the Yukawa.

**What this means for G2.** Gap G2 of Sec. 8.1 is sharpened to the more precise statement: *the Higgs construction goes through, but the Yukawa is a free input*, not closed to either "GeoVac contains a Higgs" (which would require a structural selection of $Y$) or "GeoVac excludes a Higgs" (which would require a structural obstruction to non-zero $Y$). This places G2 in the same status as G1 (three generations are formally definable but not selected by GeoVac structure): definable but not derivable.

**What this means for G3.** Gap G3 (hypercharge / electroweak unification, requiring identification of the GV chirality grading $\gamma_{\mathrm{GV}}$ with the SM weak-isospin chirality) is *not* closed by H1. In our construction $\gamma_{\mathrm{GV}}$ is the chirality-doubling grading $\sigma_x \otimes \mathbb{1}$ of the chirality-doubled model truncation (Proposition 5(b); not $\mathrm{sign}(D_{\mathrm{GV}})$, which commutes with $D_{\mathrm{GV}}$), while the SM weak-isospin chirality is a separate $\mathbb{Z}_2$-grading on the matter sector of $\mathcal{H}_F$. The two are added independently; no GeoVac mechanism currently identifies them.

**What this means for the operator-system non-multiplicative-closure caveat.** Remark 1's observation that $\mathcal{O}_{n_{\max}}$ is not closed under multiplication (witness pair $M^2_{2,1,0}$, 14.9% residual at $n_{\max} = 2$) is a separate, finite-resolution feature of the operator system itself. Inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, where the multiplier matrices $a_i, b_i$ are individual elements of $M_{\dim H}(\mathbb{C})$ (which is an algebra over $\mathbb{C}$); the lack of closure of *their span* $\mathcal{O}_{n_{\max}}$ under multiplication does not block the fluctuation calculus. On the finite factor, order zero $[a, JbJ^{-1}] = 0$ holds because $a$ acts on matter and $JbJ^{-1}$ on antimatter; on the combined triple the order-zero residual is the GV factor's operator-system residual (Table 1; §8.4, G4a).

**Scope and limitations.** H1 establishes the constructive existence of the AC extension at finite $n_{\max}$ on the truthful CH GeoVac triple. H1 does *not* establish a Higgs vev as a derived quantity (that would require a continuum spectral-action computation; deferred to Track 1 R2.5 L4 closure for the rigorous interpretation), does *not* include the SM color factor $M_3(\mathbb{C})$ (deferred per Sec. 8.1 G4), and does *not* identify the GV and weak-isospin chiralities (that identification is what Sprint G3 closed in the negative, §8.3). Per the project's rhetoric rule, no claim is made that GeoVac contains the Higgs: the claim is that it admits a Higgs given an imposed Yukawa, and that the Yukawa is not derived from GeoVac structure.

**Files.** Implementation `geovac/almost_commutative.py`; tests `tests/test_almost_commutative.py` (56 passing as of 2026-09-03); data `debug/data/h1_falsifier.json`; memo `debug/h1_ac_extension_memo.md`.

**Follow-on records.** The follow-on sprint records attached to this section — the thermal extension of the H1 verdict (Sprint TD Track 3), the modular-propinquity reformulation of its falsifier (Sub-sprint M-H1, which leaves the positive-thin verdict unchanged), and the W1c/W1d/W1e chemistry-wall arc tested against the M-Z partition (owned by Paper 19 [52]) — are kept verbatim in `docs/chronicles/paper_32_chronicle.tex`.

### 8.3 Sprint G3: Electroweak chirality co-location on $S^3$

Sprint H1 (Sec. 8.2) sharpened G2 to the Yukawa-undetermined verdict and left G3 (electroweak unification, requiring identification of $\gamma_{\mathrm{GV}}$ with the SM weak-isospin chirality on $\mathcal{H}_F$) as the next physics question. The G3 sprint (May 2026) is a constructive test: build $\gamma_{\mathrm{GV}}$ on the GeoVac side and $\gamma_F$ on the AC side as separate $\mathbb{Z}_2$ gradings each satisfying the standard NCG axioms, lift both to the tensor-product Hilbert space $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, and ask whether the residual

$$\Delta := \gamma_{\mathrm{GV}} \otimes \mathbb{1}_F - \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F \tag{5}$$

vanishes (or vanishes on the physical sector under standard left/right convention swap). $\Delta = 0$ would close G3 in the positive; $\Delta \neq 0$ robustly would close G3 in the negative on $S^3$ alone and push electroweak unification onto G4 cross-manifold territory.

**Constructions.** The sprint executed three parallel tracks (memos `debug/g3a_chirality_memo.md`, `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`). Track G3-A constructed $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}_w$ on the R3.5 full Dirac operator system — the chirality-doubled model, not the true Camporesi–Higuchi window, whose truncation is odd and carries no such grading (Proposition 5) — at $n_{\max} \in \{1, 2, 3\}$, with $w = \mathrm{spinor_dim}(n_{\max})$ and the basis ordering $[\chi = +1, \chi = -1]$ from `full_dirac_basis`. All three $\mathbb{Z}_2$-grading axioms are bit-exact at every tested $n_{\max}$: $\gamma_{\mathrm{GV}}^2 = \mathbb{1}$, $\{\gamma_{\mathrm{GV}}, D_{\mathrm{truthful}}\} = 0$, and $\gamma_{\mathrm{GV}} M \gamma_{\mathrm{GV}} = M$ as a matrix equality for every scalar multiplier $M \in \mathcal{O}_{n_{\max}}$. The natural $\sigma_z$ candidate (the $D$-eigenvalue sign) commutes with $D_{\mathrm{truthful}}$ rather than anticommuting and is exposed only as a diagnostic. Module `geovac/chirality_grading.py` (50/50 tests passing).

Track G3-B added `ElectroweakFiniteTriple.chirality_F()` to `geovac/almost_commutative.py` with the standard Connes–Marcolli SM convention at KO-dim 6 of $\mathcal{T}_F$:

$$\gamma_F^{\mathrm{matter}} = \mathrm{diag}(+1, +1, -1, -1), \qquad \gamma_F^{\mathrm{anti}} = -\gamma_F^{\mathrm{matter}}. \tag{6}$$

The antimatter sign-flip is forced by $\{J_F, \gamma_F\} = 0$: the naive "identical on both copies" convention would put $\mathcal{T}_F$ at KO-dim 0 and silently break the H1 module's $\{J_F, D_F\}$ axiom suite. All grading axioms ($\gamma_F^2 = \mathbb{1}$, $\{\gamma_F, D_F\} = 0$ at any imposed Yukawa, $\gamma_F \pi_F(a) \gamma_F = \pi_F(a)$, $\gamma_F = \gamma_F^*$, $\{J_F, \gamma_F\} = 0$) hold to bit-exact zero residual; 15 new tests in `TestChiralityF`, all passing. The finite factor therefore carries the KO-dim 6 signs $(\varepsilon_F, \varepsilon'_F, \varepsilon''_F) = (+, +, -)$; **[SYMBOLIC + MEASURED]** the combined real structure on H1's combined Dirac carries the sign pair $(\varepsilon, \varepsilon') = (-, +)$, the GV factor's pair — with $\varepsilon'' = +1$ under the finite grading, and no finite-cutoff KO label attached (§8.2). The additive rule $3 + 6 \equiv 1 \pmod 8$ does not apply to H1's ungraded sum $D_{\mathrm{GV}} \otimes \mathbb{1}_F + s_{\mathrm{GV}} \otimes D_F$; it governs the graded product of Dabrowski–Dossena [4], which the paper does not use.

**Tensor-product diagnostic.** Track G3-C computed $\Delta$ at $n_{\max} \in \{1, 2, 3\}$ using the production $\gamma_{\mathrm{GV}}$ and $\gamma_F$ from G3-A and G3-B. Sanity invariants are bit-exact: $\gamma_{\mathrm{GV}}^2 = \mathbb{1}$, $\gamma_F^2 = \mathbb{1}$, and $[\gamma_{\mathrm{GV}} \otimes$

$\mathbb{1}_F, \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F] = 0$ (the two operators commute because they act on disjoint tensor factors). The residual itself is the structurally cleanest possible non-zero object:

**Convention swaps do not absorb the residual.** Three standard convention swaps were tested at every $n_{\max}$: (a) global sign-flip $\gamma_F \to -\gamma_F$ (left/right convention), (b) restriction to the matter sector of $\mathcal{H}_F$, and (c) restriction to the Weyl chirality of $\mathcal{H}_{\mathrm{GV}}$. Swap (a) leaves the baseline invariant. Swaps (b) and (c) reduce $\|\Delta\|_F$ by dropping basis vectors, but $\|\Delta\|_{\mathrm{op}}$ remains 2 (or 1 under swap (c) with $\sigma_x$-$\gamma_{\mathrm{GV}}$, where the Weyl restriction collapses $\gamma_{\mathrm{GV}}|_w$ to $0_w$ and $\Delta_c = -\mathbb{1}_w \otimes \gamma_F$, still non-zero). Combined swaps yield the same conclusion. No standard NCG sign convention absorbs $\Delta$.

**Robustness.** A $\sigma_x$-vs-$\sigma_z$ cross-check at $n_{\max} = 2$ confirms that the spectrum of $\Delta$ is identical under both $\gamma_{\mathrm{GV}}$ conventions (both produce $\{-2{:}32,\ 0{:}64,\ +2{:}32\}$, since $\sigma_x$ and $\sigma_z$ share eigenvalues $\pm 1$ with identical multiplicities and both commute with $\gamma_F$). The structural verdict is therefore convention-independent.

**Sanity check on $\gamma_5 := \gamma_{\mathrm{GV}} \otimes \gamma_F$.** The product operator $\gamma_5 := \gamma_{\mathrm{GV}} \otimes \gamma_F$ is the canonical Connes–Chamseddine combined chirality on $\mathcal{T}_{\mathrm{AC}}$ and satisfies $\gamma_5^2 = \mathbb{1}$ to bit-exact precision at every $n_{\max}$, with spectrum $\{-1, +1\}$ in equal multiplicities. The product works as an involution; the *difference* does not. $\gamma_5$ is a well-defined $\mathbb{Z}_2$ operator, but it is a *grading of $D$* only for the production $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$ at vanishing Yukawa: with $s_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ in its place one has $\|\{\gamma_5, D\}\|_F = 17, 52, 110$ at $n_{\max} = 1, 2, 3$, and with the production grading $4.7, 9.3, 14.8$ at $Y \neq 0$. $\gamma_{\mathrm{GV}}$ and $\gamma_F$ do not identify across the tensor product.

**Verdict: NEGATIVE on $S^3$ alone.** The two chirality gradings $\gamma_{\mathrm{GV}}$ (the chirality-doubling grading $\sigma_x \otimes \mathbb{1}$ of Proposition 5(b)) and $\gamma_F$ (weak-isospin grading on the AC factor) are *independent commuting* $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$. They cannot be identified by any standard NCG sign convention because, by construction, $\gamma_{\mathrm{GV}}$ acts as identity on $\mathcal{H}_F$ and $\gamma_F$ acts as identity on $\mathcal{H}_{\mathrm{GV}}$: they are different operators, acting on different tensor factors. Electroweak chirality co-location does not close on $S^3$ alone.

**The G2–G3 corollary.** Sprint H1 left G2 (Higgs as inner fluctuation) at the verdict "construction admits a Higgs given an imposed Yukawa $Y$, but no GeoVac mechanism selects $Y$." G3's NEGATIVE offers a structural reading of why. The Yukawa sits in the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block of $D_F$ that anticommutes with $\gamma_F$; the candidate GeoVac sources for that block (offdiag CH chirality bridging, Sturmian inter-shell coupling) have no kinematic relation to $\gamma_F$, because $\gamma_{\mathrm{GV}}$ is structurally independent of $\gamma_F$. Equivalently: no GeoVac-side data fixes $Y$ because the GeoVac chirality lives on a different $\mathbb{Z}_2$ grading from the one the Yukawa flips. This is a *sharpening* of H1's G2 verdict (the Yukawa-undetermined gap is named structurally), not a new gap.

**Files.** Modules `geovac/chirality_grading.py` (G3-A, 50/50 tests) and the `chirality_F` additions to `geovac/almost_commutative.py` (G3-B); tests `tests/test_chirality_grading.py` and `tests/test_almost_commutative.py` (56 passing, `TestChiralityF` block); diagnostic `debug/g3c_tensor_chirali` data `debug/data/g3a_chirality.json`, `debug/data/g3c_tensor_chirality.json`; memos `debug/g3a_chirality` `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`.

### 8.4 Sprint G4: Cross-manifold scoping (G4a built, positive-thin; G4b NCG-framework-blocked)

With G2 sharpened to the Yukawa-undetermined verdict and electroweak unification on $S^3$ alone closed in the negative, G4 (cross-manifold) was the last route scoped for a GeoVac-internal SM construction (G4a since built and closed positive-thin; G4b framework-blocked). A scoping pass (May 2026, memo `debug/g4_cross_manifold_scoping_memo.md`) sharpens G4 into two sub-gaps with sharply different reachability.

**G4a (built): Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$.** The minimal extension of Sprint H1's electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ to the full Connes SM finite algebra is built as an extension of `geovac/almost_commutative.py`. The $M_3(\mathbb{C})$ summand supplies SU(3) at the inner-fluctuation level on the existing GeoVac $S^3$ triple, without recourse to the $S^5$ Bargmann construction. Outcome: *positive-thin*, in the sense of Sprint H1. Inner fluctuations decompose into a gauge-1-form sector (recovering $U(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ at the operator level *for the admitted* $\mathcal{A}_F$ — an organizing reading plus the adopted Upgrade B axiom, not a derivation — with the abelian and SU(2) gauge content already exhibited by H1) plus a Higgs sector that is non-trivial whenever the Yukawa-Dirac off-diagonal blocks are imposed. The G2–G3 corollary applies verbatim: G4a defines the construction, but no GeoVac mechanism selects the Yukawa, hypercharge assignments, or generations. G4a therefore does not close the SM-derivation question; it relocates the open data-selection items ($Y$, hypercharge, generation count) from "electroweak slice" (H1, G3) to "full Connes finite algebra," without changing their status. Its value is that it makes Sprint ST-SU3's Bargmann SU(3) construction structurally redundant for SM-derivation purposes (one $S^3$ construction with $M_3(\mathbb{C})$ delivers the same gauge content); it was built and closed positive-thin (below).

**G4b (NCG-framework-blocked): genuine cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$.** The structurally deeper version of G4 asks whether the GeoVac $S^3$ Coulomb triple and the Paper 24 $S^5$ Bargmann lattice can be combined into a single spectral triple with both as commuting factors, so that the SU(3) of Sprint ST-SU3 is honestly cross-manifold-located rather than recapitulated on $S^3$. This sub-gap is blocked at the *NCG-framework level*, not at the GeoVac level. Connes' tensor product of spectral triples [2] requires both factors to be Riemannian spectral triples in the same sense. $\mathcal{T}_{S^3}$ carries the Camporesi–Higuchi Riemannian spinor Dirac (KO-dim 3). Paper 24's $S^5$ Bargmann construction is built from the holomorphic Hardy sector $H^2(S^5)$ restricted to symmetric SU(3) irreps $(N, 0)$; its natural diagonal operator is the first-order complex Euler operator $\hat{N} = \sum_i z_i \partial_i$, not a Riemannian Dirac. Two candidate $\mathcal{T}_{S^5}$ exist, and neither preserves both standard NCG axioms and the GeoVac-specific content of Sprints 25–30 / ST-SU3:

- *Option A (Riemannian round-$S^5$).* Take $D_{S^5}$ to be the standard Riemannian spinor Dirac on round $S^5$ (KO-dim 5). The tensor product is a textbook Connes object with combined KO-dim $3 + 5 = 8 \equiv 0 \pmod 8$, but it discards the $(N, 0)$-tower structure that made Paper 24's $S^5$ "natural," and the bit-exact $\pi$-free certificate of Paper 24 §III is irrelevant to round-$S^5$ (calibration $\pi$ reappears via standard Weyl asymptotics).
- *Option B (Bargmann–Euler-based).* Promote the Euler operator $\hat{N}$ on the Bargmann graph to a Dirac-like object via chiral doubling. This preserves Sprint ST-SU3 SU(3) Wilson content and the $\pi$-free certificate, but the resulting object is *not* a Riemannian spectral triple in the published sense: KO-dimension is not classically defined for a Hardy-sector first-order complex operator, and there is no published prescription in the NCG literature for tensoring a Riemannian spectral triple with a Hardy-sector / Berezin–Toeplitz first-order complex object.

The closest analogs (Berezin–Toeplitz quantisation of the unit disk; Hekkelman 2022 [15] on the circle; Hekkelman–McDonald 2025 [16] on general spectrally truncated triples) stay within Berezin–Toeplitz on a single manifold or within Riemannian on a single manifold, never crossing.

The structural reading is that G4b is the Coulomb/HO asymmetry of Paper 24 §V resurfacing at the spectral-triple level. Paper 24 §5.3 records these layers of this asymmetry (seven catalogued, six standing, Paper 31), the standing ones all Coulomb-specific; the first four: (i) spectrum-computing role of $L_0$ (*withdrawn*: $L_0$ is spectrally inert on both spheres); (ii) calibration $\pi$; (iii) Wilson gauge with natural matter coupling (argued), Sec. 8.1; and (iv) posited wedge-generator Pythagorean orthogonality on the $m_j$-reflection wedge, a chirality-parity identity (Paper 42 §11.7 [46]; Remark 22 above). G4b is a *sibling* of these layers, not a lone additional layer: of the four ingredients listed next, the Coulomb $S^3$ side carries a Riemannian spectral triple in the standard NCG sense and supports the finite-cutoff wedge period closure (for the chosen integer-spectrum generator; signature-blind, so the $(3,1)$ Krein carrier adds no Lorentzian content), while the Bargmann $S^5$ side has none of the four ingredients (no spinor bundle on the $(N,0)$ Hardy tower; no second-order-vs-first-order distinction with a separate Dirac; no half-integer wedge; no calibration $\pi$ or natural Wilson matter coupling). The Connes tensor-product framework cannot bridge them as currently published. Closing G4b is therefore not a sprint-scale GeoVac task; it requires either an extension of NCG to handle Riemannian–Hardy mixed tensor products, or the identification of a unifying ambient manifold whose GeoVac sub-structure contains both $S^3$ Coulomb and $S^5$ Bargmann as natural sub-objects. Both directions are recorded as honest open structural questions, not as sprint targets.

**Net.** After Sprints H1 + G3 + G4 scoping, the four-gap analysis of Sec. 8.1 compresses to:

- G2 is the Yukawa-undetermined verdict, which G3 (NEGATIVE) reads structurally: $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$ gradings on the tensor product.
- G4 splits cleanly: G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $M_3(\mathbb{C})$) is **closed at POSITIVE-THIN** (2026-05-31): $\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$ recovered from inner fluctuations, the $J$-relations, self-adjointness and $\gamma_F^2 = I$ bit-exact (the combined order-zero/one residuals carry the GV factor's operator-system residual; see below), Yukawa/hypercharge/generations remain imposed; G4b (genuine cross-manifold) is blocked at the NCG-framework level by the Coulomb/HO structural asymmetry.
- G1 (three generations) is unchanged from H1: formally definable, not selected by GeoVac structure.

GeoVac does not autonomously generate the Standard Model; the construction admits SM gauge content given imposed Yukawa, hypercharge, and generation choices, and the GeoVac data is silent on each of those choices. This is a clean structural verdict, in the sense of the project's rhetoric rule: the framework is consistent with a GeoVac SM construction, but the framework does not select one.

**G4a closure (Sprint G4a, 2026-05-31).** The full Connes SM finite algebra $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ has been implemented and verified on $\mathcal{T}_{S^3}$ at $n_{\max}\in\{1,2,3\}$ with $\dim\mathcal{H}_F = 32$ (16 matter + 16 antimatter; $\dim\mathcal{H}_{\mathrm{total}} = 1280$ at $n_{\max}=3$). **[PANEL-VERIFIED]** The axioms $J^2 = -I$, $JD = +DJ$, $D = D^\dagger$ and $\gamma_F^2 = I$ hold at machine zero at every tested $n_{\max}$. With a linear $*$-representation of the direct sum (`geovac/standard_model_triple.py`) the finite-algebra order-zero and order-one

residuals are exactly zero, while the combined-triple residuals (1.49 and 1.26 at $n_{\max} = 2$) are the GV factor's operator-system residual (read as the compression's boundary term) (the operator-system multipliers do not commute; cf. the order-zero/one rows of Table 1), factorising through the operator-system commutator with residual 0 (`tests/test_standard_model_triple.py`); the gauge census still returns $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ at $n_{\max} = 2$ (colour now read from the antiquark block); and the combined grading $s_{\mathrm{GV}} \otimes \gamma_F$ built from $s_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ does *not* anticommute with $D$ ($\|\{\gamma, D\}\|_F = 34, 104, 220$ at $n_{\max} = 1, 2, 3$); with the production grading $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}$ the anticommutator vanishes. Inner fluctuations decompose into $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ gauge 1-forms plus a Higgs scalar that is non-trivial whenever the Yukawa-Dirac off-diagonal blocks are imposed and identically zero otherwise. In the one $D_F$ the module constructs, the matter–antimatter and lepton–quark cross-sectors decouple exactly; the general admissible $D_F$ does not (Theorem 7: twelve of the sixteen real Majorana moduli are lepton–antiquark entries). **[PANEL-VERIFIED]** The predicted POSITIVE-THIN verdict is confirmed: the construction admits the full SM gauge content with imposed Yukawa, hypercharge, and generation choices, and the GeoVac data is silent on each of those choices. Module `geovac/standard_model_triple.py` ($\sim 530$ lines), 47 tests passing.

**H1 Higgs inner fluctuation (Sprint H1, May 2026).** The full fluctuated Dirac $D_A = D + A + JAJ^{-1}$ has been constructed at $n_{\max} \in \{2, 3\}$ ($\dim \mathcal{H}_{\mathrm{total}} = 512, 1280$). Inner fluctuations decompose cleanly into gauge ($\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, matching G4a) plus Higgs (complex SU(2) doublet $\Phi$ in the L–R off-diagonal block, non-trivial iff Yukawa $Y \neq 0$). The spectral-action Mexican-hat sign structure $V(\Phi) = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$ is the standard CCM result; the checks $\mathrm{Tr}(D_F^2) > 0$ and $\mathrm{Tr}(D_F^4) > 0$ hold for any nonzero Hermitian $D_F$ **[SYMBOLIC]**, so they are consistency checks, not independent confirmation.

**Theorem 6** (H1 Yukawa non-selection)**.** ***[PROOF-BY-ARGUMENT]*** *Within the almost-commutative spectral triple $(\mathcal{A}_{\mathrm{GV}} \otimes \mathcal{A}_F, \mathcal{H}_{\mathrm{total}}, D)$ with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, the space of admissible finite Dirac operators $D_F$ satisfying all Connes axioms (Hermiticity, chirality grading, $J$-reality, order-one condition) carries a finite-dimensional real moduli space. (With a linear $*$-representation of $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ the matter-sector projection of the full-axiom moduli space has rank 16 and the full-axiom dimension is 32 — see Theorem 7. The Yukawa non-selection conclusion is independent of this count.) For one generation, $Y = \mathrm{diag}(y_\nu, y_e, y_u, y_d)$ accounts for 8 of these. Given the imposed $\mathcal{A}_F$, the gauge sector $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ follows from its inner automorphisms; the Yukawa values are **not selected** by any morphism in $\mathcal{A}_{\mathrm{GV}}$.*

*Proof sketch.* The parameter count follows from direct enumeration on the chirality-graded $J$-real subspace of Hermitian operators on $\mathcal{H}_F$. The non-selection of the Yukawa is the G3 structural finding (Sprint G3, §8.3): $\gamma_{\mathrm{GV}}$ and $\gamma_F$ are independent commuting $\mathbb{Z}_2$'s on $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, so no GeoVac-side data couples to the $\gamma_F$ flip that carries the Yukawa. The full fluctuated Dirac $D_A = D + A + JAJ^{-1}$ was constructed at $n_{\max} \in \{2, 3\}$ ($\dim \mathcal{H}_{\mathrm{total}} = 512, 1280$) and the inner fluctuations decompose cleanly into the gauge piece fixed by $\mathcal{A}_F$ (matching G4a) plus the Higgs piece (complex SU(2) doublet $\Phi$ in the L–R off-diagonal block, non-trivial iff $Y \neq 0$). The Mexican-hat sign structure is the standard CCM result (the trace positivity checks are automatic for nonzero Hermitian $D_F$). See `debug/h1_higgs_inner_fluctuation_memo.md`. □

**The forced/free seam (forcing-catalogue Door 4, 2026-06-01).** **[OBSERVATION]** The Yukawa non-selection theorem admits a sharper structural reading. The almost-commutative factorization $D^2 = D_{\mathrm{GV}}^2 \otimes 1 + 1 \otimes D_F^2$ (the cross term vanishes by chirality anticommutation *for the graded product* $D = D_{\mathrm{GV}} \otimes 1 + \gamma_{\mathrm{GV}} \otimes D_F$ with $\{\gamma_{\mathrm{GV}}, D_{\mathrm{GV}}\} = 0$; for the module's $s_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$

in its place the cross term is $2|D_{\mathrm{GV}}| \otimes D_F \neq 0$, norms $4.3, 13.2, 28.0$ at $n_{\max} = 1, 2, 3$) factorizes the master Mellin engine into an *outer* ring — the M1/M2/M3 transcendentals of the GeoVac sector — tensored with an *inner* Yukawa Dirichlet ring $\mathbb{Q}[y_i^{-2s}]$ generated by the Yukawa eigenvalues. The two rings are read as sharing no generator beyond $\mathbb{Q}$: the $\eta$-trivialization ($\{\gamma_F, D_F\} = 0$ annihilates the odd-$k$ heat traces) removes the M3 slot from the inner factor, and the G3 commuting-$\mathbb{Z}_2$ result leaves no GeoVac-side datum coupling to the Yukawa flip; neither proves the two rings disjoint (an Observation-level reading). This is the precise content of the structural-skeleton scope (§1.5 of the project guidelines): the bone (the forced, integer/rational skeleton) and the calibration data (free) are read as living in rings sharing no generator beyond $\mathbb{Q}$, and one reading (an Observation, not a derivation) is that the Yukawa sits on the free side because it generates its own ring. That reading is consistent with the H1, W3, and Koide attempts to select the Yukawa all returning clean negatives: each probed *inside* the almost-commutative category, on the free side of the seam.

**The inner algebra shape under the Hopf-tower reading (Door 4b, 2026-06-01).** The seam places the Yukawa *values* on the free side; it does *not* place the inner *algebra structure* there. **[PANEL-VERIFIED]** A finite-algebra enumeration shows that among all finite semisimple real $*$-algebras with at most three summands and factor size at most three, exactly *two* reproduce the gauge group $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ as their unitary content after per-factor unimodular reduction ($\mathrm{U}(k) \to \mathrm{SU}(k)$ on each matrix factor with $k \geq 2$, not the CCM global unimodularity condition): the Standard Model finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ and $\mathbb{C} \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C})$. Under the Bertrand $\times$ complex-Hopf-tower reading at $n \leq 3$ (Door 4b; an organizing reading recorded in the forcing catalogue, not an argument given in Sec. 8.1, which is a catalogue of gauge constructions on distinct manifolds), one factor is assigned per Hopf rung $n = 1, 2, 3$, which fixes the factor count (3), the $n = 1$ factor ($\mathbb{C}$), and the $n = 3$ factor ($M_3(\mathbb{C})$); the residual non-uniqueness collapses to a single binary fork — $\mathbb{H}$ versus $M_2(\mathbb{C})$ at the $n = 2$ rung. This fork was conjectured (Door 4b) to have a GeoVac-native handle: the outer triple's $J^2_{\mathrm{GV}} = -1$ (the continuum factor's KO-3 sign; no KO label is attached at finite cutoff) (the real-structure result of Sec. 3, the pseudoreal $\mathrm{Spin}(3) = \mathrm{SU}(2)$ two-spinor sign — the *same* $S^3 = \mathrm{SU}(2)$ fact that produces the gauge SU(2)) might select $\mathbb{H}$ over $M_2(\mathbb{C})$. **[SYMBOLIC PROOF]** A direct $J$ sign-table audit (Door 4c, 2026-06-01) closed this conjecture *negative*: $J^2_{\mathrm{GV}} = -1$ *admits* but does not *force* $\mathbb{H}$. The decisive fact is that over $\mathbb{C}$ the two candidates are the same complex algebra ($\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$); the $\mathbb{H}$-vs-$M_2(\mathbb{C})$ distinction is a choice of *real form*, fixed by the sign of an *internal* $\mathbb{C}^2$ antilinear involution ($J^2_{\mathrm{int}} = -1 \Leftrightarrow \mathbb{H}$; $J^2_{\mathrm{int}} = +1 \Leftrightarrow M_2(\mathbb{C})$), which lives on the algebra side and is invisible to the combined $J = J_{\mathrm{GV}} \otimes J_F$. The combined $J^2$ and the combined $(\varepsilon, \varepsilon')$ signs are *bit-identical* for both candidates at $n_{\max} \in \{1, 2, 3\}$, and the Chamseddine–Connes–Marcolli order-one-with-Yukawa selector is satisfied by both (matter/antimatter decoupling). The $\mathbb{H}$ choice is therefore a *literature import* (the CCM complex-fermion-representation / second-order condition on the finite factor), not a GeoVac forcing; the fork stays open. The honest ledger: *fixed under the Hopf-tower reading* are the factor count and the $\mathbb{C}$ and $M_3(\mathbb{C})$ summands; *admitted but not forced* is the $n = 2$ real form ($\mathbb{H}$ vs $M_2(\mathbb{C})$); *free* are the generation count (invisible to inner automorphisms), the inner KO-dimension, and the Yukawa values. The net sharpening is real but bounded and conditional on the Hopf-tower reading: it fixes the factor count and two of the three matrix factors — more internal algebraic *shape* than the gauge group alone, which then follows from the inner automorphisms of the admitted $\mathcal{A}_F$ — while the third factor's real form, the generation multiplicity, and the Yukawa values remain unselected. Sources: `debug/door4_gauge_yukawa_boundary_memo.md`, `debug/door4b_inner_algebra_forcing_memo.md`, `debug/door4c_j_signtable_audit_memo.md`.

**A geometric handle on the $n = 2$ fork (Door 4d, 2026-06-02).** A second handle, structurally orthogonal to the combined-$J$ sign table of Door 4c, is the *Division-Algebra-of-the-Sphere* (DAS) criterion: *at each Hopf rung $n$, the inner algebra is the natural associative real normed division algebra whose unit-norm sphere IS the rung's Hopf bundle total space $S^{2n-1}$.* The identifications are bit-exact at all three rungs: $S^1$ is the unit-norm sphere of $\mathbb{C}$ under complex multiplication (closure $1.1 \times 10^{-16}$); $S^3$ is the unit-norm sphere of $\mathbb{H}$ under quaternion multiplication and is bit-exactly isomorphic to $\mathrm{Sp}(1) = \mathrm{SU}(2)$ via the standard $\mathbb{H} \to M_2(\mathbb{C})$ embedding (closure $2.2 \times 10^{-16}$, SU(2)-isomorphism error $3.3 \times 10^{-16}$); and by Hurwitz's classification of finite-dimensional real normed division algebras (only $\mathbb{R}, \mathbb{C}, \mathbb{H}$ are associative, with unit spheres $S^0, S^1, S^3$ respectively), $S^5$ has no associative-division-algebra realization, so the inner algebra at $n = 3$ falls back to a matrix algebra. DAS agrees with the factors fixed under the Door 4b reading at $n = 1$ ($\mathbb{C}$) and $n = 3$ ($M_3(\mathbb{C})$), and *would* select $\mathbb{H}$ at $n = 2$ if adopted as an axiom (the Door 4e sharpening, Upgrade B, adopts it; absent that axiom $\mathbb{H}$ is admitted, not forced) — the direct realization with $\mathrm{U}(\mathbb{H}) = \mathrm{Sp}(1) = \mathrm{SU}(2)$ on the 3-dimensional $S^3$, in contrast to the $M_2(\mathbb{C})$ realization whose unit-norm sphere is the 4-dimensional U(2) with SU(2) reached only via the $\mathrm{U}(2)/\mathrm{U}(1)$ unimodularity quotient. The DAS argument never invokes any $J$ or sign-table data; it is a purely geometric criterion identifying the rung-$n$ sphere with the unit-norm subset of the rung's associative division algebra. The verdict is *partial-door*: the existing Hopf-rung reading (Door 4b) extracts only the gauge group from the Hopf-rung sphere; adopting DAS as a forcing requires the additional structural theorem that the AC tensor product $\mathcal{H} = \mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$ transfers the rung-$n$ sphere's division-algebra structure to a constraint on $\mathcal{H}_F$'s inner algebra (tested negative in Door 4e below: no standard AC compatibility condition forces $\mathbb{H}$). Had it closed, the $n = 2$ real form would move from "admitted, with $\mathbb{H}$ imported from the literature (CCM second-order condition)" to "selected by a GeoVac-coherent geometric criterion" — still relative to the Hopf-tower organizing reading and the adopted Upgrade B axiom, not a derivation — leaving only the generation count, the inner KO-dimension, and the Yukawa values outside the forcing. Source: `debug/door4d_division_algebra_sphere_memo.md`.

**Falsifier A′ on DAS (Door 4e, 2026-06-02).** The DAS handle introduced above is partial-door because the existing Hopf-rung reading (Door 4b) extracts only the gauge group from the Hopf-rung sphere. The natural test was whether any standard AC tensor-product compatibility condition fills the gap, i.e., distinguishes $\mathbb{H}$ from $M_2(\mathbb{C})$ as the $n = 2$ inner algebra. Three candidates were tested literally: (C1) SU(2)-equivariance under the Ad action $g \cdot m \cdot g^{-1}$; (C2) Hopf U(1)-equivariance under scalar phase $e^{i\theta} I$ acting from the right; (C3) principal-bundle compatibility with the Hopf U(1) fiber. All three returned *negative*: (C1) both $\mathbb{H}$ and $M_2(\mathbb{C})$ are SU(2)-Ad-invariant real $*$-subalgebras of $M_2(\mathbb{C})$ (the former is the unique minimal non-abelian such, the latter is the maximal; the distinction requires a minimality criterion that is itself extra input); (C2) every $2 \times 2$ complex matrix commutes with the scalar phase $e^{i\theta} I$ trivially, so the condition is vacuous; (C3) both candidate actions are principal-bundle compatible at machine precision by associativity of matrix multiplication and centrality of scalar phases. *No standard AC tensor-product compatibility condition forces $\mathbb{H}$ over $M_2(\mathbb{C})$.* The substantive content of the test is not the negative outcome (which was the expected one) but the *import comparison* it enables: the standard Chamseddine–Connes–Marcolli derivation of $\mathbb{H}$ at the $n = 2$ factor invokes *three* axioms beyond the bare Connes spectral-triple data (second-order condition with nonzero finite Dirac, complex chiral fermion representation, and the $2N^2 = 32$ dimension count), whereas DAS invokes *one* (the rung-$n$ sphere's division-algebra structure is inherited by the inner algebra). Both are imports relative to the bare axioms; DAS is leaner. The path to upgrading partial-door to door is therefore not a

theorem about existing CCM machinery but a paper-level structural axiom: adopting "the inner algebra at rung $n$ is the $*$-algebra whose unit group IS the rung-$n$ sphere $S^{2n-1}$ as a Lie group, when one exists; else fallback $M_n(\mathbb{C})$." This reads the Hopf-rung sphere fully (Lie-group structure plus topology) where the existing construction reads it partially (gauge group only), and gives $\mathbb{C}$ at $n = 1$ (via $S^1 = \mathrm{U}(1) = \mathrm{U}(\mathbb{C})$), $\mathbb{H}$ at $n = 2$ (via $S^3 = \mathrm{Sp}(1) = \mathrm{U}(\mathbb{H})$), and $M_3(\mathbb{C})$ at $n = 3$ (since $S^5$ is not even an H-space, by Adams 1958 [61]). Source: `debug/door4e_falsifier_A_prime_memo.md`.

**Closure of the Door 4 series (Door 4f, 2026-06-02; net axiom savings audited).** The natural next test was whether adopting Upgrade B introduces any new downstream constraint on existing GeoVac observables. Twelve sub-tests were run: inner KO-dimension (T1), chirality grading $\gamma_F$ (T2), Yukawa eigenvalues (T3), generation count (T4), Connes order-zero (T5), order-one (T6), and second-order (T7) conditions, bosonic spectral-action gauge-sector coefficient (T8), Higgs vacuum manifold (T9), cross-rung Yukawa coupling (T10), extensibility to rung $n \geq 4$ ($S^7$ is an H-space but not a Lie group; $S^{2n-1}$ for $n \geq 5$ is not even an H-space [61]) (T11), and consistency with the Bertrand $\times$ Hopf-tower reading of Door 4b (T12). *All twelve* return either SILENT (Upgrade B unconstrained on the target), COMPATIBLE (the pre-existing constraint passes for $\mathbb{H}$, the mechanism unaffected), IDENTICAL to CCM (same observable from either $\mathbb{H}$ or $M_2(\mathbb{C})$ post-unimodularity), or a CLEAN extensibility/strengthening rule (T11, T12). Zero sub-tests introduce a new constraint; zero contradict any existing GeoVac result. Upgrade B is therefore a *clean reformulation* of the CCM $\mathbb{H}$-selection with no new predictive content.

**Honest axiom-savings audit.** The naive comparison in the preceding paragraph (CCM 3 axioms vs DAS 1 axiom) over-counts. Sub-test T7 (Connes second-order condition) shows that the second-order condition is *still* required in the Upgrade-B formulation, because it constrains the off-diagonal Yukawa structure of $D_F$ (which $\mathbb{H}$-selection alone does not determine). Upgrade B therefore replaces *two* of CCM's three $\mathbb{H}$-selection axioms — the complex chiral fermion representation requirement and the $2N^2 = 32$ dimension count — with the single sphere-Lie-group axiom; the second-order condition persists in both formulations. Net axiom savings: *one*. The Door 4 series is structurally complete at **partial-door (final)**. The remaining open questions are paper-level (whether to adopt Upgrade B's axiom-minimality advantage as the working construction principle) and the deep wall (force $N_{\mathrm{gen}}$ and the inner KO-dim from a packing principle; no known handle; deferred). Source: `debug/door4f_falsifier_A_double_prime_memo.md`.

**The deep wall is a relative non-emission statement, not a gap (Direction 2 scoping, 2026-06-03).** The "deep wall" named above — forcing $N_{\mathrm{gen}}$ and the inner KO-dimension from a packing principle — is sharper than an open gap: relative to the packing construction of Paper 0 as presently formulated, the construction emits no such datum. One established fact carries this, beside one Observation-level reading. The reading first, the Door 4 seam: the inner Yukawa/real-structure data lives in a ring (and a set of $(\mathcal{H}_F, D_F, J_F, \gamma_F)$ data) sharing no generator with any forced outer ring (an Observation-level reading, §8.4). The fact, Paper 0 §VII: the packing construction is *kinematic* — it emits the label set $(n, \ell, m, s)$ and the graph topology, and emits no operator, no real structure, no coupling, and no Hilbert-space multiplicity. But $N_{\mathrm{gen}}$ is exactly a Hilbert-space multiplicity (the factor in $\mathcal{H}_F = \mathbb{C}^{N_{\mathrm{gen}}} \otimes \mathcal{H}_F^{(1)}$, invisible to every algebra- and automorphism-level mechanism), and the inner KO-dimension is exactly a $(J_F, \gamma_F)$ datum — both of the type the packing construction does not produce. The packing axiom therefore cannot constrain either: the residue is packing-inaccessible by the kinematic fact alone (the seam reading is consistent with it). The single logically-open escape is a *packing-native multiplicity* — a datum the construction emits on the outer labels that would map to a factor-$N_{\mathrm{gen}}$ tripling of $\mathcal{H}_F$. It is empty: Paper 0's

entire multiplicity inventory is the $2\ell + 1$ angular slots (odd, and *growing* with $\ell$, hence not a generation-universal multiplicity) together with a single $\mathbb{Z}_2$ already consumed by spin; no constant factor of three appears anywhere. This sharpens the structural-skeleton scope for the $N_{\text{gen}}$/inner-KO-dimension residue into a *relative non-emission statement* [**PROOF-BY-ARGUMENT**]: the current packing construction emits no such multiplicity. It offers one structural reading of why the H1, W3, and Koide selection attempts failed: all probed inside the almost-commutative category, on the free side of the seam (an Observation-level partition; the period rings on its two sides share no generator beyond $\mathbb{Q}$, which records the partition rather than forbidding selection), where the packing axiom has no reach. The qualifier "relative" is load-bearing: the statement forecloses the residue for the *current* packing construction, not for every conceivable discrete one. Lifting it would require a *sibling axiom* that emits fiber multiplicities — a new primitive Paper 0 does not contain — recorded as the open (long-range) direction with no present handle. The inner *algebra shape* $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is reached only relative to an organizing reading: under the Bertrand $\times$ complex-Hopf-tower reading plus the adopted Upgrade B axiom (Door 4b/4d, via the Hopf tower and Hurwitz's classification) the factor count and the $\mathbb{C}$ and $M_3(\mathbb{C})$ summands are fixed, while $\mathbb{H}$ at $n = 2$ is admitted, not forced; outside that reading the residue is precisely the multiplicity- and real-structure-valued data. Source: `debug/seam_packing_scoping_memo.md`.

**The Forced-Count Theorem (synthesis, 2026-06-03).** The eight preceding paragraphs (Sprint H1 through Door 4f and the Direction 2 scoping) compose into a single Bertrand-shaped statement of the structural content those sub-sprints established (Door 4 itself is partial-door). [**SYMBOLIC + MEASURED**] Stated as a theorem it makes explicit what the distributed prose shows in pieces: given the admitted $\mathcal{A}_F$ (an organizing reading plus the adopted Upgrade B axiom) and the imported CCM representation, the standard axioms fix the *dimension* of the $D_F$ moduli space, but do not (and at the current packing primitive cannot) fix the points within it.

**Theorem 7** (Forced $D_F$ moduli dimension)**.** *Let $\mathcal{T} = (\mathcal{A}_{\text{GV}} \otimes \mathcal{A}_F,\ \mathcal{H}_{\text{GV}} \otimes \mathcal{H}_F,\ D_{\text{GV}} \otimes 1 + \gamma_{\text{GV}} \otimes D_F,\ J,\ \gamma)$ be the GeoVac almost-commutative spectral triple of Sec. 3 (the graded form is displayed for reference; the modules build the ungraded sum $D_{\text{GV}} \otimes 1 + s_{\text{GV}} \otimes D_F$, §8.2; the count below concerns $D_F$ alone and is the same for both), assuming:*

1. *The Bertrand-with-complex-Hopf-tower reading $n \leq 3$ (Door 4b; an organizing reading, not an argument of Sec. 8.1), with the Upgrade B sphere-Lie-group axiom of Door 4e (an adopted axiom), selecting $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$.*
2. *The canonical Chamseddine–Connes–Marcolli SM representation of $\mathcal{A}_F$ on $\mathcal{H}_F$, with generation multiplicity $N_{\text{gen}}$ imported as an external input (Direction 2: not packing-derived).*
3. *The standard real-spectral-triple axioms: Hermiticity, KO-dim-6 chirality grading $\{\gamma_F, D_F\} = 0$, $J$-reality $[J_F, D_F] = 0$, and Connes order-one.*

*Then the space of admissible finite Dirac operators $D_F$ has real dimension forced by these inputs alone, with the full-axiom reduction chain at $N_{\text{gen}} = 1$ (on the full 32-dimensional finite geometry):*

$$\dim_{\mathbb{R}} \mathcal{M}(D_F) :\ 2048 \xrightarrow{\text{Herm.}} 1024 \xrightarrow{\{\gamma_F, D_F\}=0} 512 \xrightarrow{[J_F, D_F]=0} 272 \xrightarrow{\textit{order-one}} 32. \tag{7}$$

***[SYMBOLIC + MEASURED]*** *The endpoint 32 is computed with a linear $*$-representation of $\mathcal{A}_F$ on $\mathbb{C}^{32}$ (particles colour-blind, $M_3(\mathbb{C})$ acting on antiquarks; the CCM algebra basis and ten random generic algebra elements give the same null space through one null-space routine, stable across tolerances, singular-value gap 12.4 vs $10^{-13}$; `tests/test_trunk_qa_forced_count_moduli.py`). Its*

*projection onto the* 16*-dimensional matter sector has rank* 16 *— the* full *lepton* $2 \times 2$ *and quark-flavour* $2 \times 2 \otimes \mathbb{M}_3$ *blocks, of which the four colour-diagonal Yukawas* $y_\nu, y_e, y_u, y_d$ *with their* $L \leftrightarrow R$ *placements are* 4 *complex, i.e.* half*, and onto the matter–antimatter (Majorana) block rank* 16*. The physical-slice dimension under restriction to diagonal Yukawa* $Y = \mathrm{diag}(y_\nu, y_e, y_u, y_d)$ *is* 8 *per generation. The* $N_{\mathrm{gen}}$*-generation count with full generation-mixing Yukawa matrices is not re-derived here.*

*Proof sketch.* The full-axiom chain in eq. (7) is the direct parameter-count of the constraint solution space (`tests/test_trunk_qa_forced_count_moduli.py`) on the full 32-dimensional finite geometry with $N_{\mathrm{gen}} = 1$: Hermiticity halves the free parameters; $\{\gamma_F, D_F\} = 0$ kills the chirality-diagonal blocks; $J$-reality identifies the matter and antimatter sectors ($512 \to 272$); the order-one condition, imposed with a linear $*$-representation of $\mathcal{A}_F$, removes 240 further parameters ($272 \to 32$): with $M_3(\mathbb{C})$ acting on the antiquark colour index, $[[D_F, a], JbJ^{-1}] = 0$ forces the quark Yukawa blocks to be colour-diagonal and colour-independent, and strongly constrains the matter→antimatter block $P$ (it does *not* annihilate it — the rank-16 real Majorana slice survives, as the next paragraph details).

*What survives, measured.* The support pattern of the solution space is the theorem's actual content. The matter block (rank 16 real = 8 complex) is *full* on the lepton $2 \times 2$ — including $\nu_L \to e_R$ and $e_L \to \nu_R$ — and full on the quark flavour $2 \times 2 \otimes \mathbb{M}_3$, including $u_L^c \leftrightarrow d_R^c$; the four named Yukawas are 4 complex, i.e. *half* of it. The Majorana block (rank 16 real) carries 2 complex $\nu_R$ entries and *six* complex lepton$_R$ ↔ antiquark-$u_R$ entries — so twelve of its sixteen real moduli are lepton–quark operators. Each constructed-triple axiom-consistency check passes at $n_{\max} \in \{2, 3\}$ in module `geovac/standard_model_triple.py` (backed by `tests/test_standard_model_triple.py`, 47 tests); those tests verify the axiom consistency of one constructed $D_F$, *not* the moduli-dimension counts of the chain, which are the direct enumeration of `tests/test_trunk_qa_forced_count_moduli.py`. (No $N_{\mathrm{gen}}$ scaling is claimed: as the statement says, the $N_{\mathrm{gen}}$-generation count is not re-derived.) The status of the inputs: (1) rests on the Door 4b Hopf-tower reading (which fixes the factor count) plus the Upgrade B sphere-Lie-group axiom of Door 4e, which selects $\mathbb{H}$ at $n = 2$ as an *adopted axiom*, not a derivation (the Door 4 record is partial-door; Door 4c found $\mathbb{H}$ admitted but not forced); (3) is the standard real-spectral-triple axiom set; (2) is the one external input beyond the adopted axiom in (1), recorded by the Direction 2 paragraph as packing-unreachable in the current Paper 0 construction. □

**Corollary 5** (Bertrand analogy)**.** *Theorem 7 is the inner-factor analogue of Bertrand's theorem on closed-orbit central forces. Bertrand forces the* form *of the central potential* ($-k/r$ *or* $kr^2$) *without forcing the coupling* $k$*; Theorem 7 forces the* dimension *of the* $D_F$ *moduli space without forcing the* $D_F$*-entries themselves.* ***[OBSERVATION]*** *The two seams are analogous: given its inputs, the construction fixes what the moduli space* is*, not where in it the realised system sits.*

**Remark 24** (Paired with the Door 4 negative half)**.** *The positive content of Theorem 7 sits beside two negative companions already recorded above:*

- ***Door 4 seam reading (Yukawa-value wall):*** *the* $\eta$*-trivialization plus tensor factorization plus the G3 commuting-*$\mathbb{Z}_2$ *result place the* $D_F$*-entries inside* $\mathcal{M}(D_F)$ *in a Dirichlet ring sharing no generator with any forced outer ring, so the framework does not derive them from the forced rings (an Observation-level reading).* ***[PANEL-VERIFIED]*** *Sprint Yukawa-PSLQ (2026-06-03) added empirical confirmation: a* 162*-cell sweep of the nine measured Yukawa values against the inner-factor pure-Tate basis* $\mathbb{Q}[\pi, \pi^{-1}]$ *suggested by the M1/M2 slot restriction (basis sharpened by the* $\eta$*-trivialization audit ruling out* $M_3$*/Dirichlet* $L$ *on the inner*

*factor) returned zero hits at $M \leq 10^3$, at both MS-bar $M_Z$ and MS-bar $\mu = 2 \times 10^{16}$ GeV, across three transforms ($y_f$, $y_f^2$, $\log y_f$). Source:* `debug/sprint_yukawa_pslq_memo.md`.

- ***Direction 2 packing-reach (non-emission) statement:*** *$N_{\text{gen}}$ and the inner KO-dimension are packing-unreachable in the current Paper 0 construction (preceding paragraph, source* `debug/seam_packing_scoping_memo.md`*).*

***[CONDITIONAL]*** *Theorem 7 thus crystallises the* full *forced/free seam at the level of $D_F$: the forced data is the moduli space and its dimension; the free data is (i) the point in the moduli space (Yukawa values, Door 4 wall), (ii) the multiplicity factor ($N_{\text{gen}}$, Direction 2), and (iii) the real-structure signature (inner KO-dimension, Direction 2). All three free data are calibration inputs in the structural-skeleton-scope sense (§1.5 of the project guidelines).*

**Theorem 8** ($N_{\text{gen}}$ non-selection under canonical CCM representation)**.** ***[PROOF-BY-ARGUMENT]*** *(Here and in the non-selection results below, "a morphism in $\mathcal{A}$" is informal: $\mathcal{A}$ is a list of axioms, and "no morphism selects $X$" means that the axioms are satisfied for every admissible value of $X$.) Let $\mathcal{A}$ denote the framework's current axiom set {Paper 0 packing axiom; standard real-spectral-triple axioms (Hermiticity, $J$-reality, KO-dim chirality, order-one); Hopf-tower truncation (Hurwitz); Bertrand's classical theorem; Upgrade B (sphere-Lie-group axiom)}, and fix the canonical Chamseddine–Connes SM representation of $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ on the inner Hilbert space.*

*For every integer $N \geq 1$, let $\mathcal{T}_N$ denote the GeoVac spectral triple with inner Hilbert-space multiplicity $N$ (i.e., $\dim \mathcal{H}_F = 32N$ under the canonical rep). Then $\mathcal{T}_N$ satisfies every axiom in $\mathcal{A}$ that $\mathcal{T}_1$ satisfies (on the GV factor order zero and order one fail at finite cutoff, Table 1, independently of $N$). Consequently, $\mathcal{A}$ contains no morphism that selects a specific value of $N$.*

*Proof sketch.* Each axiom in $\mathcal{A}$ is independent of $N$:

- The Paper 0 packing axiom outputs $(n, l, m, m_s)$ labels for a single $S^3$ outer graph; $N$ is not in this output.
- The standard real-spectral-triple axioms constrain the relations $(D, J, \gamma)$ on $\mathcal{H} = \mathcal{H}_{\text{GV}} \otimes \mathcal{H}_F$. $\dim \mathcal{H}_F$ scales linearly with $N$, but Hermiticity, $J$-reality, KO-dim chirality, and order-one are structural relations independent of multiplicity.
- Hopf-tower truncation (Hurwitz, Doors 4b/4d/4e) operates on $\mathcal{A}_F$; the algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is fixed independent of $N$.
- Bertrand's classical theorem selects the outer manifold $S^3$; $N$-independent.
- Upgrade B constrains the inner-algebra structure via the sphere-Lie-group identification of Door 4e; $N$-independent.
- The canonical CCM SM representation distributes 32 fermion degrees of freedom per generation across the three algebra factors; the multiplicity $N$ enters as an external Hilbert-space multiplicity rather than as a constraint emergent from the rep structure.

The Read 2 scoping (`debug/sprint_read2_n_gen_scoping_memo.md`, 2026-06-03) provides three independent obstructions to any candidate Hopf-tower-to-representation shortcut that would correlate $N$ with the algebra factor count under canonical representation: the SM fermion DOF count per generation distributes across all three algebra factors; the Yukawa-Higgs cross-generation coupling structure required for CKM/PMNS phenomenology prevents algebra-factor reorganisation; and the alternative "3 factors = 3 generations" assignment gives 12 DOF per gen versus the 32 DOF of the SM. Each obstruction independently closes the candidate shortcut.

Therefore $\mathcal{T}_N$ is a valid GeoVac spectral triple for every $N \geq 1$ under canonical representation; no axiom in $\mathcal{A}$ has been exhibited that selects a specific value. □ □

**Remark 25** (Scope and the unconditional question)**.** *Theorem 8 is conditional on the canonical CCM SM representation choice. The unconditional question — whether any SM-phenomenologically-consistent representation of $\mathcal{A}_F$ on $\mathcal{H}_F$ could correlate $N$ with the algebra factor count under an alternative rep — remains the open long-range structural- research target named by the Read 2 sharpest falsifier: exhibit a representation in which the* 32 *fermion DOF per generation are distributed across the three algebra factors in a way that makes the three generations the three rep blocks of the algebra, and the cross-generation Yukawa-Higgs coupling (CKM/PMNS) is naturally encoded as the algebra cross-factor coupling. Absent such a construction, Theorem 8 stands as the conditional impossibility result corresponding to the Read 2 NO-GO scoping verdict.*

*The Theorem identifies $N_{\text{gen}}$ as Class 1 calibration data (external input on which the framework's outer-factor mechanism is structurally silent), formalising the Direction 2 packing-reach and Read 2 representation-level NO-GO scopings into a single structural statement. It is the multiplicity-side analogue of the H1 Yukawa non-selection theorem: both are non-selection results for inner-factor content, both conditional on the canonical rep, and together with the Forced-Count Theorem (7) they characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level.*

***[CONDITIONAL]*** *Paper 57 §6.1 (the Forced/Free Seam observations paper) records this theorem as the C3 closure of the empirical Direction 2 / Read 2 NO-GO scopings, and as a proof-by-argument non-selection result alongside H1 Yukawa non-selection (the Forced-Count Theorem, listed with them there, is a dimension count, not a non-selection result).*

**Theorem 9** (Inner KO-dim is external calibration data under canonical-rep convention)**.** ***[PROOF-BY-ARGUMENT]*** *(The combined triple's own sign data is the measured $(-,+,+)$ of §8.2; no additive KO rule applies to it, since for the graded product Dabrowski–Dossena's Table 5 admits no consistent $\varepsilon'$ with a KO-6 finite factor. The statement below concerns only the* inner *factor's label.) Let $\mathcal{A}$ denote the framework's current axiom set (as defined in* Theorem 8*). The Paper 0 packing axiom is kinematic (Paper 0 § VII.B "What the construction does not provide"): it outputs the labels $(n, l, m, m_s)$ and the $S^3$ graph topology, but neither a Dirac operator, nor a real structure, nor a chirality grading. The inner KO-dimensional signature is a property of the real-structure data $(D_F, J_F, \gamma_F)$, none of which is in the packing axiom's output.*

*The remaining axioms in $\mathcal{A}$ (standard real-spectral-triple axioms, Hopf-tower truncation, Bertrand's classical theorem, Upgrade B) constrain the consistency of $(D_F, J_F, \gamma_F)$ with the outer GeoVac triple's signs (KO-dimension* 3 *for the continuum factor; sign triple $(-,+,+)$ and no label at finite cutoff), but output no $(D_F, J_F, \gamma_F)$, and so no inner KO-dim signature. The canonical Chamseddine–Connes SM representation (inner KO-dim* 6*) is supplied as external input to the framework.* ***[OPEN]*** *No alternative inner datum with a different KO-dimension that satisfies every axiom of $\mathcal{A}$ together with the outer signs is exhibited here; the theorem asserts non-selection, not the existence of alternatives.*

*Proof sketch.* The argument is a direct composition of two established facts:

**Fact 1 (Paper 0 §VII.B, explicit scope statement).** The packing construction is *kinematic*: it produces labels and graph topology, and explicitly does *not* produce operators, real structures, couplings, or fiber-Hilbert-space multiplicities.

**Fact 2 (Door 4f T1, Door 4b Q3).** The inner KO-dimensional signature is defined as a property of the real-structure data $(D_F, J_F, \gamma_F)$. The chirality-grading axiom $\{\gamma_F, D_F\} = 0$ and the $J_F$-reality conditions $J_F^2 = \pm\mathbf{1}$, $J_F D_F = \pm D_F J_F$, $J_F \gamma_F = \pm \gamma_F J_F$ specify the residue classes

mod 8 that are consistent, but do not select one (alternatives satisfying every axiom of $\mathcal{A}$ are not exhibited here) in the absence of additional constraints (such as the phenomenological constraints imposed by SM fermion doubling and Majorana neutrino-mass structure).

Composition. $\mathcal{A}$ produces no $(D_F, J_F, \gamma_F)$ data autonomously (Fact 1), and the inner KO-dim is a function of that data (Fact 2). Therefore $\mathcal{A}$ does not autonomously select an inner KO-dim signature. The canonical CCM SM rep supplies a specific $(D_F, J_F, \gamma_F)$ triple (KO-dim 6) as external input. □ □

**Remark 26** (The full inner-factor structural-skeleton boundary)**.** *Theorems 7, the H1 Yukawa non-selection (Theorem 6), 8, and 9 together characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level:*

- ***Forced data:*** *the moduli space $\mathcal{M}(D_F)$ and its dimension (full-axiom:* 32 *at $N_{\text{gen}} = 1$, matter-sector projection rank* 16*,* 8 *per generation on the diagonal-Yukawa slice; Theorem 7).*
- ***Free data:*** *(i) the point in $\mathcal{M}(D_F)$ (Yukawa values — H1 non-selection), (ii) the Hilbert-space multiplicity ($N_{\text{gen}}$ — Theorem 8), (iii) the real-structure signature (inner KO-dim — Theorem 9).*

*These four theorems together formalise the structural-skeleton scope pattern at the inner-factor level for the GeoVac framework's current axiom set. Each is conditional on the canonical CCM SM representation. The unconditional versions — whether alternative phenomenologically-consistent representations could correlate any of the three free data with framework-derivable structure — remain long-range structural-research targets; Read 2 sharpest falsifier names the candidate construction for $N_{\text{gen}}$, and analogous targets exist for KO-dim (a non-canonical rep that admits a different KO-dim while retaining SM phenomenology) and for Yukawa values (a packing-derivable structure on $\mathcal{A}_F$ that selects within the diagonal slice).*

*Paper 57 §6.3 (Forced/Free Seam observations paper) records the trio plus Forced-Count Theorem as the C3 closure of the inner-factor non-selection arc.*

**Definition 4** (Multi-cutoff spectral action structure)**.** *A multi-cutoff spectral action on a real spectral triple $\mathcal{T} = (\mathcal{A}, \mathcal{H}, D)$ is a functional of the form*

$$S^{\text{multi}}(D, \{\Lambda_i\}_{i=1}^k, \{f_i\}_{i=1}^k) \;=\; \sum_{i=1}^{k} c_i \operatorname{Tr} f_i(D/\Lambda_i)$$

*or more generally any functional of $D$ depending on $k \geq 2$ independent energy scales $\Lambda_1, \ldots, \Lambda_k$ in a manner not reducible to a function of a single scale by reparametrization.*

**Observation 3** (Single-cutoff spectral action)**.** ***[OBSERVATION]*** *(true by stipulation, plus an inspection of the listed axioms; not a proof over all derivable constructions) The framework's current axiom set $\mathcal{A}$ (as defined in* Theorem 8*) specifies the Chamseddine–Connes spectral action $S(D, \Lambda) = \operatorname{Tr} f(D/\Lambda)$ as a single-cutoff functional in the sense of* Definition 4*, and none of the constructions inspected below introduces a second independent scale.*

*Inspection.* The argument is a structural inspection of each axiom in $\mathcal{A}$:

1. **Spectral action axiom (CCM).** The standard CC spectral action $S(D, \Lambda) = \operatorname{Tr} f(D/\Lambda)$ specifies a single-cutoff functional by stipulation. The argument $D/\Lambda$ is a dimensionless ratio against a single energy scale.

2. **Paper 0 packing axiom + standard real-spectral-triple axioms + Hopf-tower truncation + Bertrand + Upgrade B.** These axioms are algebraic, kinematic, and topological: they specify $(\mathcal{A}_F, \mathcal{H}_F, D_F, J_F, \gamma_F)$ and the outer manifold geometry, but introduce no energy-scale data.

3. **Inner fluctuations $D \to D + A$ for $A = a[D, b]$.** These preserve the single-Dirac structure of the underlying spectral triple. The fluctuated operator $D' = D + A$ enters the spectral action as $S(D', \Lambda)$, still a single-cutoff functional.

4. **Tensor products $\mathcal{T}_1 \otimes \mathcal{T}_2$ (Paper 39).** These produce a new spectral triple with combined Dirac $D = D_1 \otimes \sigma_1 \otimes 1 + 1 \otimes \sigma_2 \otimes D_2$ (the Clifford-doubled product of two odd triples), on which the spectral action $S(D, \Lambda)$ remains single-cutoff with the new combined $D$. No second cutoff is introduced by the tensor-product construction.

None of these introduces a second independent energy scale. The first item is a stipulation, and the list is an inspection of the named constructions, not an enumeration of everything derivable from $\mathcal{A}$. □ □

**Corollary 6** (Multi-loop renormalization counterterms are external calibration data)**.** ***[CONDITIONAL]*** Premise (not argued here): *multi-loop QED renormalization counterterms (wavefunction renormalization $Z_2$, mass counterterm $\delta m$, vertex counterterm $Z_1$, etc.) require a second independent scale $\mu_R$ beside the ultraviolet cutoff $\Lambda_{UV}$. (An on-shell scheme, whose subtraction point is the physical mass already carried by $D$, is a candidate counterexample to the premise.) Given the premise and Observation 3, the multi-loop QED counterterms are external calibration data, supplied to the framework rather than derived from it.*

*Under that premise the following catalogue entries from Paper 57 [49] are classified as multi-cutoff renormalization calibration data:*

- ***E7:*** *two-loop self-energy counterterms $Z_2, \delta m$.*
- ***E8:*** *multi-loop QED corrections beyond the one-loop LS-7 closure.*
- ***G3:*** *HF-5 multi-loop QED on hyperfine.*
- ***G4:*** *LS-8a two-loop SE renormalization (identical to E7 under the cross-domain unity of Paper 57 §4).*

*In Paper 36 (LS-8a) the framework's iterated Chamseddine–Connes spectral action reproduces the two-loop UV-divergent integrand at the right prefactor, sign, and divergence rate $\sim N^{3.79}$, but has not generated the $Z_2, \delta m$ counterterms required for finite multi-loop closure. The corollary offers a structural reading of that wall, conditional on its premise.*

**Remark 27** (Honest scope)**.** *Observation 3 holds by stipulation within the framework's current axiom set $\mathcal{A}$. Two scope clarifications:*

***(i) The observation does not claim "no multi-cutoff structure exists."*** *It records that none of the inspected constructions from $\mathcal{A}$* as currently stated *introduces a second scale. A sibling axiom adding renormalization-group flow machinery would extend the axiom set and could in principle produce multi-cutoff observables. Such a sibling axiom is the analog of the $N_{\text{gen}}$ sibling-axiom direction (*Theorem 8 *unconditional version): a logically-possible escape, named here, that requires new primitive content beyond $\mathcal{A}$.*

***(ii) The observation does not address the spatial-composition sub-sector of the multi-focal-composition wall.*** *The catalogue entries G1 (HF-3 recoil cross-register $V_{eN}(\hat{r}_e, \hat{R}_N)$),*

*G2 (HF-4 Zemach magnetization density), and G5 (W1e chemistry inner-region overattraction) lie outside this observation's scope: they are spatial × spatial composition walls, structurally distinct from the multi-cutoff renormalization sub-sector.* ***[OPEN]*** *A general multi-focal-composition wall theorem unifying both sub-sectors remains the named open structural-research target.*

*Paper 57 §4 (cross-domain unity) and Paper 57 §6.3 record this observation and its corollary as a conditional classification of the renormalization sub-sector of the multi-focal-composition wall.*

**Theorem 10** (Tensor-product radial coupling does not autonomously match physical two-body profiles)**.** ***[PROOF-BY-ARGUMENT, with an empirical leg: the Pearson check of step (2.4) (***$0.846$***,*** $0.796$ ***at*** $n_{\max} = 2, 3$***) is a measurement, so the argument covers only the structural part (2.1)–(2.3)]*** *Let* $\mathcal{T}_a = (\mathcal{A}_a, \mathcal{H}_a, D_a^{\mathrm{CH}})$ *and* $\mathcal{T}_b = (\mathcal{A}_b, \mathcal{H}_b, D_b^{\mathrm{CH}})$ *be two GeoVac spectral triples on the same outer manifold* $S^3$ *at distinct focal lengths* $\lambda_a \neq \lambda_b$*, equipped with the Camporesi–Higuchi spinor Dirac operators. The framework's natural tensor-product spectral triple* $\mathcal{T}_a \otimes \mathcal{T}_b$ *[39] produces composed two-body operators whose:*

1. ***Angular selection rules*** *are forced: Gaunt compatibility and* $m$*-conservation hold structurally for the composed algebra (Paper 54 [48], angular-structure theorem). The interaction is pure* $k = 0$ *monopole at* $n_{\max} \in \{2, 3\}$ *(measured there), so the* $k \geq 1$ *multipole channels of the physical Coulomb interaction are* not *reproduced at those cutoffs.*
2. ***Radial coupling profile*** *is determined autonomously by the combined Dirac structure (the Clifford-doubled graded product of Paper 39, since the naive* $D_a^{\mathrm{CH}} \otimes 1 + \gamma_a \otimes D_b^{\mathrm{CH}}$ *is unrealisable with* $\gamma_a = 0$ *on* $S^3$*), but does not autonomously coincide with the physical 1/r Coulomb form, the dipole-magnetization form, or other specific two-body radial profiles required by empirical observables.*

*The framework's axiom set* $\mathcal{A}$ *contains no morphism enforcing that the composed radial coupling profile matches a specific physical two-body observable; matching a given observable requires external specification.*

*Proof sketch.* The two assertions of the theorem stand on independent structural content:

**(1) Angular selection rules.** Paper 54's angular-structure theorem [48] establishes that the composed algebra structure on $\mathcal{A}_a \otimes \mathcal{A}_b$ acts on the two-body Hilbert space through Wigner 3j coefficients, so the Gaunt and $m$-conservation selection rules hold structurally; the multipole content is measured there to be pure $k = 0$ at $n_{\max} \in \{2, 3\}$, which is not the multipole hierarchy of the physical Coulomb interaction. The selection rules are a direct consequence of the SU(2) representation-theoretic structure of the angular momentum operators on $S^3$, which the framework inherits via the Fock projection (Paper 7) and the Hopf-tower truncation.

**(2) Radial coupling non-reproduction.** The structural reason is the non-commutativity of the Fock-projection conformal factor with the tensor-product construction:

1. Paper 7's chordal-distance identity maps continuum $-Z/r$ on $\mathbb{R}^3$ to the Laplace–Beltrami operator on the unit $S^3$ (whose harmonics the graph's nodes label; the graph Laplacian is not its discretisation and does not converge to it, Paper 7) via a focal-length- dependent stereographic conformal factor $c(\lambda)$. Two different focal lengths $\lambda_a, \lambda_b$ give two different conformal factors $c_a, c_b$.
2. The tensor-product spectral triple $\mathcal{T}_a \otimes \mathcal{T}_b$ has continuum limit on the product manifold $S^3_a \times S^3_b$. Under Fock projection, this corresponds to a six-dimensional space $\mathbb{R}^3_a \times \mathbb{R}^3_b$ parametrised by two independent position vectors, with combined conformal factor $c_{ab} \neq c_a \cdot c_b$ in general.

3. Physical two-body observables ($\mathrm{V}_{eN}(\hat{r}_e, \hat{R}_N)$, Zemach magnetization, chemistry inner-region $V_{ee}(r_{12})$) are functions of the *relative* coordinate $\vec{r}_a - \vec{r}_b$ on $\mathbb{R}^3$ (three dimensions, not six). The reduction from the six-dimensional product manifold to the three-dimensional relative manifold is not in the framework's axiom set.
4. Empirical verification: Sprint resolvent_two_body (2026-06-01, `debug/sprint_resolvent_two_body_memo.md`) tested four weightings of the resolvent $(D_a^{\mathrm{CH}})^{-2}$ against the physical 1/r Coulomb profile and found Pearson correlations of at most 0.846 ($n_{\max} = 2$) and 0.796 ($n_{\max} = 3$) for the best weighting (values after the 2026-08-29 ERI-evaluator fix), decreasing with $n_{\max}$. Paper 54 verified the angular selection rules but reported only partial radial matching (connected fraction 77% at $n_{\max} = 2$, 32% at $n_{\max} = 3$). Together these indicate that the framework's natural composition mechanisms (tensor product, resolvent, gauged spectral action) do not autonomously match physical two-body radial profiles.

No axiom in $\mathcal{A}$ supplies the relative-coordinate reduction from the six-dimensional product manifold to the three-dimensional relative manifold, nor enforces a specific composed radial profile on the tensor-product spectral triple. Therefore the framework specifies the angular selection rules of two-body operators autonomously (via the SU(2) representation theory of $S^3$) but does not autonomously match the radial structure of any specific physical two-body observable. □ □

**Corollary 7** (Spatial-composition catalogue entries are calibration data)**.** *The following catalogue entries from Paper 57 [49] are classified as spatial-composition calibration data:*

- ***G1:*** *HF-3 recoil cross-register $V_{eN}(\hat{r}_e, \hat{R}_N)$ (Track NI cross-register, Sprint HF 2026-05-07);*
- ***G2:*** *HF-4 Zemach magnetization density (Sprint HF 2026-05-07);*
- ***G5:*** *W1e chemistry inner-region overattraction (Sprint W1c arc 2026-05-23).*

*Each requires a specific physical radial coupling profile that Theorem 10 establishes is not autonomously produced by the framework's tensor-product composition mechanism. The angular selection rules are forced; the radial content (and, at the measured cutoffs, the $k \geq 1$ multipoles) is external calibration.*

**Remark 28** (Multi-focal-composition wall: spatial half bound by argument, renormalization half conditional)**.** *Together with Observation 3 + Corollary 6 (E7, E8, G3, G4), Theorem 10 + Corollary 7 (G1, G2, G5) classify seven catalogue entries of the multi-focal-composition wall family (Paper 57 §4 cross-domain unity): the spatial-composition entries at proof-by-argument level with an empirical leg, the renormalization entries conditionally on the premise of Corollary 6:*

- ***Renormalization sub-sector*** *(E7, E8, G3, G4): classified by Observation 3 (by stipulation) and its conditional corollary.*
- ***Spatial-composition sub-sector*** *(G1, G2, G5): closed by Theorem 10 — tensor-product composition carries the angular selection rules but not the radial structure; no relative-coordinate-reduction morphism in $\mathcal{A}$.*

*An eighth catalogue entry (G6) is the H1 Yukawa non-selection, closed by Theorem 8's companion result (Theorem 6). The entries are finer than the instances: all six independent multi-focal-composition wall instances of Paper 57 §4 (H1, LS-8a, HF-3/4/5, W1e; G6 = F1 is not independent) are characterised by one of the three results (LS-8a and HF-5 only conditionally).*

*The spatial-composition sub-sector of the multi-focal-composition wall is bound by argument at the canonical-representation level; the renormalization sub-sector is classified conditionally. Together with the inner-factor results (the Forced-Count Theorem 7, a dimension count, and the non-selection theorems 8, 9 and H1), the corpus carries five proof-by-argument results covering the inner-factor structural-skeleton boundary and the spatial-composition sub-sector, and a stipulation-level Observation with a conditional corollary for the multi-loop QED renormalization sub-sector.*

*Paper 57 §6.3 records this classification.*

**Observation 4** (No single-mechanism generation of the combination rule $K = \pi(B + F - \Delta)$ has been found)**.** ***[OBSERVATION]*** *Twelve candidate mechanisms for single-principle generation of the empirical combination rule $K = \pi(B + F - \Delta)$ for $\alpha^{-1}$ (Paper 2) have been empirically eliminated (Paper 2 §IV.G; Phases 4B–4I, April 2026, eight mechanisms; Sprint A $\alpha$-SP, 2026-04-18, Connes–Chamseddine direct spectral-action derivation; Sprint K-CC, 2026-05-03, three further mechanisms via PSLQ at 100 dps against a 244-element basis, a natural $\Lambda$ search from $S^3$ spectral data, and a unified CC heat-kernel route blocked by the algebraic "T9 obstruction" of Paper 2 — not Paper 28's Theorem T9, the $\zeta_{D^2}$ closed form). The three ingredients have separate spectral homes ($B = 42$ a finite Casimir trace, $F = \pi^2/6$ a Fock-degeneracy Dirichlet value, $\Delta = 1/40 = (g_3^{\mathrm{Dirac}})^{-1}$ an inverse Dirac mode degeneracy; Observation 1), and no single mechanism is currently exhibited that produces their combination with the weights $(+1, +1, -1)$ and the external $\pi$ prefactor.*

**Remark 29** (E6 scope: why this is an Observation, not a theorem)**.** *The two-fact structural argument for a theorem does not stand. A sub-ring partition assigning the rationals $B = 42$ and $\Delta = 1/40$ to different Mellin sub-rings fails, since the sub-rings are not exclusive: $\mathbb{Q}$ lies in both the $M_1$ ring $\mathbb{Q}[\pi, \pi^{-1}]$ and the $M_2$ ring $\bigoplus_k \pi^{2k}\mathbb{Q}$, so $B + F - \Delta$ lies wholly in the $M_2$ ring and no mixing of sub-rings occurs. Nor can the $\pi$-source case-exhaustion (Theorem 4) be invoked for* every *transcendental: that theorem is [CONDITIONAL] and covers only the first fifteen of the twenty-eight Paper 34 projections; it also sits in tension with the remark that the spectral action engages $M_1$ and $M_2$ jointly (Sec. 8). What survives is the empirical record of twelve eliminations, recorded above as an Observation. It does not derive $K$, and it does not show that $K$ cannot be generated by a single mechanism from some extended axiom set.*

*Per the project's standing hard prohibition, the Paper 2 combination rule is labeled an* Observation *(not a conjecture or derivation); nothing in this section promotes or demotes that label. Paper 2's status as Observations-folder paper (post 2026-05-02 curve-fit-audit migration) is unchanged. Catalogue entry E6 (the combination rule for $\alpha^{-1}$) is classified as calibration data with no single mechanism found: the framework provides three structural homes for the ingredients but no principle for their combination.*

**Theorem 11** (Cutoff function $f$ and its Mellin moments are external calibration data)**.** ***[PROOF-BY-ARGUMENT]*** *The framework's axiom set $\mathcal{A}$ specifies the Chamseddine–Connes spectral action functional $S(D, \Lambda, f) = \operatorname{Tr} f(D/\Lambda)$ on a real spectral triple. The cutoff function $f$ enters as a Schwartz-class test function on $[0, \infty)$ (Paper 51) but is not constrained by any axiom in $\mathcal{A}$. Consequently the Mellin moments*

$$\varphi(s) \;=\; \int_0^\infty x^{s-1} f(x)\, dx$$

*are external calibration data: no morphism in $\mathcal{A}$ selects the moment values $\varphi(0), \varphi(1), \varphi(2), \ldots$.*

*The Wald two-term-exactness ⇒ pure Einstein ⇒ action-G ≡ entropy-G argument (Paper 51 §G7) forces* relations *among the moments via the sector-wise Mellin moment map (tip ↔ $\varphi(0)$, Einstein–Hilbert ↔ $\varphi(1)$, $\Lambda_{cc}$ ↔ $\varphi(2)$); these relations follow by argument within $\mathcal{A}$ (the Wald step itself is a theorem). The moment* values *are not constrained by $\mathcal{A}$.*

**Corollary 8** (CC fine-tuning is external moment selection)**.** *The observed cosmological-constant problem in the framework's sector-wise Mellin moment formulation (Paper 51 §G4-5) requires*

$$\Lambda_{cc} \cdot G_{\text{eff}} \;\approx\; 36\pi \cdot \frac{\varphi(2)}{\varphi(1)^2} \;\approx\; 10^{-122}$$

*(dimensionless), which is consistent with the framework's structural relations between $G_N$, $\Lambda_{cc}$, and $S_{\text{BH}}$ (Wald, theorem-grade) provided $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ with $\varphi(0), \varphi(1) = O(1)$. No natural cutoff function $f$ produces this fine-tuned moment ratio, and no axiom in $\mathcal{A}$ constrains the moment values to it.*

*Catalogue entries D5 (cutoff function moments $\varphi(0), \varphi(1), \varphi(2)$) and D6 (cosmological constant value $\Lambda_{cc}$) are external calibration data inheriting from the cutoff function selection.*

**Remark 30** (D5/D6 scope and the sibling-axiom direction)**.** *Theorem 11 formalises the precise location of the cosmological-constant problem in the GeoVac framework: the gravitational* relations *are forced (Wald, Paper 51 §G7); the cutoff-function* moments *are external. The CC problem in this framing is a moment-selection problem on the cutoff function $f$, structurally similar to the Higgs–Yukawa selection problem (*Theorem *H1) but on a different calibration object.*

*A sibling axiom that constrains the cutoff function or its moments (e.g., a renormalisation-group flow axiom, a regulator-independence principle, an asymptotic-safety constraint) would extend $\mathcal{A}$ and could in principle select the moment values. None is currently exhibited; the long-range structural-research direction is named here as the natural analog of the inner-factor sibling-axiom directions.*

*Paper 57 §6.3 records this theorem as the gravity sector closure of the C-arc.*

**Remark 31** (Chemistry-side $\eta$-trivialization analog: open structural-research target)**.** *The SM-side $\eta$-trivialization theorem (Paper 18* `thm:eta_trivialization`*) restricts the inner factor to the $M_1/M_2$ Mellin* slots *(the $M_3$ slot vanishes) on any Krajewski-class even finite spectral triple satisfying the chirality grading axiom $\{\gamma_F, D_F\} = 0$: for all odd $k$, $\mathrm{Tr}(D_F^k \cdot e^{-tD_F^2}) \equiv 0$, so $M_3$ (vertex-parity Hurwitz / Catalan G / Dirichlet L) vanishes on the inner factor. One reading (an Observation, not a derivation) is that this is why F1 (Yukawa values) and F4–F7 (Higgs VEV, CKM, PMNS, neutrino masses) live outside the master Mellin engine's structural skeleton: the inner-factor Yukawa values occupy the M1/M2 slots, but their generators are tied to the couplings $y_i$ and are not among those of the rings the framework's outer-factor mechanism generates (an Observation-level reading; the rings are not disjoint, since both contain $\mathbb{Q}$).*

*The chemistry-side analog would be a structural property of the FrozenCore $Z_{\text{eff}}(r)$ pipeline that mirrors $\{\gamma_F, D_F\} = 0$ on the chemistry inner-factor side: a chirality-style grading on the Clementi–Raimondi $Z_{\text{eff}}(r)$ structure that imposes a similar slot restriction on chemistry inner-factor data. Such a property is not currently exhibited. The most concrete candidate direction would identify a discrete grading on the FrozenCore radial profile pipeline (e.g., parity of multi-zeta exponents, partition of core orbital basis into chirality-graded blocks) that produces an $\eta$-trivialization-style theorem for chemistry inner-factor data.*

*Sprint-scale path is unclear: unlike the SM-side $\eta$-trivialization, which follows directly from a single axiom ($\{\gamma_F, D_F\} = 0$, the standard grading axiom of even finite spectral triples, used in Krajewski's 1998 classification [20] and canonically adopted in Chamseddine–Connes–Marcolli 2007 [62]), the chemistry-side analog would need to identify the analog grading axiom from scratch. This is a substantial NCG-research target.*

*Catalogue entries H6 (Clementi–Raimondi $Z_{\text{eff}}$ exponents) and H7 (multi-zeta coefficients in second-row valence basis) remain empirical-only pending this structural-research target's completion.*

*The empirical-grade content (W1e period-class diagnostic, **debug/sprint_w1e_period_class_memo.md**, 2026-06-04) found no relation placing H6, H7 in $M_1 \cup M_2 \cup M_3$ at audit ceiling 100 (a PSLQ null result); the open question is whether they sit in a structurally distinct chemistry-side ring forced by a hypothetical analog axiom.*

*Paper 57 §6.3 records this remark as the named substantial structural-research target for the chemistry sector; the C-arc's sprint-scale run closes at v3.103.0 with this target named but not attempted.*

**Remark 32** (The C-arc terminal state)**.** *After Sprints C2 + C3 + F3 + E7/E8 + G1/G2/G5 + E6 + D5/D6 + Chemistry-Analog (all 2026-06-08, v3.97.0–v3.103.0), the corpus carries* six *proof-by-argument results (the Forced-Count dimension count and five non-selection results), one stipulation-level Observation (the single-cutoff spectral action) and one empirical non-selection Observation across the following sectors:*

***Inner-factor sector** (closed at canonical-rep level):*

1. *Forced-Count Theorem (7).*
2. *H1 Yukawa non-selection (Theorem 6).*
3. *$N_{\text{gen}}$ non-selection (8).*
4. *Inner KO-dimension non-selection (9).*

***Multi-focal-composition wall** (one sub-sector theorem with an empirical leg, and one stipulation-level Observation):*

5. *Single-cutoff spectral action (3).*
6. *Spatial-composition radial wall (10).*

$\alpha$ ***combination rule sector** (single-mechanism generation not found; empirical, not closed):*

7. *No single-mechanism generation of $K = \pi(B + F - \Delta)$ found (Observation 4; twelve mechanisms eliminated).*

***Gravity sector** (cutoff function external-selection closed):*

8. *Cutoff function $f$ and its moments are external (11).*

*The chemistry-side $\eta$-trivialization analog (Remark 31) is the named substantial structural-research target; catalogue entries H6, H7 remain empirical-only pending its completion.*

*The remaining empirical-only catalogue entries (F4–F7 Higgs VEV / CKM / PMNS / neutrino mass values, structurally inheriting from F1 and F4; I-entries foundational calibration) are subsidiary residuals inheriting from theorems above (Higgs admission B7 + H1 Yukawa non-selection) or sitting outside the structural-skeleton scope (Born rule via Gleason).*

*The full forced/free seam (Paper 57) is therefore in a stable terminal state for the path-#1 vein C arc: six proof-by-argument results, one stipulation-level and one empirical Observation across four sectors, one named substantial follow-on (chemistry $\eta$-trivialization analog), all subsidiary residuals accounted for.*

**Files.** Scoping memo `debug/g4_cross_manifold_scoping_memo.md` ($\sim 5\,100$ words). G4a sprint memo `debug/sprint_g4a_connes_sm_memo.md`. H1 sprint memo `debug/h1_higgs_inner_fluctuation_memo.md`. Sprint C3 $N_{\text{gen}}$ non-selection theorem memo `debug/sprint_c3_n_gen_non_selection_memo.md` (2026-06-08). G4b is recorded as an open structural question.

### 8.5 Frontier-of-field framing: GeoVac-internal walls vs. spectral-action open problems

[**OBSERVATION**] The five walls catalogued in the multi-focal-composition audit (`debug/multifocal_phase_a_s` May 2026) are not all GeoVac-specific. Reading them through the spectral-triple construction of Sec. 3 and the Marcolli–van Suijlekom lineage of Sec. 9, three are GeoVac-internal (two architectural gaps, and one, W1c, built without closing the chemistry wall it targeted), and two sit at the broader frontier of the spectral-action / NCG programs. We name them here for honesty and to fix the scope of the open questions of Sec. 10.

**GeoVac-internal walls (W1a, W1b, W1c). [OPEN]**

- **W1a (cross-register coordinate operator):** no two-body spatial operator $V(\hat{\mathbf{r}}_e, \hat{\mathbf{R}}_n)$ has been implemented across the nuclear and electronic registers of the composed nuclear–electronic Hamiltonian (Paper 23 [30] §VI; `geovac/nuclear/nuclear_electronic.py`). The framework couples the registers through spin-spin hyperfine and a classical finite-size shift but evaluates spatial operators at the classical proton position. This is an architectural gap, not a no-go theorem; the calibrated atomic-physics target is the Pachucki–Patkoš–Yerokhin recoil Hamiltonian (PRL 130, 023004, 2023), which is exact in the mass ratio at $(Z\alpha)^6$.
- **W1b (magnetization-distribution operator):** no operator on the proton register represents the Zemach radius $r_Z$ as a Layer-2 magnetization focal length distinct from the charge radius $r_p$ (Paper 34 [36] § III; Sprint HF-4, May 2026). Architectural gap; QCD-internal magnetization density is input data, not GeoVac-internal.
- **W1c (cross-center frozen-core screening):** built (Paper 19; the W1c sprint arc is in the chronicle file): cross-center screening of bare $V_{ne}$ through the multipole expansion is implemented and bit-faithful to quadrature, yet the NaH PES of that pipeline has no internal minimum. The residual is the W1d/W1e hierarchy of Paper 19 (block-diagonal h1; inner-region overattraction), which five sprint-scale closure mechanisms did not remove; Paper 58 localizes W1e to the Hamiltonian specification (the neglected inter-center overlap), not the correlation treatment: with genuine end-to-end integrals NaH binds, at the cost of the sparsity.

[**OBSERVATION**] W1a and W1b are tooling gaps inside the present framework: the missing operators exist in principle and require no new mathematics in the broader NCG literature, though each is a substantial program rather than a bounded sprint. W1c is different: it was built, and the chemistry wall persists (W1d/W1e). None of the three undermines the spectral-triple framing of Sec. 3.

**Frontier-of-field walls (W2a, W2b). [OPEN]**

- **W2a (multi-loop UV/IR composition):** the bare iterated Connes–Chamseddine spectral action on the Camporesi–Higuchi spectrum reproduces UV-divergent integrands of multi-loop QED (Sprint LS-8a, May 2026; Paper 36 [38] §VII) but cannot autonomously generate $Z_2$, $\delta m$, $Z_3$ counterterms for finite extraction at two loops. This is not GeoVac-specific. The Fathizadeh–Marcolli periods-and-motives analysis of Robertson–Walker spectral actions [7] partially addresses spectral-action coefficient structure on homogeneous compact backgrounds, and the subsequent Hekkelman–McDonald 2025 noncommutative integral

approximation [16] provides a single-cutoff Szegő-style asymptotic theory in the Connes–van Suijlekom paradigm; neither closes the multi-cutoff matching question that renormalization counterterm generation requires. The published QFT archetype is the matched effective-field-theory chain (e.g. HQET → NRQED → pNRQED → SCET; Fontes and Szafron, JHEP 05 (2025) 171), where each matching step is at a single physical scale and imports $\overline{MS}$ counterterms by hand. No published spectral-action framework reproduces this autonomously. W2a is therefore an open question of the spectral-action program, of which GeoVac is one specific instance; it is not a defect of the present construction relative to the lineage.

- **W2b (cross-manifold spectral-triple composition):** the tensor product of two infinite metric spectral triples $(\mathcal{T}_{S^3}, d_{D_{S^3}}) \otimes (\mathcal{T}_{S^5}, d_{D_{S^5}})$, needed to compose the GeoVac Coulomb sector (Sec. 3) with the Bargmann–Segal HO sector of Paper 24 [31], has no published Latrémolière-propinquity convergence theorem. The almost-commutative case (one infinite Riemannian factor + one finite factor) is addressed by Latrémolière 2026 [19] (*Spectral continuity of almost commutative manifolds for the $C^1$ topology on Riemannian metrics*), in the sense of metric continuity of almost-commutative models $C(M) \otimes \mathfrak{B}$ as the metric varies; the two-infinite case is not handled in that paper (implicitly, by its construction), nor in the closely related Farsi–Latrémolière–Packer 2024 [63] inductive-sequence work. This is the structural content of the G4b cross-manifold blocker (Sec. 8.4): not a GeoVac-specific limitation but a genuinely open NCG question. Closing it would itself be a substantial mathematical contribution, parallel to but separate from Paper 38's SU(2) state-space GH convergence theorem.

**Forecast (without claiming closure).** We do not claim W1a can be closed without surprises; the multi-$\lambda$ Shibuya–Wulfman algebraic check that gates the W1a attack (multipole termination across mismatched exponents) is not in the published Avery-school literature and could fail to preserve Gaunt-style closure (Track 3 literature memo, May 2026, §1). W1b is bounded by external QCD input. W2a and W2b are open in the broader literature, not just in the GeoVac record. The honest forecast is that closing W1a/b is internal-effort plausible as a substantial program per wall, while the W1e residual of W1c is localized to the Hamiltonian specification (Paper 58) and has no sparsity-preserving closure; closing W2a or W2b would be a contribution to the spectral-action / NCG programs rather than to GeoVac alone, and we do not commit to it as a deliverable here.

**Implication for the Marcolli–vS lineage placement.** Sec. 9 read the present construction as a graph spectral triple in the lineage of [6, 10, 11]. The frontier-of-field framing above sharpens this: the lineage placement is correct, and the W2a/W2b walls are the lineage's own open problems, not GeoVac's separate concerns. This is consistent with Observation 5 (each structural ingredient has published precedent; what is not duplicated is the $\alpha$ observation: $K$ matches $\alpha^{-1}$ to $4.8 \times 10^{-7}$, and the root of the cubic $1/\alpha = K - \alpha^2$ matches to $8.8 \times 10^{-8}$). The W1 walls remain GeoVac-side because they are about the architectural extension of the present triple to multi-particle real-space coupling, which the SM-style almost-commutative construction does not address.

### 8.6 Lorentzian extension (descoped)

The Lorentzian sprint record of this section (the Sprint L0 transfer audit of the master Mellin engine under signature change, the Krein carrier and Krein-space Dirac operator of Sprints L2-B/C, the Connes axiom audit at BBB $(m, n) = (4, 6)$ of Sprint L2-D, the Krein-carrier period closure of Sprint L2-E, and the finite-cutoff modular-Hamiltonian construction of Sprint L1) is kept verbatim in `docs/chronicles/paper_32_chronicle.tex`. What survives of it: the finite-cutoff $2\pi$ period

closure is automatic for any integer-spectrum generator and identifies no Bisognano–Wichmann flow (Paper 42; `tests/test_modular_period_closure_automatic.py`); compressing a Krein-self-adjoint product Dirac to the Krein-positive cone annihilates the spatial Dirac, and the Lorentzian programme is descoped (Paper 45).

## 9 Marcolli–van Suijlekom Placement: Instance + Extension

The construction of Sec. 3 is an *instance* of the gauge networks framework of Marcolli and van Suijlekom [6] (with the Perez-Sanchez correction [11] that the continuum limit is Yang–Mills without Higgs) at the data level, together with three structural additions: the choice of label set (Bertrand's theorem and Fock's projection single out $S^3$, both classical results), the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ at $k \in \{0, 1, 2\}$ tagging the framework's transcendentals (a **[CONDITIONAL]** case-check over the first fifteen projections), and the cosmic-Galois layer $U^*$ constructed in Paper 56 [57]. Marcolli–van Suijlekom [6] does not predict a Tannakian dual; the $U^*$ sits on top of the gauge-network scaffolding as additional structure, not as an MvS corollary. The key parallels at the data level:

- *Vertex algebra.* Marcolli–vS take $\mathcal{A}_V = C(V)$ on a vertex set $V$; we take $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ on the Fock-projected $S^3$ vertex set (Sec. 3).
- *Edge connection.* Marcolli–vS take a connection on edges valued in a compact unitary group (edge maps in $U(N)$); Paper 25 takes $G = U(1)$ and Paper 30 takes $G = \mathrm{SU}(2)$.
- *Spectral action.* Marcolli–vS show that the spectral action of the resulting almost-commutative spectral triple equals a Wilson lattice gauge action on the connection; Paper 30 verifies this explicitly for SU(2) at $n_{\max} = 3, 4$.
- *Continuum limit.* Per Perez-Sanchez, the continuum limit of the gauge-network spectral action is Yang–Mills without a Higgs sector. Paper 28's heat-kernel Seeley–DeWitt expansion is the spectral-action expansion of the Camporesi–Higuchi Dirac on $S^3$, not of the gauge-network action.

**Observation 5** (Lineage placement)**.** *Each structural ingredient of the GeoVac spectral triple has published precedent: finite almost-commutative spectral triples ([20, 21, 22]), graph spectral triples ([6, 10]), spectral truncations ([12, 15, 16]), and the Standard Model spectral action ([5]). What is not duplicated in the literature is the $\alpha$ observation ($K$ matches $\alpha^{-1}$ to $4.8 \times 10^{-7}$; the root of the cubic $1/\alpha = K - \alpha^2$ matches to $8.8 \times 10^{-8}$) from a discrete spectral construction; that remains the original empirical Observation of Paper 2.*

This is part of what the working hypothesis WH1 in the project register rests on (working-hypothesis register; its commutative-factor convergence leg proved as of 2026-06-10, the almost-commutative structure's convergence open): the structural framing is in the published lineage; the empirical claim is not duplicated.

## 10 Open Questions

The construction of Sec. 3 and the audit of Sec. 4 leave the following structural questions genuinely open.

**Q1. Order one beyond rank 1.** **[OPEN]** The order-one condition holds vacuously for the diagonal $\mathcal{A}_{\mathrm{GV}}$ (whose commutators with $D_{\mathrm{GV}}$ vanish, Proposition 2), *fails* at finite $n_{\max} \geq 2$ on the Fock/Weyl model of the operator-system representative (Table 1) and, from $n_{\max} = 1$, on the true window (audit item 6), and non-trivially constrains the rank-1 non-abelian extension $\mathcal{A}_{\mathrm{GV}} \otimes M_2(\mathbb{C})$ of Paper 30 in the gauge-network setting. At rank $\geq 2$ (e.g., $M_3(\mathbb{C})$ or sums of matrix algebras as in the Standard Model triple), order-one becomes a strong constraint and typically requires modification of the Dirac operator (*first-order condition* $\Rightarrow$ Higgs sector in [5]). Whether GeoVac admits a natural rank-$\geq 2$ extension—and what its spectral action would predict—is an open question. The Bargmann–Segal $S^5$ HO triple of Paper 24 was probed for an SU(3) extension in Sprint 5 (project development-frontier log) with negative outcome; that is one negative data point, not the full picture. The electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (rank up to 2 via $\mathbb{H} \cong M_2(\mathbb{C})$ over $\mathbb{C}$, a real $*$-subalgebra) is the most reachable test, and Sprint H1 addresses it; see Sec. 8.2 below for the verdict.

**Q2. Real structure at finite $n_{\max}$ (resolved for the $J$-relations; order zero/one open).** **[PANEL-VERIFIED]** Proposition 4 establishes a real structure on the continuum spinor bundle that restricts to the truncation. The axiom audit of Sec. 4 pins the finite-$n_{\max}$ behaviour down to machine precision: the restricted $J_{\mathrm{GV}}^{(n_{\max})}$ satisfies $J^2 = -\mathbb{1}$ (*not* $+\mathbb{1}$), $JD = +DJ$ and $J\mathcal{O}J^{-1} = \mathcal{O}$ bit-exactly at every $n_{\max}$ examined in the audit (`test_real_structure`), so for these three relations the finite-versus-limit question is closed in favour of finite-$n_{\max}$ exactness. No KO-dimension label is attached at finite cutoff: the chirality-doubled model truncation is even (Proposition 5(b)) with sign triple $(-,+,+)$, the true window that converges is odd like the continuum (measured, $n_{\max} \leq 3$; Proposition 5(c)) with only the pair $(\varepsilon, \varepsilon') = (-,+)$ defined, and the KO-3 label belongs to the continuum factor. **[MEASURED]** **[OPEN]** The order-zero and order-one conditions on the Fock/Weyl model of the operator-system representative fail at finite $n_{\max}$ with residuals that grow with the cutoff (order zero over $n_{\max} = 2, 3, 4$, order one over $n_{\max} = 2, 3$). **[MEASURED]** On the true Camporesi–Higuchi window of the convergence theorem the three $J$-relations hold exactly and order zero and order one fail from $n_{\max} = 1$, with residuals growing toward the generators' sup-norm scale, identified as a Toeplitz boundary term confined to the top Dirac levels (audit item 6, `tests/test_true_window_real_structure.py`). Open: the mechanism on the model, and the sense (if any) in which the continuum identities are recovered — on the true window the norm residual does not decay, while its normalised Hilbert–Schmidt weight falls over $n_{\max} = 2$–$5$ (measured in the audit driver, not test-pinned).

**Q3. KO-dimension and the four-way coincidence.** **[OPEN]** Working hypothesis WH4 asks: is the four-way $S^3$ coincidence (Sec. 1) forced by the KO-dimension 3 structure of the present triple's continuum factor, or is it independent geometric data? The relevant test is whether any other rank-1 spectral triple of KO-dimension 3 fails to produce all four roles. $S^3$ is unique among compact rank-1 non-abelian simply-connected groups, so the question reduces to: within the Marcolli–vS gauge-network setting, does KO-dimension 3 + non-abelian fiber + non-trivial Hopf bundle uniquely select $S^3$? The lineage literature does not appear to address this directly.

**Q4. Spectral-action selection of $\Lambda$.** **[OPEN]** Paper 28's spectral-action computation produced an exact two-term form $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = (\sqrt{\pi}/2)\Lambda^3 - (\sqrt{\pi}/4)\Lambda + O(e^{-\pi^2\Lambda^2})$ but did not autonomously select $\Lambda$. Setting the trace equal to $K/\pi$ gave a closed-form Cardano solution $\Lambda_\infty = 3.7102\ldots$ (50 dps), which was tautological in the sense that it was determined by demanding the equality. Whether there is a structural principle that selects $\Lambda$ is open; if so, $K/\pi$

would become a derived quantity rather than a coincidence. This is the most direct route by which Paper 2 could be promoted from observation to derivation, and it is also the one most resistant to incremental progress.

**Q5. Higher-form structure (Breit / rank-2 connections).** [**OPEN**] Paper 25 §VII.B asked whether the Breit interaction (rank-2 in electron-electron coupling) lifts the $U(1)$ 1-form picture to a higher-form gauge theory. In spectral-triple language this is the question whether the GeoVac triple admits a natural extension to a rank-2 fiber that produces the Breit retarded integrals. Paper 30's SU(2) result is rank 1 (vector); rank 2 (tensor) is a distinct extension; the answer is unknown.

**Q6. Convergence beyond the commutative factor.** [**OPEN**] Theorem 5 proves state-space convergence of the commutative factor on the true window, under the translation seminorm. Two convergence statements are not proved: the same convergence in the Dirac-commutator seminorm, and convergence of the almost-commutative (gauge-network, $\mathcal{A} \otimes M_n(\mathbb{C})$) structure that WH1 names (Remark 21).

# 11 Conclusion

[**INTERNAL THEOREM**] This paper has constructed an explicit finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected $S^3$ graph and audited it against the standard Connes axioms (Table 1, Theorem 2). The diagonal triple is a spectral, not a metric, object ($[D_{\mathrm{GV}}, \pi(a)] = 0$); the metric role and the continuum limit belong to the operator-system representative on the true Camporesi–Higuchi window (Paper 38), metrized by the left-translation seminorm: convergence of that commutative factor is proved, with the bound $d_{\mathrm{GH}} \leq \gamma_{n_{\max}}$ of Paper 38 [**INTERNAL THEOREM**], while Dirac-seminorm convergence and convergence of the almost-commutative structure are [**OPEN**] (Q6). At finite cutoff, on the Fock/Weyl model objects, the three $J$-relations hold exactly, the model truncation is even (sign triple $(-,+,+)$, no KO label), and order zero and order one fail with residuals that grow with the cutoff ([**MEASURED**], mechanism open); on the true window the three $J$-relations also hold exactly and order zero and order one also fail, by a Toeplitz boundary term of the compression ([**MEASURED**]). [**OBSERVATION**] Papers 25, 28, 30, 31 are read as projections of the resulting structure (Table 2), Papers 25 and 30 through the gauge-network Dirac rather than $D_{\mathrm{GV}}$. The construction sits inside the published Marcolli–van Suijlekom gauge-network framework with the Perez-Sanchez correction. The combination rule $K = \pi(B + F - \Delta)$ is phrased as a sum across three sectors of distinct type in the same triple (Observation 1), without changing its status as an observation. [**OBSERVATION**] The Coulomb/HO asymmetry of Paper 31 is the statement that the GeoVac triple's continuum limit is one specific real odd KO-dimension-3 triple, not a generic class (Observation 2); its true-window truncation is odd like the limit (measured, $n_{\max} \leq 3$), and only the chirality-doubled model truncation is even (Proposition 5).

Four things this paper did not do. It did not derive $\alpha$; Paper 2 stays in Observations. It did not claim the Standard Model: the G4a construction admits the SM gauge content only with imposed Yukawa, hypercharge and generation data (§8.4). It did not prove that the four-way $S^3$ coincidence is forced by KO-dimension 3; that is open question Q3. It did not identify a Bisognano–Wichmann modular flow at finite cutoff: the finite-cutoff period closure is automatic for the chosen integer-spectrum generator, and deriving the generator from a wedge-restricted state is open (Remark 18).

Three things this paper did do. It made the spectral-triple framing explicit and auditable. It placed the framework in a published mathematical lineage. It made the open questions about

$\alpha$, the fiber rank, and the four-way coincidence into precise spectral-triple questions instead of free-floating conjectures.

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| Axiom | Status at finite $n_{\max}$ | Status in continuum limit |
|---|---|---|
| Faithful $*$-representation | Holds rigorously | Holds rigorously |
| Bounded commutators | Holds ($[D, \pi(a)] = 0$ on $\mathcal{A}_{\mathrm{GV}}$) | Holds rigorously |
| Compact resolvent | Auto (finite dim) | Holds (CH summability) |
| Finite summability | Auto | $3^+$-summable |
| Order one, $\mathcal{A}_{\mathrm{GV}}$ (diagonal) | Holds vacuously | — |
| Real structure $J$ | Holds (audited) | Holds |
| $J^2 = -\mathbb{1}$ | Holds exactly $\forall n_{\max}$ | Holds |
| $JD = +DJ$ (truthful CH) | Holds exactly $\forall n_{\max}$ | Holds |
| $J\mathcal{O}J^{-1} = \mathcal{O}$ | Holds exactly $\forall n_{\max}$ | Holds |
| Order zero, $\mathcal{O}_{n_{\max}}$ (model) | Fails, $n_{\max} \geq 2$ (grows) | Holds |
| Order one, $\mathcal{O}_{n_{\max}}$ (model) | Fails, $n_{\max} \geq 2$ | Holds |
| $J^2$, $JD$, $J\mathcal{O}J^{-1}$ (true window) | Hold exactly, $n_{\max} \leq 4$ (pinned) | Holds |
| Order zero (true window) | Fails, $n_{\max} \geq 1$ (grows; boundary term) | Holds |
| Order one (true window) | Fails, $n_{\max} \geq 1$ (grows; boundary term) | Holds |
| $\mathbb{Z}_2$-grading (model) | Even ($\sigma_x \otimes \mathbb{1}$) | Absent (odd) |
| $\mathbb{Z}_2$-grading (true window) | None (commutant of $\mathcal{O}$ = scalars, $n_{\max} \leq 3$) | None (odd) |

Table 1: Axiom audit for the GeoVac triple at signature $(3, 0)$. Rows marked $\mathcal{A}_{\mathrm{GV}}$ refer to the diagonal representative, rows marked $\mathcal{O}_{n_{\max}}$ to the Fock/Weyl *model* of the operator-system representative (Remark 1), not the true Camporesi–Higuchi window that converges; rows marked *true window* are measured on the true Camporesi–Higuchi window (audit item 6), all other finite-cutoff entries on model objects. The continuum column lists properties of the *target* triple, the round $S^3$ with the Camporesi–Higuchi Dirac (real, odd, KO-dimension 3, classical facts); the convergence theorem identifies the limiting state space with that target's, up to isometry, and does not by itself deliver these properties. At finite $n_{\max}$ the three $J$-relations hold exactly (the rows marked $\forall\, n_{\max}$ additionally follow from the construction of Proposition 4); **[MEASURED]** order zero and order one on the model fail and the failure grows (order zero in residual and failing fraction over $n_{\max} = 2, 3, 4$; order one in failing fraction, $22\% \to 44\%$ over $n_{\max} = 2, 3$; normalised by the generator norms its max residual is nearly flat, a measurement not frozen in the tests; mechanism open), and on the true window they fail from $n_{\max} = 1$ and grow, as a boundary term of the compression; the chirality-doubled model truncation is even (Proposition 5(b)), with sign triple $(-, +, +)$ and no KO label at finite cutoff, while the true window is odd (measured, $n_{\max} \leq 3$; Proposition 5(c)). The finite-cutoff Krein-carrier sign audit at BBB $(m, n) = (4, 6)$ (signature-blind; the Lorentzian programme is descoped, Paper 45) is kept in `docs/chronicles/paper_32_chronicle.tex`.

| Paper | Object in this paper's language | Sector |
|---|---|---|
| Paper 7 | Fock projection / label set | Manifold sector[a] |
| Paper 22 | Selection rules from $\mathcal{A}_{\mathrm{GV}}$ | $\mathcal{A}$ sector (universal) |
| Paper 23 | Nuclear shell model on HO labels | Different $\mathcal{A}$ and $D$ |
| Paper 24 | Different $\mathcal{A}$: $S^5$ holomorphic | Distinct triple (HO) |
| Paper 25 | $U(1)$ gauge-network fluctuation[b] | Cartan torus |
| Paper 28 | Tr $f(D_{\mathrm{GV}}^2/\Lambda^2)$ | Spectral action (one loop) |
| Paper 30 | SU(2) gauge network, $M_2$ fiber[b] | Non-abelian extension |
| Paper 31 | $\mathcal{A}/D$ split (spectral data) | Structural partition |

Table 2: **[OBSERVATION]** Sub-sector identification of the major framework papers as projections of the GeoVac spectral triple. [a]The continuum limit is carried by the operator-system representative on the true Camporesi–Higuchi window, $\mathcal{O}^{\mathrm{CH}}_{n_{\max}}$ (Theorem 5), not by $\mathcal{A}_{\mathrm{GV}}$. [b]Via the gauge-network (edge-hopping) Dirac of Papers 25/30, not the diagonal $D_{\mathrm{GV}}$, which has no inner fluctuations.

| Layer | Coulomb (this paper) | Harmonic oscillator (Paper 24) |
|---|---|---|
| Manifold algebra $\mathcal{A}$ | Functions on $S^3$ Fock graph | Functions on $S^5$ holomorphic Hardy graph |
| Projection order | Second-order (scalar Laplace–Beltrami)* | First-order (complex Euler) |
| Spectrum | Quadratic $n^2-1$ (scalar LB)* | Linear $E_N = N + 3/2$ |
| Calibration constant | $\pi$ (volume slot $\mathrm{Vol}(S^2)/4$) | None ($\pi$-free in exact arithmetic) |
| Wilson + natural matter | SU(2), matter rep = link rep (argued) | Pure gauge only; matter needs CG |
| $L_0$ spectrum (withdrawn) | No ($L_0 \not\to \Delta_{S^3}$) | No (spectrum diagonal) |

Table 3: **[OBSERVATION]** The Coulomb/HO asymmetry from Paper 31, in spectral-triple language. The two systems are distinct spectral triples sharing only the algebra type (functions on a graph) and the trace structure of the spectral action. *The rows describe the Fock projection's scalar sector; this paper's Dirac operator $D_{\mathrm{GV}}$ is first order with spectrum $\pm(n+3/2)$, and the order contrast of the row is between the two *projections* (conformal vs. holomorphic).

| **Construction** | $G$ | **Host manifold** | **Vertex Hilbert space $\mathcal{H}_v$** | $\frac{1}{4N_c}$ |
|---|---|---|---|---|
| Paper 25 | $U(1)$ | $S^3$ (Hopf total space) | $\mathbb{C}$ (scalar) | $\frac{1}{4}$ |
| Paper 30 | SU(2) | $S^3 = \mathrm{SU}(2)$ | $\mathbb{C}^2$ (Dirac) | $\frac{1}{8}$ |
| ST-SU3 | SU(3) | $S^5$ Bargmann–Segal | $\bigoplus_N \mathbb{C}^{\dim(N,0)}$ | $\frac{1}{12}$ |

Table 4: The three Wilson lattice gauge constructions as instances of the Marcolli–van Suijlekom gauge-network framework on GeoVac sub-manifolds. The kinetic coefficient $\frac{1}{4N_c}$ in the rightmost column is the weak-coupling expansion coefficient of the Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{N_c}\mathrm{Re}\,\mathrm{tr}\,U_P\big)$ around the trivial vacuum, in the standard normalisation $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. **[SYMBOLIC + MEASURED]** The two non-abelian rows verify this coefficient independently: Paper 30 §5.2 and Sprint ST-SU3 §4 to machine precision; the abelian Paper 25 row is a separate $U(1)$ construction whose coefficient $\frac{1}{4}$ coincides with the formula at $N_c = 1$ — it is not a limit of the non-abelian ones (SU(1) is trivial).

| $n_{\max}$ | $\dim \mathcal{H}_{\mathrm{GV}}$ | $\dim \mathcal{H}_{\mathrm{total}}$ | $\lVert\Delta\rVert_{\mathrm{op}}$ | $\lVert\Delta\rVert_F$ | spec($\Delta$) |
|---|---|---|---|---|---|
| 1 | 4 | 32 | 2.000 | 8.000 | $\{-2{:}8,\ 0{:}16,\ +2{:}8\}$ |
| 2 | 16 | 128 | 2.000 | 16.000 | $\{-2{:}32,\ 0{:}64,\ +2{:}32\}$ |
| 3 | 40 | 320 | 2.000 | 25.298 | $\{-2{:}80,\ 0{:}160,\ +2{:}80\}$ |

Table 5: Tensor-product chirality residual $\Delta = \gamma_{\mathrm{GV}} \otimes \not{K}_F - \not{K}_{\mathrm{GV}} \otimes \gamma_F$ at $n_{\max} \in \{1, 2, 3\}$, computed in `debug/g3c_tensor_chirality.py` from the production $\gamma_{\mathrm{GV}}$ (G3-A, $\sigma_x \otimes \not{K}_w$) and $\gamma_F$ (G3-B, KO-dim 6). Operator norm is exactly 2 at every $n_{\max}$; eigenvalue spectrum is $\{-2, 0, +2\}$ with multiplicities in ratio 1:2:1, $n_{\max}$-independent. Frobenius norm scales as $\sqrt{2 \cdot \dim \mathcal{H}_{\mathrm{total}}}$ exactly. Data `debug/data/g3c_tensor_chirality.json`.